MAT7108 RIEMANNIAN GEOMETRY

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Preface

These are my personal lecture notes for **Prof. Ma's MAT 7108 Riemannian Geometry course** at SUSTech in the **2026 Spring** semester.

The course and these notes mainly follow the second edition of *Introduction to Riemannian Manifolds* by John M. Lee [4].

These notes are not endorsed by the lecturer. I have revised and expanded the handwritten notes after each lecture, so they should not be read as a verbatim record. Added explanations and proofs are intended to make the arguments easier to follow; advanced results used without proof are identified explicitly. Any remaining errors are my responsibility.

Classification is a central theme in mathematics: one chooses an equivalence relation and seeks invariants or structural theorems that distinguish its classes. Algebraic topology provides tools for studying spaces up to homotopy or homeomorphism, while Riemannian geometry relates a smooth manifold's curvature and metric to its global geometry and topology. The results in this course illustrate these connections; they do not give a classification of all smooth manifolds.

Corrections and suggestions are welcome. If you find typos or errors in these notes, please contact me at:

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ChatGPT-6 Astra is used for editorial purposes.

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Chapter 1

Basic Riemannian Geometry

1.1 Review of Smooth Manifolds

This first section is a short review of smooth manifolds and the language that will be used throughout the notes. The basic reference for this background is Chapter 0 of do Carmo's book [3]; for a more systematic treatment, one may also consult Lee's book [4]. The purpose here is not to prove every foundational theorem, but to fix the notation and recall the main objects before introducing Riemannian metrics, connections, geodesics, and curvature.

The guiding idea is simple: a smooth manifold is a space that may be curved globally, but locally looks like an open subset of Euclidean space. Therefore, calculus on a manifold is done by using local coordinates and then checking that the result is independent of the chosen coordinates.

Definition 1.1.1 (Smooth manifold)**.** An n-dimensional **smooth manifold** is a Hausdorff, second countable topological space M equipped with a smooth atlas. More explicitly, an atlas is a collection of charts

$$(U_\alpha, \varphi_\alpha), \quad \varphi_\alpha : U_\alpha \to \varphi_\alpha(U_\alpha) \subset \mathbb{R}^n,$$

where $\{U_\alpha\}$ is an open cover of M, each φ_α is a homeomorphism onto an open subset of $\mathbb{R}^n$, and for every pair of overlapping charts the transition map

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \longrightarrow \varphi_\beta(U_\alpha \cap U_\beta)$$

is smooth. We usually take the maximal atlas generated by these charts and call it the smooth structure of M.

The Hausdorff condition allows us to separate points, while second countability prevents pathological "too large" examples. These topological assumptions are also important in the appendix on partitions of unity, where they are used to prove paracompactness. In this course, unless stated otherwise, all manifolds are smooth, finite-dimensional, and without boundary.

Example 1.1.2. Every open subset $U \subset \mathbb{R}^n$ is a smooth manifold with one global chart. The sphere

$$\mathbb{S}^n = \{x \in \mathbb{R}^{n+1} : |x| = 1\}$$

is also a smooth manifold; it can be described by stereographic charts, or viewed as an embedded submanifold of $\mathbb{R}^{n+1}$. Later, when we put the induced Riemannian metric on $\mathbb{S}^n$, it becomes one of the most important examples for geodesics, curvature, and comparison theorems.

Definition 1.1.3 (Smooth maps)**.** Let M and N be smooth manifolds. A continuous map $F : M \to N$ is called **smooth** if for every chart (U, φ) on M and every

chart (V, ψ) on N, the local expression

$$\psi \circ F \circ \varphi^{-1} : \varphi(U \cap F^{-1}(V)) \longrightarrow \psi(V)$$

is smooth as a map between open subsets of Euclidean spaces. A smooth map with a smooth inverse is called a **diffeomorphism**. A smooth real-valued function on M is an element of $C^\infty(M)$.

Thus, smoothness is still the usual Euclidean smoothness, but tested in charts. This viewpoint will be used repeatedly: for example, a Riemannian metric is smooth precisely because its local coefficient functions are smooth, and a connection is smooth because its action on local frames has smooth coefficients.

Definition 1.1.4 (Tangent space)**.** Let $p \in M$. A tangent vector at p may be defined as a derivation

$$v : C^\infty(M) \to \mathbb{R}$$

satisfying

$$v(af + bg) = av(f) + bv(g), \qquad v(fg) = f(p)v(g) + g(p)v(f).$$

The vector space of all tangent vectors at p is denoted by T_pM and is called the **tangent space** of M at p.

Given a chart $(U, (x^1, \cdots, x^n))$ around p, there is a natural basis of T_pM:

$$\left.\frac{\partial}{\partial x^1}\right|_p, \cdots, \left.\frac{\partial}{\partial x^n}\right|_p.$$

Every tangent vector $v \in T_pM$ can be written uniquely as

$$v = \sum_{i=1}^{n} v^i \left.\frac{\partial}{\partial x^i}\right|_p.$$

This coordinate expression explains why tangent vectors should be thought of as directional derivatives. The disjoint union

$$TM = \bigsqcup_{p \in M} T_pM$$

is called the **tangent bundle**. It is the first and most important example of a vector bundle, and it will reappear in the definitions of Riemannian metrics, connections, geodesics, curvature, and Jacobi fields.

Definition 1.1.5 (Differential)**.** Let $F : M \to N$ be a smooth map. The **differen-**

tial of F at p is the linear map

$$dF_p : T_pM \to T_{F(p)}N$$

defined by

$$dF_p(v)(h) = v(h \circ F), \qquad h \in C^\infty(N).$$

In local coordinates, dF_p is represented by the Jacobian matrix of the coordinate expression of F.

The differential is the bridge between smooth maps and tangent spaces. In the next section, if $F : M \to N$ is an immersion and h is a Riemannian metric on N, then the pullback metric is defined by

$$(F^*h)_p(X, Y) = h_{F(p)}(dF_p(X), dF_p(Y)).$$

This is exactly the same idea as the first fundamental form for surfaces in $\mathbb{R}^3$.

Definition 1.1.6 (Vector fields)**.** A **smooth vector field** on M is a smooth section of the tangent bundle, namely a smooth map

$$X : M \to TM$$

such that $X_p \in T_pM$ for every $p \in M$. The space of smooth vector fields is denoted by $\mathscr{X}(M)$ or $\Gamma(TM)$.

In a local chart, a vector field has the form

$$X = \sum_{i=1}^{n} X^i \frac{\partial}{\partial x^i},$$

where the coefficient functions X^i are smooth. A vector field acts on smooth functions by

$$X(f)(p) = X_p(f), \qquad f \in C^\infty(M).$$

This means that vector fields are global versions of first-order directional derivatives.

Definition 1.1.7 (Lie bracket)**.** For $X, Y \in \mathscr{X}(M)$, the **Lie bracket** $[X, Y]$ is the vector field defined by

$$[X, Y](f) = X(Y(f)) - Y(X(f)), \qquad f \in C^\infty(M).$$

The Lie bracket measures the failure of two directional derivatives to commute. It is not defined using a Riemannian metric; it belongs to the smooth structure

itself. Later, one will learn the torsion-free condition for the Levi-Civita connection:

$$\nabla_X Y - \nabla_Y X = [X, Y].$$

Definition 1.1.8 (Differential forms)**.** The **cotangent space** at $p \in M$ is the dual vector space $T_p^* M = (T_p M)^*$. A smooth k-form on M is a smooth section of the bundle $\Lambda^k T^* M$, namely a smoothly varying alternating k-linear map

$$\omega_p : T_p M \times \cdots \times T_p M \to \mathbb{R}.$$

The space of all smooth k-forms is denoted by $\Omega^k(M)$. In particular, $\Omega^0(M) = C^\infty(M)$.

For a smooth function $f \in C^\infty(M)$, its differential $df \in \Omega^1(M)$ is defined by

$$df_p(v) = v(f), \qquad v \in T_p M.$$

In local coordinates $(x^1, \cdots, x^n)$, the 1-forms $dx^1, \cdots, dx^n$ form the basis dual to $\frac{\partial}{\partial x^1}, \cdots, \frac{\partial}{\partial x^n}$, and a k-form can be written as

$$\omega = \sum_{i_1 < \cdots < i_k} \omega_{i_1 \cdots i_k} \, dx^{i_1} \wedge \cdots \wedge dx^{i_k}.$$

The wedge product combines a k-form and an l-form into a $(k+l)$-form, and the exterior derivative

$$d : \Omega^k(M) \to \Omega^{k+1}(M)$$

extends the operation $f \mapsto df$ and satisfies $d^2 = 0$. Differential forms are the natural objects for integration on manifolds. Once a Riemannian metric is chosen, it also determines a Riemannian density, and on an oriented coordinate neighborhood this is represented by the volume form

$$d\,\mathrm{vol}_g = \sqrt{\det(g_{ij})}\, dx^1 \wedge \cdots \wedge dx^n,$$

which is used whenever we discuss volume, integration, or analytic constructions on Riemannian manifolds.

If $F : M \to N$ is smooth and $\omega \in \Omega^k(N)$, the **pullback** $F^*\omega \in \Omega^k(M)$ is defined by

$$(F^*\omega)_p(v_1, \cdots, v_k) = \omega_{F(p)}(dF_p(v_1), \cdots, dF_p(v_k)).$$

This is the covariant counterpart of the differential dF_p and is one reason differential forms fit naturally with smooth maps.

Definition 1.1.9 (Immersion, submersion, embedding)**.** Let $F : M \to N$ be a smooth map.

- F is an **immersion** if $dF_p : T_pM \to T_{F(p)}N$ is injective for every $p \in M$.
- F is a **submersion** if $dF_p : T_pM \to T_{F(p)}N$ is surjective for every $p \in M$.
- F is an **embedding** if it is an immersion and a homeomorphism onto its image with the subspace topology.

These three notions control how manifolds sit inside or project onto one another. Immersions are used to pull back Riemannian metrics, submersions lead to Riemannian submersions and horizontal/vertical decompositions, and embeddings are the basic model for submanifolds. In later sections, these ideas will be combined with metric information to study induced metrics, second fundamental forms, and examples such as the sphere and projective spaces.

Definition 1.1.10 (Embedded submanifold)**.** Let M be an n-dimensional smooth manifold. A subset $S \subset M$ is called an **embedded k-dimensional submanifold** if for every $p \in S$ there is a chart $(U, (x^1, \cdots, x^n))$ of M around p such that

$$S \cap U = \{q \in U : x^{k+1}(q) = \cdots = x^n(q) = 0\}.$$

Such coordinates are called adapted coordinates for S.

In these notes, when we say submanifold without further qualification, we usually mean embedded submanifold. Immersed submanifolds are also important, but embedded submanifolds are better behaved topologically because the subspace topology agrees with the manifold topology.

In adapted coordinates, S inherits a smooth structure from M, and its tangent space is naturally identified with a subspace of the ambient tangent space:

$$T_pS = \operatorname{span}\left\{ \left.\frac{\partial}{\partial x^1}\right|_p, \cdots, \left.\frac{\partial}{\partial x^k}\right|_p \right\} \subset T_pM.$$

If $i : S \hookrightarrow M$ denotes the inclusion map, then $di_p : T_pS \to T_pM$ is injective. Therefore, when M has a Riemannian metric g, the pullback i^*g gives a Riemannian metric on S. This is the induced metric that will be used later in the discussion of Riemannian submanifolds and the second fundamental form.

Theorem 1.1.11 (Regular level set theorem)**.** *Let $F : M^n \to N^m$ be a smooth map, and let $y \in N$ be a regular value, meaning that $dF_p : T_pM \to T_yN$ is surjective for every $p \in F^{-1}(y)$. Then $F^{-1}(y)$ is either empty or an embedded submanifold of M of dimension $n - m$, and at every point of a nonempty level set,*

$$T_p(F^{-1}(y)) = \ker(dF_p).$$

This theorem is the standard way to produce submanifolds. For example, the sphere is the regular level set of $F : \mathbb{R}^{n+1} \to \mathbb{R}$, $F(x) = |x|^2$, at the value 1. Fibers of submersions are also regular level sets, which is why submersion theory appears naturally before Riemannian submersions. After adding a Riemannian metric, which is a pointwise inner product structure, one can decompose along a submanifold

$$T_pM = T_pS \oplus (T_pS)^{\perp},$$

and this tangent–normal decomposition is the starting point of the second fundamental form.

To summarize, the smooth structure gives us charts, smooth functions, tangent spaces, vector fields, differential forms, smooth maps, and submanifolds. Riemannian geometry begins by adding one extra ingredient: an inner product g_p on each tangent space T_pM, varying smoothly with p. Once this metric is fixed, we can define length, distance, volume, geodesics, curvature, and the comparison theorems studied in the rest of these notes.

1.2 Riemannian Metrics and Riemannian Manifolds

A smooth structure lets us differentiate functions and maps. To measure lengths and angles, we need an inner product on each tangent space. For instance, a sphere and an ellipsoid are diffeomorphic, but their induced metrics need not be isometric: a diffeomorphism preserves the smooth structure, whereas an isometry also preserves the metric.

Definition 1.2.1 (Riemannian Metric)**.** Let M be a smooth manifold. A **Riemannian metric** on M is a smooth choice of inner product on each tangent space T_pM. More precisely, for every $p \in M$, it is a bilinear form

$$g_p : T_pM \times T_pM \longrightarrow \mathbb{R}$$

satisfying:

1. Symmetry: $g_p(X_p, Y_p) = g_p(Y_p, X_p)$.
2. $\mathbb{R}$-linearity in each variable.
3. Positive definiteness: $g_p(X_p, X_p) > 0$ for every nonzero $X_p \in T_pM$.

The word "smooth" means that for any smooth vector fields $X, Y \in \mathscr{X}(M)$, the function

$$p \longmapsto g_p(X_p, Y_p)$$

is smooth on M.

For tangent vectors $v, w \in T_pM$, the metric defines

$$|v|_g = \sqrt{g_p(v, v)}, \qquad \cos \angle_g(v, w) = \frac{g_p(v, w)}{|v|_g |w|_g}$$

when $v, w \neq 0$. Smoothness ensures that these inner products vary compatibly with the smooth structure.

Example 1.2.2. Let $U \subset \mathbb{R}^n$ be an open subset. Since the tangent bundle of U is naturally trivial, we may identify each tangent space T_pU with $\mathbb{R}^n$. For $v = (v_1, \cdots, v_n)$ and $w = (w_1, \cdots, w_n)$, define

$$(g_0)_p(v, w) = \sum_{k=1}^{n} v_k w_k.$$

This is the standard Euclidean metric, also called the flat metric. We will use the notation g_0 for this metric throughout the notes.

Local coordinates and change of coordinates. In coordinates $(x^1, \dots, x^n)$, write $\partial_i = \partial/\partial x^i$ and

$$g_{ij} = g(\partial_i, \partial_j).$$

For $X = X^i \partial_i$ and $Y = Y^j \partial_j$, bilinearity gives

$$g(X, Y) = \sum_{i,j=1}^{n} g_{ij} X^i Y^j, \qquad g = \sum_{i,j=1}^{n} g_{ij}\, dx^i \otimes dx^j.$$

Thus, a metric is represented in each chart by a smooth symmetric positive definite matrix (g_{ij}). Repeated indices are summed when the summation symbol is omitted.

On the overlap with another chart $(y^1, \dots, y^n)$, the chain rule gives

$$\frac{\partial}{\partial x^i} = \sum_{\alpha=1}^{n} \frac{\partial y^\alpha}{\partial x^i} \frac{\partial}{\partial y^\alpha}.$$

Consequently, the matrices in the two charts satisfy

$$g^{(x)}_{ij} = \sum_{\alpha,\beta=1}^{n} \frac{\partial y^\alpha}{\partial x^i} \frac{\partial y^\beta}{\partial x^j} g^{(y)}_{\alpha\beta}.$$

This transformation law explains how the coordinate matrices represent a single global tensor. Choosing a positive definite matrix independently in every chart need not produce a metric unless the matrices agree by this rule on overlaps.

Proposition 1.2.3. *Let $F : M \to N$ be an immersion, and let h be a Riemannian metric on N. Then the formula*

$$g_p(X, Y) = h_{F(p)}(dF_p(X), dF_p(Y))$$

defines a Riemannian metric $g = F^*h$ on M, called the pullback metric.

Proof. Symmetry, bilinearity, and smoothness follow from the corresponding properties of h and the smoothness of F. If $X \neq 0$, then $dF_p(X) \neq 0$ because F is an immersion; hence $g_p(X, X) > 0$. □

For a general smooth map, the same pullback formula defines a smooth symmetric tensor, but it can be degenerate. Injectivity of dF_p is precisely what ensures positive definiteness.

Example 1.2.4 (The first fundamental form)**.** For a regular surface parametrization $X(u, v)$ in $\mathbb{R}^3$, set

$$E = \langle X_u, X_u \rangle, \quad F = \langle X_u, X_v \rangle, \quad G = \langle X_v, X_v \rangle.$$

The induced metric is

$$g = E\,du \otimes du + F(du \otimes dv + dv \otimes du) + G\,dv \otimes dv.$$

Thus $E = g_{11}$, $F = g_{12} = g_{21}$, and $G = g_{22}$: the first fundamental form is exactly a pullback metric. For the upper hemisphere parametrized by

$$X(u, v) = (u, v, \sqrt{1 - u^2 - v^2}), \qquad u^2 + v^2 < 1,$$

putting $D = 1 - u^2 - v^2$ gives

$$E = 1 + \frac{u^2}{D}, \qquad F = \frac{uv}{D}, \qquad G = 1 + \frac{v^2}{D}.$$

Proposition 1.2.5 (Existence of Riemannian Metrics)**.** *Every smooth manifold admits a Riemannian metric.*

Proof. One way to prove this is to use the Whitney embedding theorem and then pull back the Euclidean metric from some $\mathbb{R}^N$. A more intrinsic and more useful proof uses a partition of unity: first define Euclidean metrics in coordinate charts, and then glue them together using a partition of unity subordinate to the chart cover. The local positivity and smoothness are preserved under this gluing process. More explicitly, if g_i is the coordinate Euclidean metric on U_i and $\{\lambda_i\}$ is a subordinate partition, then $g = \sum_i \lambda_i g_i$, with each summand extended by zero, is smooth and positive definite. The full construction is given in Appendix B. □

1.2.1 Riemannian Submersion

Recall that a submersion is a smooth map $\pi : \tilde{M} \to M$ such that $d\pi_{\tilde{p}} : T_{\tilde{p}}\tilde{M} \to T_{\pi(\tilde{p})}M$ is surjective for every $\tilde{p} \in \tilde{M}$. Two basic results from smooth manifold theory are especially useful here:

- *Regular Value Theorem*: if $y \in M$ is a regular value of π, then $\pi^{-1}(y)$ is an embedded submanifold of $\tilde{M}$ with dimension $\dim \tilde{M} - \dim M$.
- *Rank Theorem*: if $\pi : \tilde{M}^{m+n} \to M^m$ is a submersion, then near each point there are coordinates $(x^1, \cdots, x^{m+n})$ on $\tilde{M}$ and $(y^1, \cdots, y^m)$ on M such that

$$\pi(x^1, \cdots, x^{m+n}) = (x^1, \cdots, x^m).$$

These facts use only the smooth structure. Once a Riemannian metric is added to $\tilde{M}$, the tangent space of $\tilde{M}$ naturally splits into vertical and horizontal parts.

Definition 1.2.6 (Vertical and horizontal spaces)**.** Let $\pi : \tilde{M} \to M$ be a submersion,

and let $\tilde{g}$ be a Riemannian metric on $\tilde{M}$. For each $\tilde{p} \in \tilde{M}$, define the **vertical space**

$$V_{\tilde{p}} := \ker(d\pi_{\tilde{p}}) \subset T_{\tilde{p}}\tilde{M}.$$

The **horizontal space** is the orthogonal complement of the vertical space:

$$H_{\tilde{p}} := V_{\tilde{p}}^{\perp} = \{w \in T_{\tilde{p}}\tilde{M} : \tilde{g}_{\tilde{p}}(w, v) = 0 \text{ for all } v \in V_{\tilde{p}}\}.$$

Lemma 1.2.7. *Every smooth vector field $X \in \mathscr{X}(\tilde{M})$ has a unique decomposition*

$$X = X^V + X^H,$$

where X^V is vertical and X^H is horizontal. Moreover, both X^V and X^H are smooth vector fields.

Proof. Pointwise, this is just the orthogonal decomposition

$$T_{\tilde{p}}\tilde{M} = V_{\tilde{p}} \oplus H_{\tilde{p}}.$$

The only point to check is smoothness. By the rank theorem, locally, the vertical space is spanned by

$$\frac{\partial}{\partial x^{m+1}}, \cdots, \frac{\partial}{\partial x^{m+n}}.$$

Apply Gram-Schmidt to the coordinate frame, listing the vertical vectors first. All divisions are with respect to positive smooth norms, so the resulting orthonormal frame is smooth. Its first n vectors $E_1, \ldots, E_n$ span the vertical space, and its remaining vectors span the horizontal space. In this frame,

$$X^V = \sum_{a=1}^{n} \tilde{g}(X, E_a)E_a, \qquad X^H = X - X^V,$$

which proves smoothness. The uniqueness of the pointwise decomposition ensures that these local formulas agree on overlaps. □

Corollary 1.2.8. *For any vector field $Y \in \mathscr{X}(M)$, there exists a unique horizontal vector field $\tilde{Y} \in \mathscr{X}(\tilde{M})$ such that*

$$d\pi_{\tilde{p}}(\tilde{Y}_{\tilde{p}}) = Y_{\pi(\tilde{p})}$$

for every $\tilde{p} \in \tilde{M}$. The vector field $\tilde{Y}$ is called the horizontal lift of Y.

Proof. Since $d\pi_{\tilde{p}}$ is surjective and $V_{\tilde{p}} = \ker(d\pi_{\tilde{p}})$, the restriction

$$d\pi_{\tilde{p}}|_{H_{\tilde{p}}} : H_{\tilde{p}} \longrightarrow T_{\pi(\tilde{p})}M$$

is a linear isomorphism. Taking its pointwise inverse yields the unique horizontal lift. Smoothness follows from the same local coordinate argument as in the previous lemma. □

Definition 1.2.9 (Riemannian Submersion)**.** Let $(\tilde{M}, \tilde{g})$ and (M, g) be Riemannian manifolds. A surjective submersion $\pi : \tilde{M} \to M$ is called a **Riemannian submersion** if, for every $\tilde{p} \in \tilde{M}$, the restricted map

$$d\pi_{\tilde{p}}|_{H_{\tilde{p}}} : H_{\tilde{p}} \longrightarrow T_{\pi(\tilde{p})}M$$

is a linear isometry.

Riemannian submersions provide a means of transferring metric information from a larger space to a quotient-like space. A common method for constructing them uses group actions.

Definition 1.2.10. Let a group G act smoothly on a Riemannian manifold $(\tilde{M}, \tilde{g})$, and suppose $\pi : \tilde{M} \to M$ is a submersion.

- The action is *vertical* if $\pi(h\tilde{p}) = \pi(\tilde{p})$ for all $h \in G$ and $\tilde{p} \in \tilde{M}$.
- The action is *transitive on fibers* if, for every $y \in M$ and every $\tilde{p}_1, \tilde{p}_2 \in \pi^{-1}(y)$, there exists $h \in G$ such that $h\tilde{p}_1 = \tilde{p}_2$.
- The action is *isometric* if every $h \in G$ acts by an isometry of $(\tilde{M}, \tilde{g})$.

Theorem 1.2.11. *Suppose a group G acts on $(\tilde{M}, \tilde{g})$ isometrically, vertically, and transitively on the fibers of a surjective submersion $\pi : \tilde{M} \to M$. Then there exists a unique Riemannian metric g on M such that*

$$\pi : (\tilde{M}, \tilde{g}) \longrightarrow (M, g)$$

is a Riemannian submersion.

Proof. Let $y \in M$ and $X, Y \in T_yM$. Choose $\tilde{p} \in \pi^{-1}(y)$ and take the unique horizontal lifts $X_{\tilde{p}}, Y_{\tilde{p}} \in H_{\tilde{p}}$ satisfying

$$d\pi_{\tilde{p}}(X_{\tilde{p}}) = X, \qquad d\pi_{\tilde{p}}(Y_{\tilde{p}}) = Y.$$

Define

$$g_y(X, Y) = \tilde{g}_{\tilde{p}}(X_{\tilde{p}}, Y_{\tilde{p}}).$$

If another point $\tilde{p}' \in \pi^{-1}(y)$ is chosen, then by transitivity there is $h \in G$ with $h\tilde{p} = \tilde{p}'$. Since the action is vertical, $d\pi_{\tilde{p}'} \circ dh_{\tilde{p}} = d\pi_{\tilde{p}}$; since the action is isometric, $dh_{\tilde{p}}$ preserves horizontal spaces and inner products. Hence, the value of $g_y(X, Y)$ is

independent of the choice of $\tilde{p}$. To check smoothness, take a local smooth section $s : U \to \tilde{M}$, which exists by the rank theorem. For vector fields X, Y on U, evaluate their smooth horizontal lifts along s; their $\tilde{g}$-inner product is the smooth function $g(X, Y)$ on U. Positive definiteness follows because a nonzero vector has a nonzero horizontal lift. The defining isometry condition also forces this formula, proving uniqueness. By construction, $d\pi$ restricts to an isometry on horizontal spaces, so π is a Riemannian submersion. □

Remark 1.2.12. Immersions allow us to pull a metric back from a target manifold, while Riemannian submersions allow us to pass a metric down to a quotient-like space. These are two fundamental ways to transport Riemannian metrics.

Example 1.2.13 (Fubini–Study Metric on $\mathbb{CP}^n$)**.**

Recall that complex projective space is defined by

$$\mathbb{CP}^n := \{\text{complex one-dimensional subspaces of } \mathbb{C}^{n+1}\}.$$

Equivalently,

$$\mathbb{CP}^n = (\mathbb{C}^{n+1} \setminus \{0\})/\mathbb{C}^*.$$

The standard projective charts show that $\mathbb{CP}^n$ is a smooth manifold of real dimension $2n$. The map

$$q : \mathbb{C}^{n+1} \setminus \{0\} \longrightarrow \mathbb{CP}^n, \qquad z \longmapsto [z],$$

is a submersion. Indeed, in the chart where $z^k \neq 0$, its coordinate functions are the ratios z^j/z^k for $j \neq k$, whose differential is surjective. Its differential has real kernel $\operatorname{span}_{\mathbb{R}}\{z, iz\}$.

Each complex line intersects the unit sphere $\mathbb{S}^{2n+1} \subset \mathbb{C}^{n+1}$ in an $\mathbb{S}^1$-orbit. Therefore, the quotient map restricts to the Hopf fibration

$$\pi : \mathbb{S}^{2n+1} \longrightarrow \mathbb{CP}^n.$$

At a unit vector z, the tangent space to the sphere is the real orthogonal complement of z. Therefore

$$\ker d\pi_z = \mathbb{R}\, iz, \qquad \operatorname{rank}_{\mathbb{R}} d\pi_z = 2n,$$

so the restriction is also a surjective submersion. Its horizontal space consists of the tangent vectors orthogonal to iz.

The group $\mathbb{S}^1 \subset \mathbb{C}$ acts on $\mathbb{S}^{2n+1}$ by

$$\lambda \cdot z = \lambda z.$$

With the metric on $\mathbb{S}^{2n+1}$ induced from the Euclidean metric on $\mathbb{C}^{n+1} \cong \mathbb{R}^{2n+2}$, this action is isometric. It is also vertical and transitive on the fibers of π. Hence,

the previous theorem yields a unique metric on $\mathbb{CP}^n$ that makes π a Riemannian submersion. This metric is the Fubini–Study metric.

$$(\mathbb{C}^{n+1} \setminus \{0\}, g_0) \xrightarrow{\text{induced}} \mathbb{S}^{2n+1} \xrightarrow{\text{quotient}} \mathbb{CP}^n.$$

1.2.2 Riemannian Distance Function and Global Metric

The metric g_p measures tangent vectors at one point. Integrating their norms gives lengths of curves; taking the infimum of these lengths gives a distance between points of the manifold.

Definition 1.2.14 (Admissible curve and length)**.** A curve $\gamma : [a, b] \to M$ is **admissible** if it is continuous and piecewise smooth, with a finite subdivision into smooth pieces. Its length is

$$L_g(\gamma) = \int_a^b |\gamma'(t)|_g \, dt, \qquad |\gamma'(t)|_g = \sqrt{g_{\gamma(t)}(\gamma'(t), \gamma'(t))}.$$

The integral is the sum of the integrals over the smooth pieces. Derivatives at the finite number of break points do not affect the value.

In coordinates, this becomes

$$L_g(\gamma) = \int_a^b \sqrt{\sum_{i,j} g_{ij}(\gamma(t))\dot{\gamma}^i(t)\dot{\gamma}^j(t)} \, dt$$

for a curve contained in one chart. A change of parameter by a smooth, monotone diffeomorphism preserves length by the chain rule and the change-of-variables formula. In particular, reversing a curve preserves its length; concatenating two curves adds their lengths.

Lemma 1.2.15 (Existence of admissible curves)**.** *If M is connected, any two points of M can be joined by an admissible curve.*

Proof. Coordinate balls are path connected, so a connected manifold is path connected. Choose a continuous path $\sigma : [0, 1] \to M$ from p to q. Cover its compact image by coordinate neighborhoods whose images are Euclidean balls. By the Lebesgue number lemma for the induced open cover of $[0, 1]$, there is a finite subdivision

$$0 = t_0 < t_1 < \cdots < t_k = 1$$

such that each $\sigma([t_{i-1}, t_i])$ lies in one such chart. Join the coordinate images of $\sigma(t_{i-1})$ and $\sigma(t_i)$ by a straight line segment. The convexity of the coordinate ball

keeps the segment within the chart. Pulling these segments back and concatenating them yields the required piecewise-smooth curve. □

Definition 1.2.16 (Riemannian distance)**.** On a connected Riemannian manifold (M, g), define

$$d_g(p, q) = \inf\{L_g(\gamma) : \gamma \text{ is admissible from } p \text{ to } q\}.$$

The preceding lemma ensures that this infimum is finite.

Lemma 1.2.17 (Uniform comparison in a compact coordinate set)**.** *Let (W, ϕ) be a coordinate chart and let $K \subset W$ be compact. There are constants $0 < c \leq C < \infty$ such that*

$$c\,|d\phi_x(v)|^2 \leq g_x(v, v) \leq C\,|d\phi_x(v)|^2$$

for every $x \in K$ and $v \in T_xM$.

Proof. In coordinates, the function

$$(x, u) \longmapsto \sum_{i,j} g_{ij}(x)u^i u^j$$

is continuous and positive on the compact set $K \times S^{n-1}$. Its minimum is positive, and its maximum is finite. Homogeneity gives the inequalities for all vectors. □

The corresponding norm bounds have constants $\sqrt{c}$ and $\sqrt{C}$. These are inequalities between quadratic forms, not entrywise inequalities between the matrix entries.

Theorem 1.2.18. *The Riemannian distance d_g is a metric on a connected manifold M, and its metric topology agrees with the original manifold topology.*

The proof is omitted here since it is not the core idea of the course, but this ensures that the metric does not introduce topological issues on the manifold.

Remark 1.2.19 (Why the Hausdorff assumption matters)*.* In a proof by the first-exit argument, one uses the fact that a compact coordinate ball is closed in the ambient manifold. On the non-Hausdorff line with two origins, the locally Euclidean metric allows paths between the two distinct origins of arbitrarily small length. Thus, the resulting infimum does not separate those points.

The construction of d_g leads to the questions emphasized in the lecture:

- Is (M, d_g) complete?
- Is the infimum defining $d_g(p, q)$ attained by a curve?
- If a minimizing curve exists, is it unique?
- Which differential equation must a smooth minimizing curve satisfy?

A smooth variation of a minimizing curve must have zero first derivative of length whenever the length functional is differentiable along that variation. This leads to an Euler-Lagrange equation. To express the resulting geodesic equation intrinsically, we next develop vector bundles and connections.

1.3 Vector Bundles and Connections

In the previous section, a Riemannian metric gave us the lengths of tangent vectors, lengths of curves, as well as a global distance function. To study this metric effectively, however, we also need a way to differentiate vector fields and tensor fields on a manifold. This is the role of a connection.

The natural framework is that of vector bundles. A vector bundle is a family of vector spaces smoothly parametrized by points of a manifold. The tangent bundle TM is the main example, but it is useful to keep the more general language in mind.

Definition 1.3.1 (Vector bundle)**.** A smooth vector bundle of rank k over M is a smooth manifold E together with a smooth surjective map

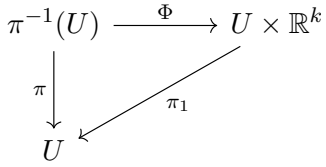
$$\pi : E \longrightarrow M$$

such that each fiber $E_x = \pi^{-1}(x)$ is a k-dimensional real vector space and such that every $x \in M$ has a neighborhood U with a diffeomorphism

$$\Phi : \pi^{-1}(U) \longrightarrow U \times \mathbb{R}^k$$

satisfying the following two conditions:

- the diagram



  commutes;

- for each $x \in U$, the restriction $\Phi|_{E_x} : E_x \to \{x\} \times \mathbb{R}^k$ is a linear isomorphism.

The map Φ is called a local trivialization.

Thus, a vector bundle is locally a product $U \times \mathbb{R}^k$, but it may be globally twisted. If Φ_α and Φ_β are two local trivializations over U_α and U_β, then on $U_\alpha \cap U_\beta$ the change of trivialization has the form

$$\Phi_\beta \circ \Phi_\alpha^{-1}(x, v) = (x, h_{\beta\alpha}(x)v),$$

where

$$h_{\beta\alpha} : U_\alpha \cap U_\beta \longrightarrow GL(k, \mathbb{R})$$

is a smooth map. These maps are called transition functions. Our index convention is that $h_{\beta\alpha}$ changes fiber coordinates from the α-trivialization to the β-trivialization.

Proposition 1.3.2. *The transition functions satisfy the cocycle identities*

$$h_{\alpha\alpha} = I_k, \qquad h_{\alpha\beta} = h_{\beta\alpha}^{-1}, \qquad h_{\gamma\alpha} = h_{\gamma\beta} h_{\beta\alpha}$$

on the corresponding nonempty overlaps.

Proof. The first two identities follow from the identity map and inverse changes of trivialization. On a triple overlap, compose $\Phi_\gamma \circ \Phi_\beta^{-1}$ with $\Phi_\beta \circ \Phi_\alpha^{-1}$ to obtain the third identity. □

Conversely, transition functions satisfying these identities can be used to reconstruct the bundle by gluing local products:

$$E \cong \coprod_\alpha (U_\alpha \times \mathbb{R}^k) / \sim,$$

where

$$(x, v)_\alpha \sim (x, w)_\beta \quad \Longleftrightarrow \quad x \in U_\alpha \cap U_\beta, \quad w = h_{\beta\alpha}(x) v.$$

This is why transition functions contain the topological information of the vector bundle.

Example 1.3.3. Important examples of vector bundles include:

- the trivial bundle $M \times \mathbb{R}^k \to M$;
- the tangent bundle $TM \to M$;
- the cotangent bundle $T^*M \to M$;
- tensor bundles such as $T^r_s M \to M$.

For the tangent bundle, a chart (U, φ) gives the local trivialization over U

$$\Phi : TU \longrightarrow U \times \mathbb{R}^n, \qquad v_p \longmapsto (p, d\varphi_p(v_p)).$$

In base coordinates this is the usual identification of TU with $\varphi(U) \times \mathbb{R}^n$ via $d\varphi$.

Definition 1.3.4 (Smooth section)**.** Let $E \to M$ be a vector bundle and let $U \subset M$ be open. A smooth section of E over U is a smooth map $s : U \to E$ such that

$$\pi \circ s = \mathrm{id}_U \,.$$

The space of smooth sections over U is denoted by $\Gamma(U, E)$.

The space $\Gamma(U, E)$ is a module over $C^\infty(U)$, with

$$(fs)(x) = f(x) s(x) \in E_x.$$

A local trivialization gives a smooth frame $e_1, \dots, e_k$ by taking the inverse images of the standard basis of $\mathbb{R}^k$. Every section over that neighborhood has a unique expression $s = s^a e_a$, where the coefficient functions s^a are smooth. When $E = TM$, its sections are precisely smooth vector fields:

$$\Gamma(M, TM) = \mathscr{X}(M).$$

Smooth functions can also be viewed as sections of the trivial line bundle $M \times \mathbb{R}$. In this sense, vector bundle sections generalize smooth functions by allowing the value at each point to lie in a vector space depending on that point.

One can also study classification problems for vector bundles of fixed rank over a given manifold. This leads to K-theory; see Atiyah's book [1] for a classical reference.

1.3.1 Connections

Why ordinary differentiation does not glue. On a trivial bundle $M \times \mathbb{R}^k$, a section is described by a vector of smooth functions, which can be differentiated componentwise. For a general vector bundle, let s_α and s_β be the column vectors of coefficients of a section in two local trivializations. On their overlap,

$$s_\beta = h_{\beta\alpha} s_\alpha.$$

For a vector field X, the product rule gives

$$X(s_\beta) = h_{\beta\alpha} X(s_\alpha) + X(h_{\beta\alpha}) s_\alpha.$$

The last term is the obstruction emphasized in the handwritten notes: the componentwise derivatives do not in general transform as a section. A connection supplies a compatible way to differentiate sections.

Definition 1.3.5 (Connection)**.** A connection on a smooth vector bundle $E \to M$ is a map

$$\nabla : \mathscr{X}(M) \times \Gamma(M, E) \longrightarrow \Gamma(M, E), \qquad (X, s) \longmapsto \nabla_X s,$$

satisfying, for smooth vector fields X, Y, sections s, s_1, s_2, and smooth functions f, g,

$$\begin{aligned} \nabla_X(s_1 + s_2) &= \nabla_X s_1 + \nabla_X s_2, \\ \nabla_{fX+gY} s &= f \nabla_X s + g \nabla_Y s, \\ \nabla_X(fs) &= f \nabla_X s + X(f) s. \end{aligned}$$

The last identity is the Leibniz rule. Taking constant functions in it shows that ∇ is $\mathbb{R}$-linear in the section as well. A connection is also called a covariant derivative.

Example 1.3.6 (The trivial connection)**.** On $M \times \mathbb{R}^k$, write $s = (s^1, \dots, s^k)$ and set

$$\nabla^0_X s = (X(s^1), \dots, X(s^k)).$$

The rules for directional derivatives verify the connection axioms. This construction uses the chosen global trivialization.

Proposition 1.3.7 (Existence and differences of connections)**.** *Every smooth vector bundle admits a connection. If ∇^1 and ∇^2 are two connections, their difference*

$$A(X, s) = \nabla^1_X s - \nabla^2_X s$$

is $C^\infty(M)$-linear in both X and s.

Proof. Choose local trivializations, use the trivial connection $\nabla^{(\alpha)}$ on each one, and take a subordinate smooth partition of unity $\{\lambda_\alpha\}$. Define

$$\nabla_X s = \sum_\alpha \lambda_\alpha \nabla^{(\alpha)}_X s,$$

extending each summand by zero outside its trivializing neighborhood. The sum is smooth and locally finite. The Leibniz rule follows from $\sum_\alpha \lambda_\alpha = 1$; the other axioms follow by linearity. Appendix B justifies this construction.

For the difference, linearity in X follows from the axioms, while

$$\begin{aligned} A(X, fs) &= f\nabla^1_X s + X(f)s - f\nabla^2_X s - X(f)s \\ &= fA(X, s). \end{aligned}$$

Thus the derivative terms cancel. □

Proposition 1.3.8 (Locality in the section)**.** *If $s_1, s_2 \in \Gamma(M, E)$ agree on an open set U, then*

$$(\nabla_X s_1)|_U = (\nabla_X s_2)|_U.$$

Proof. Set $s = s_1 - s_2$, so $s = 0$ on U. For $p \in U$, choose a smooth cutoff χ supported in U and equal to 1 near p. The section χs is globally zero, and hence

$$0 = \nabla_X(\chi s) = \chi \nabla_X s + X(\chi)s.$$

At p this gives $(\nabla_X s)(p) = 0$. Since p is arbitrary, the result follows. □

Proposition 1.3.9 (Pointwise dependence in the direction)**.** *If $X_1(p) = X_2(p)$, then*

$$(\nabla_{X_1} s)(p) = (\nabla_{X_2} s)(p)$$

for every smooth section s.

Proof. First observe that if a global vector field Z vanishes on a neighborhood of p, a cutoff χ supported there with $\chi(p) = 1$ gives

$$0 = \nabla_{\chi Z} s = \chi \nabla_Z s,$$

so $(\nabla_Z s)(p) = 0$. Thus the direction may be replaced by any global vector field agreeing with it near p.

Now set $X = X_1 - X_2$ and choose coordinates on U around p. Write $X = X^i \partial_i$ on U. Choose a cutoff χ supported in U and equal to 1 on a smaller neighborhood V of p. Extend

$$F_i = \chi \partial_i, \qquad f^i = \chi X^i$$

by zero to all of M. These extensions are smooth. The global field $Y = \sum_i f^i F_i$ agrees with X on V, and $f^i(p) = X^i(p) = 0$. Therefore

$$(\nabla_X s)(p) = (\nabla_Y s)(p) = \sum_i f^i(p)(\nabla_{F_i} s)(p) = 0.$$

The cutoff extensions are necessary here because the original definition of ∇ uses global vector fields and sections. □

These properties let us apply ∇ to local sections and vector fields: multiply by cutoffs to extend them near the point of interest, and use locality to see that the result is independent of the extensions. For a fixed section s and $v \in T_pM$, choose X with $X(p) = v$ and define

$$\nabla_v s = (\nabla_X s)(p).$$

This gives the well-defined linear map

$$(\nabla s)_p : T_pM \longrightarrow E_p, \qquad v \longmapsto \nabla_v s.$$

Its target is the fiber E_p; for a general vector bundle it is not a scalar-valued functional. Dependence on the direction is pointwise, whereas dependence on the section involves its behavior near p.

1.3.2 Covariant Differentiation Along Curves

Definition 1.3.10 (Sections along a curve)**.** Let $\gamma : I \to M$ be a smooth curve. A smooth section of E along γ is a smooth map $V : I \to E$ satisfying

$$\pi \circ V = \gamma, \qquad V(t) \in E_{\gamma(t)}.$$

Following the lecture notation, write $\mathscr{X}(\gamma, E)$ for this space. It is a $C^\infty(I)$-module under $(fV)(t) = f(t)V(t)$. Equivalently, it is the space $\Gamma(\gamma^* E)$ of sections of the pullback bundle.

For $E = TM$, the velocity $\dot\gamma$ is a vector field along γ. A section s of E near the image of γ gives another example, $V = s \circ \gamma$. Such a section along the curve is called extendible; if this holds on a subinterval, it is locally extendible there. A general section along a curve need not arise from a single section on M.

Theorem 1.3.11 (Covariant differentiation along a curve)**.** *A connection ∇ on E determines a unique $\mathbb{R}$-linear operator*

$$D_t : \mathscr{X}(\gamma, E) \longrightarrow \mathscr{X}(\gamma, E)$$

with the following properties:

1. *$D_t(fV) = f'V + fD_tV$ for every $f \in C^\infty(I)$.*
2. *If $V = s \circ \gamma$ on a subinterval J, where s is a local section near $\gamma(J)$, then*

$$(D_tV)(t) = \nabla_{\dot\gamma(t)} s, \qquad t \in J.$$

Proof. Work first on a subinterval J whose image lies in a trivializing neighborhood with frame $e_1, \dots, e_k$. Write

$$V(t) = \sum_{a=1}^{k} V^a(t) e_a(\gamma(t)).$$

Linearity and the two required properties force the formula

$$D_tV = \sum_a \dot V^a(t) e_a(\gamma(t)) + \sum_a V^a(t) \nabla_{\dot\gamma(t)} e_a. \tag{1.1}$$

As with the cutoff argument above, the Leibniz rule implies locality on the parameter interval, so this computation also proves uniqueness for a globally defined operator.

To verify independence of the frame, let $\widehat{e}_b = A_b{}^a e_a$ be another smooth frame

and write $V = \widehat{V}^b \widehat{e}_b$. Then $V^a = \widehat{V}^b (A_b{}^a \circ \gamma)$, and

$$\nabla_{\dot\gamma} \widehat{e}_b = \frac{d}{dt}(A_b{}^a \circ \gamma) e_a \circ \gamma + (A_b{}^a \circ \gamma) \nabla_{\dot\gamma} e_a.$$

Substituting this into the formula in the new frame and using the product rule gives exactly (1.1). Hence the local formulas agree and define a smooth operator along all of γ.

The same formula immediately gives $\mathbb{R}$-linearity and the Leibniz rule. If $V = s \circ \gamma$ and $s = s^a e_a$, the chain rule gives $\dot V^a = \dot\gamma(s^a)$, so the formula equals $\nabla_{\dot\gamma} s$, as required. □

Example 1.3.12. For the trivial connection on $M \times \mathbb{R}^k$, the constant standard frame has zero covariant derivative. Thus, if $V(t) = (V^1(t), \dots, V^k(t))$, then

$$D_t V = (\dot V^1, \dots, \dot V^k).$$

Christoffel symbols and the coordinate formula. Return to a connection on TM and choose local coordinates $(x^1, \dots, x^n)$. The smooth functions Γ_{ij}^k defined by

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k$$

are its **Christoffel symbols**. For $X = X^i \partial_i$ and $Y = Y^j \partial_j$, the axioms give

$$\begin{aligned} \nabla_X Y &= \sum_{i,j} X^i \nabla_{\partial_i}(Y^j \partial_j) \\ &= \sum_{i,j} X^i (\partial_i Y^j) \partial_j + \sum_{i,j,k} X^i Y^j \Gamma_{ij}^k \partial_k \\ &= \sum_k \left(\sum_i X^i \partial_i Y^k + \sum_{i,j} \Gamma_{ij}^k X^i Y^j \right) \partial_k. \end{aligned}$$

Consequently, for a field $V(t) = V^j(t) \partial_j|_{\gamma(t)}$ along a curve in this chart,

$$D_t V = \sum_k \left(\dot V^k + \sum_{i,j} \Gamma_{ij}^k(\gamma(t)) \dot\gamma^i V^j \right) \partial_k|_{\gamma(t)}.$$

Christoffel symbols depend on the coordinate frame and are not tensor components. The formulas above nevertheless define intrinsic covariant derivatives.

1.3.3 Levi-Civita Connection

For the tangent bundle of a Riemannian manifold, the metric selects a distinguished connection by two conditions.

Definition 1.3.13 (Torsion-free and metric-compatible)**.** A connection ∇ on TM is **torsion-free** if

$$\nabla_X Y - \nabla_Y X = [X, Y]$$

for all smooth vector fields X, Y. It is **compatible with the metric** g if

$$X(g(Y, Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$$

for all X, Y, Z.

Example 1.3.14. On Euclidean space $(\mathbb{R}^n, g_0)$, the standard connection $\nabla^0_X Y = (X(Y^1), \ldots, X(Y^n))$ is torsion-free because $[X, Y]^k = X(Y^k) - Y(X^k)$. It is metric-compatible by the ordinary product rule applied to $\sum_k Y^k Z^k$.

Theorem 1.3.15 (Fundamental theorem of Riemannian geometry)**.** *Every Riemannian manifold (M, g) has a unique connection on TM that is torsion-free and compatible with g. It is called the* ***Levi-Civita connection****.*

Proof. Suppose first that such a connection exists. Write metric compatibility in the three cyclic orders:

$$\begin{aligned} Xg(Y, Z) &= g(\nabla_X Y, Z) + g(Y, \nabla_X Z), \\ Yg(Z, X) &= g(\nabla_Y Z, X) + g(Z, \nabla_Y X), \\ Zg(X, Y) &= g(\nabla_Z X, Y) + g(X, \nabla_Z Y). \end{aligned}$$

Add the first two identities and subtract the third. Using torsion-freeness to interchange the covariant derivatives gives the **Koszul formula**

$$\begin{aligned} 2g(\nabla_X Y, Z) = {} & Xg(Y, Z) + Yg(Z, X) - Zg(X, Y) \\ & - g(X, [Y, Z]) + g(Y, [Z, X]) + g(Z, [X, Y]). \end{aligned} \tag{1.2}$$

The right-hand side is determined by g and the vector fields. Nondegeneracy of g therefore proves uniqueness.

For existence, denote the right-hand side of (1.2) by $K(X, Y, Z)$. The product rules for differentiation and Lie brackets give

$$\begin{aligned} K(X, Y, fZ) &= fK(X, Y, Z), \\ K(fX, Y, Z) &= fK(X, Y, Z), \\ K(X, fY, Z) &= fK(X, Y, Z) + 2X(f)g(Y, Z). \end{aligned}$$

In particular, $K(X, Y, \cdot)$ is a smooth one-form. The metric identifies it with a unique smooth vector field $\nabla_X Y$ through $2g(\nabla_X Y, \cdot) = K(X, Y, \cdot)$. Additivity and

the displayed identities give the connection axioms. Finally,

$$K(X,Y,Z) - K(Y,X,Z) = 2g([X,Y],Z),$$
$$K(X,Y,Z) + K(X,Z,Y) = 2Xg(Y,Z)$$

give torsion-freeness and metric compatibility, respectively. □

Solving for the Christoffel symbols. In a coordinate frame, $[\partial_i, \partial_j] = 0$, so torsion-freeness is equivalent to $\Gamma^k_{ij} = \Gamma^k_{ji}$. Metric compatibility gives

$$\partial_i g_{j\ell} = \Gamma^a_{ij} g_{a\ell} + \Gamma^a_{i\ell} g_{ja},$$
$$\partial_j g_{i\ell} = \Gamma^a_{ji} g_{a\ell} + \Gamma^a_{j\ell} g_{ia},$$
$$\partial_\ell g_{ij} = \Gamma^a_{\ell i} g_{aj} + \Gamma^a_{\ell j} g_{ia}.$$

Adding the first two equations and subtracting the third cancels the cross terms:

$$\sum_a g_{\ell a}\Gamma^a_{ij} = \frac{1}{2}\left(\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}\right).$$

For fixed i, j, this is a linear system for the unknowns Γ^a_{ij}. Let $(g^{k\ell}) = (g_{k\ell})^{-1}$. Multiplying by $g^{k\ell}$ and summing over ℓ yields

$$\Gamma^k_{ij} = \frac{1}{2}\sum_\ell g^{k\ell}\left(\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}\right). \tag{1.3}$$

As emphasized in the handwritten notes, (g^{ij}) is the inverse of the *whole matrix* (g_{ij}). In general, g^{ij} is not the reciprocal of the entry g_{ij}.

1.3.4 Induced Connection

Let $M \subset \widetilde{M}$ be an embedded submanifold of a Riemannian manifold $(\widetilde{M}, \widetilde{g})$, and let g be the induced metric on M. At each $p \in M$,

$$T_p\widetilde{M} = T_pM \oplus N_pM, \qquad N_pM = (T_pM)^\perp.$$

The orthogonal projection onto T_pM, denoted by $(\cdot)^T$, varies smoothly with p, as follows by choosing smooth local tangent and normal frames.

Definition 1.3.16 (Induced connection)**.** Let $\widetilde{\nabla}$ be the Levi-Civita connection of $\widetilde{g}$. For $X, Y \in \mathscr{X}(M)$, choose local ambient extensions $\widetilde{X}, \widetilde{Y}$ and define

$$\nabla_X Y = (\widetilde{\nabla}_{\widetilde{X}}\widetilde{Y})^T\big|_M.$$

Such extensions exist locally by using coordinates adapted to the embedded submanifold.

Proposition 1.3.17. *This expression is independent of the extensions, defines a connection on TM, and equals the Levi-Civita connection of the induced metric g.*

Proof. Independence of the extensions. At $p \in M$, the ambient covariant derivative depends only on $\widetilde{X}(p) = X(p)$ in its first argument. To deal with the second argument, choose a smooth curve γ in M with $\gamma(0) = p$ and $\dot{\gamma}(0) = X(p)$. Then

$$(\widetilde{\nabla}_{\widetilde{X}}\widetilde{Y})(p) = D_t(\widetilde{Y} \circ \gamma)\Big|_{t=0}.$$

Two ambient extensions of Y agree along γ, so their covariant derivatives there agree. This proves independence of the second extension as well. Taking tangential projections preserves this independence.

Connection axioms. Smoothness and additivity follow from the ambient connection and the smooth projection. For $f \in C^\infty(M)$, take a local ambient extension $\widetilde{f}$. Then $\widetilde{f}\widetilde{X}$ extends fX, giving

$$\nabla_{fX}Y = f\nabla_X Y.$$

Since $\widetilde{Y}|_M = Y$ is tangent and $\widetilde{X}(\widetilde{f})|_M = X(f)$, the ambient Leibniz rule gives

$$\begin{aligned}\nabla_X(fY) &= \big(\widetilde{X}(\widetilde{f})\widetilde{Y} + \widetilde{f}\widetilde{\nabla}_{\widetilde{X}}\widetilde{Y}\big)^T\big|_M \\ &= X(f)Y + f\nabla_X Y.\end{aligned}$$

Identification with the Levi-Civita connection. Ambient extensions that are tangent along M satisfy $[\widetilde{X}, \widetilde{Y}]|_M = [X, Y]$, as can be checked in adapted coordinates. Thus

$$\nabla_X Y - \nabla_Y X = \big(\widetilde{\nabla}_{\widetilde{X}}\widetilde{Y} - \widetilde{\nabla}_{\widetilde{Y}}\widetilde{X}\big)^T\big|_M = [X, Y].$$

For tangent fields Y, Z, the normal components are orthogonal to them. Ambient metric compatibility consequently gives

$$Xg(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z).$$

The induced connection is torsion-free and metric-compatible, so uniqueness identifies it with the Levi-Civita connection of (M, g). □

Example 1.3.18. For a regular surface $M \subset \mathbb{R}^3$, differentiate a tangent field in the ambient Euclidean space and project to the tangent plane. More generally, for a

tangent vector field V along a curve in an embedded Euclidean submanifold,

$$D_t V = \left(\frac{dV}{dt} \right)^T .$$

This recovers the classical covariant derivative on a surface and prepares for the geodesic equation in the next section.

1.4 Geodesics

Geodesics are the Riemannian analog of straight lines. More precisely, they are curves whose velocity vector is parallel along the curve. This differential equation is local, but it is closely related to the global question of length minimization: sufficiently short geodesic segments minimize length; longer geodesic segments may fail to minimize length.

Definition 1.4.1 (Geodesic)**.** Let (M, g) be a Riemannian manifold with Levi-Civita connection ∇. A smooth curve $\gamma : I \to M$ is called a **geodesic** if

$$D_t\dot{\gamma} = 0.$$

The equation $D_t\dot{\gamma} = 0$ says that the acceleration of γ vanishes intrinsically. In Euclidean space, this is exactly the equation $\ddot{\gamma} = 0$, whose solutions are straight lines. On a curved manifold, the same idea must be expressed using the Levi-Civita connection.

Proposition 1.4.2 (Geodesic equation)**.** *Let $(U, (x^1, \cdots, x^n))$ be a local coordinate chart and write*

$$\gamma(t) = (x^1(t), \cdots, x^n(t)).$$

Then γ is a geodesic if and only if its coordinate functions satisfy

$$\ddot{x}^k(t) + \sum_{i,j=1}^{n} \Gamma_{ij}^k(\gamma(t))\dot{x}^i(t)\dot{x}^j(t) = 0, \qquad k = 1, \cdots, n.$$

Proof. In local coordinates,

$$\dot{\gamma} = \sum_k \dot{x}^k \frac{\partial}{\partial x^k}.$$

Applying the connection along γ gives

$$D_t\dot{\gamma} = \sum_k \left(\ddot{x}^k + \sum_{i,j} \Gamma_{ij}^k \dot{x}^i \dot{x}^j \right) \frac{\partial}{\partial x^k}.$$

Therefore, $D_t\dot{\gamma} = 0$ is equivalent to the displayed system of ordinary differential equations. □

Proposition 1.4.3 (Local existence and uniqueness)**.** *Let (M, g) be a Riemannian manifold. For every $p \in M$ and every $v \in T_pM$, there exists an interval I containing 0 and a geodesic $\gamma : I \to M$ such that*

$$\gamma(0) = p, \qquad \dot{\gamma}(0) = v.$$

If $\tilde{\gamma} : \tilde{I} \to M$ is another geodesic with the same initial data, then $\gamma = \tilde{\gamma}$ on $I \cap \tilde{I}$.

Proof. Put $y^k = \dot{x}^k$. The geodesic equation becomes a first-order system

$$\dot{x}^k = y^k, \qquad \dot{y}^k = -\sum_{i,j} \Gamma^k_{ij}(x) y^i y^j,$$

with initial data $(x(0), y(0)) = (x(p), (v^1, \dots, v^n))$. Its right-hand side is smooth, so local existence, uniqueness, and smooth dependence on the initial data follow from the ODE theorem. To obtain uniqueness on the whole common interval, the set of times at which the positions and velocities of two solutions agree is nonempty, closed, and open by local uniqueness. Since the common interval is connected, they agree throughout it. □

Example 1.4.4 (Geodesics on the sphere)**.** Let $\mathbb{S}^n \subset \mathbb{R}^{n+1}$ carry the induced metric. For $p \in \mathbb{S}^n$ and $v \in T_p\mathbb{S}^n$, with $v \neq 0$, define

$$\gamma(t) = \cos(t|v|)p + \frac{\sin(t|v|)}{|v|} v.$$

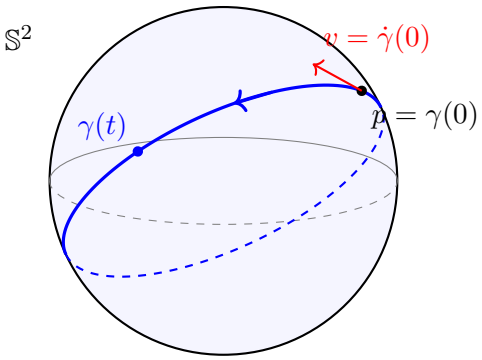
If $v = 0$, take $\gamma(t) = p$. Since $p \perp v$ and $|p| = 1$, the curve lies in $\mathbb{S}^n$. Differentiating twice gives

$$\ddot{\gamma}(t) = -|v|^2 \gamma(t).$$

The vector $\gamma(t)$ is normal to $\mathbb{S}^n$ at $\gamma(t)$, so the tangential component of $\ddot{\gamma}$ is zero. Hence

$$D_t \dot{\gamma} = (\ddot{\gamma})^T = 0.$$

The initial data are $\gamma(0) = p$ and $\dot{\gamma}(0) = v$. By uniqueness, every nonconstant geodesic on the round sphere is a portion of a great circle parametrized with constant speed.



a geodesic on $\mathbb{S}^2$ is a great circle

The sphere example is global: the geodesic is defined for all $t \in \mathbb{R}$. On a general Riemannian manifold, local geodesics always exist, but they may or may not

extend for all time. This naturally leads to the exponential map and the question of completeness.

Proposition 1.4.5 (Constant speed and rescaling)**.** *Every geodesic has constant speed. If γ_v is the maximal geodesic with initial velocity $v \in T_pM$, then*

$$\gamma_{cv}(t) = \gamma_v(ct)$$

whenever the right-hand side is defined. For $c \neq 0$, their maximal domains satisfy $I_{cv} = \{t : ct \in I_v\}$; for $c = 0$, the geodesic is constant and defined on $\mathbb{R}$.

Proof. Metric compatibility gives

$$\frac{d}{dt}|\dot{\gamma}|_g^2 = 2g(D_t\dot{\gamma}, \dot{\gamma}) = 0.$$

For $\sigma(t) = \gamma_v(ct)$, one has $\sigma(0) = p$, $\dot{\sigma}(0) = cv$, and $D_t\dot{\sigma} = c^2(D_t\dot{\gamma}_v)(ct) = 0$. Uniqueness gives the identity. Applying the same argument with $1/c$ proves the assertion about maximal intervals. □

1.4.1 Exponential Map

For $p \in M$ and $v \in T_pM$, let γ_v denote the maximal geodesic satisfying

$$\gamma_v(0) = p, \qquad \dot{\gamma}_v(0) = v.$$

The word "maximal" means that γ_v is defined on the largest open interval containing 0 to which the local ODE solution can be extended.

Definition 1.4.6 (Domain of the exponential map)**.** Define

$$\mathcal{E}_p = \{v \in T_pM : \gamma_v(1) \text{ is defined}\}.$$

Then $\mathcal{E}_p \subset T_pM$ is the pointwise domain of the exponential map at p.

Definition 1.4.7 (Pointwise exponential map)**.** The exponential map at p is

$$\exp_p : \mathcal{E}_p \longrightarrow M, \qquad \exp_p(v) = \gamma_v(1).$$

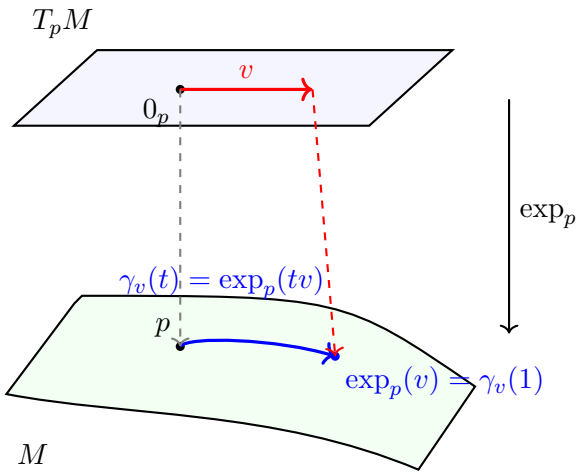
Equivalently, by rescaling the geodesic parameter,

$$\gamma_v(t) = \exp_p(tv)$$

whenever both sides are defined. Thus $\exp_p$ sends a tangent vector v to the point reached at time 1 by the geodesic starting at p with initial velocity v.

The domains can be glued together inside the tangent bundle:

$$\mathcal{E} = \bigcup_{p \in M} \mathcal{E}_p \subset TM.$$



The tangent space T_pM is a linear model for the manifold near p. The exponential map sends the radial line $t \mapsto tv$ in T_pM to the geodesic $t \mapsto \gamma_v(t)$ in M.

This construction is intrinsic and requires no Euclidean embedding. Its first-order relation to the tangent space is the identity

$$(d\exp_p)_0 = \mathrm{id}_{T_pM},$$

proved below.

Definition 1.4.8 (Exponential map)**.** The global exponential map is

$$\exp : \mathcal{E} \longrightarrow M, \qquad (p, v) \longmapsto \exp_p(v).$$

Proposition 1.4.9 (Basic properties of the exponential map)**.** *The exponential map has the following properties:*

- $\mathcal{E} \subset TM$ *is open and* $\exp : \mathcal{E} \to M$ *is smooth.*
- $\mathcal{E}$ *contains the zero section, and each* $\mathcal{E}_p$ *is star-shaped about* 0*.*
- *If* $v \in \mathcal{E}_p$*, then* $tv \in \mathcal{E}_p$ *for every* $t \in [0, 1]$*, and* $\gamma_v(t) = \exp_p(tv)$*.*
- *The differential at the origin satisfies the canonical identification*

$$(d\exp_p)_0 : T_0(T_pM) \longrightarrow T_pM, \qquad (d\exp_p)_0(w) = w.$$

Proof. The maximal flow domain of the geodesic ODE is open in $\mathbb{R} \times TM$, and its solution depends smoothly on time and initial data. Restricting to time 1 proves openness of $\mathcal{E}$ and smoothness of exp. The zero initial velocity gives the constant geodesic, so $0_p \in \mathcal{E}_p$ for every p. If $1 \in I_v$, then $[0,1] \subset I_v$, and the rescaling identity shows that $tv \in \mathcal{E}_p$ and $\exp_p(tv) = \gamma_v(t)$ for $0 \leq t \leq 1$. More generally, $tv \in \mathcal{E}_p$ if and only if $t \in I_v$.

Under the canonical identification $T_0(T_pM) \cong T_pM$, the vector w is represented by the curve $s \mapsto sw$. Therefore

$$(d\exp_p)_0(w) = \left.\frac{d}{ds}\right|_{s=0} \exp_p(sw) = \dot{\gamma}_w(0) = w.$$

□

The pointwise map $\exp_p$ is smooth on an open neighborhood of 0 and has an invertible differential there. We can therefore use the inverse function theorem to construct coordinates.

construction of normal coordinates Choose an orthonormal basis of T_pM and use it to identify T_pM with $\mathbb{R}^n$ by a linear isometry $\iota : T_pM \to \mathbb{R}^n$. Since $(d\exp_p)_0$ is an isomorphism, the inverse function theorem gives neighborhoods $V_p \subset T_pM$ of 0 and $U_p \subset M$ of p such that

$$\exp_p : V_p \longrightarrow U_p$$

is a diffeomorphism. The corresponding normal coordinate map is

$$x := \iota \circ \exp_p^{-1} : U_p \longrightarrow \iota(V_p) \subset \mathbb{R}^n.$$

The diagram summarizes the construction.

$$\begin{array}{ccc} V_p \subset T_pM & \xrightarrow[\sim]{\exp_p} & U_p \subset M \\ \iota \downarrow & & \downarrow x = \iota \circ \exp_p^{-1} \\ \iota(V_p) \subset \mathbb{R}^n & = & \iota(V_p) \subset \mathbb{R}^n. \end{array}$$

In words, to find the coordinates of $q \in U_p$, first pull q back to T_pM by $\exp_p^{-1}$ and then take its components in the chosen orthonormal basis.

Proposition 1.4.10 (Properties of normal coordinates)**.** *Let $(U, (x^1, \cdots, x^n))$ be normal coordinates centered at p. Then:*

- $x^1(p) = \cdots = x^n(p) = 0$;
- $g_{ij}(p) = \delta_{ij}$;

- $\Gamma_{ij}^k(p) = 0$;
- *if* $v = \sum_i v^i \frac{\partial}{\partial x^i}\big|_p$, *then the geodesic with initial velocity* v *is represented in these coordinates by*

$$t \longmapsto (tv^1, \cdots, tv^n)$$

for t *sufficiently small.*

Proof. Write the chosen orthonormal basis as $E_1, \ldots, E_n$. Since $\exp_p(0) = p$ and $(d\exp_p)_0 = \mathrm{id}$, we have $x(p) = 0$ and $\partial_i|_p = E_i$, hence $g_{ij}(p) = \delta_{ij}$. The rescaling identity gives

$$x(\gamma_v(t)) = t\,\iota(v) = (tv^1, \ldots, tv^n).$$

Substituting this straight-line coordinate expression into the geodesic equation at $t = 0$ yields

$$\sum_{i,j} \Gamma_{ij}^k(p) v^i v^j = 0 \qquad \text{for every } v \in T_pM.$$

Torsion-freeness gives $\Gamma_{ij}^k = \Gamma_{ji}^k$. Taking $v = E_i$ and then $v = E_i + E_j$ proves that all these coefficients vanish at p. Finally, metric compatibility gives

$$\partial_k g_{ij} = \sum_a \big(\Gamma_{ki}^a g_{aj} + \Gamma_{kj}^a g_{ia}\big),$$

so $\partial_k g_{ij}(p) = 0$ as well. □

Remark 1.4.11. The vanishing $\Gamma_{ij}^k(p) = 0$ is one of the main advantages of normal coordinates. With an orthonormal initial basis, the first derivatives of the metric also vanish at p:

$$\partial_k g_{ij}(p) = 0.$$

The second derivatives of g_{ij} encode the curvature tensor.

1.4.2 Variations and the First Variation of Length

We now return to the length problem, following the order of the lecture. For the variation arguments in this subsection, we restrict to piecewise regular admissible curves: the velocity is nonzero on each closed smooth piece, so arc-length parametrization is available. This is an additional hypothesis, not a change to the definition of admissible curves in the previous section. Constant curves are treated separately when needed.

Definition 1.4.12 (Variation and variational vector field)**.** A variation of admissible curves is a continuous map $F : J \times [a, b] \to M$ with a fixed partition $a = a_0 < \cdots < a_m = b$, such that F is smooth on each strip $J \times [a_{i-1}, a_i]$ and each $\gamma_s(t) = F(s, t)$

is admissible. Smoothness on a closed strip means smooth extendibility near that strip. If $0 \in J$ and $F(0,t) = \gamma(t)$, it is a variation of γ, with variational field

$$V(t) = \partial_s F(0,t) \in T_{\gamma(t)}M.$$

It has fixed endpoints if $F(s,a) = \gamma(a)$ and $F(s,b) = \gamma(b)$ for all s.

The fields $\partial_s F, \partial_t F$ are vector fields along F. We write D_s, D_t for covariant differentiation along its parameter curves on each smooth strip.

Lemma 1.4.13 (Two differentiation identities)**.** *On each smooth strip, $D_s\partial_t F = D_t\partial_s F$. For fields X, Y along a smooth curve,*

$$\frac{d}{dt}g(X,Y) = g(D_tX, Y) + g(X, D_tY).$$

Proof. In coordinates,

$$(D_s\partial_t F)^k = \partial_s\partial_t F^k + \Gamma^k_{ij}(F)\partial_s F^i \partial_t F^j.$$

Equality of mixed partials and $\Gamma^k_{ij} = \Gamma^k_{ji}$ prove the first identity. The second is metric compatibility along the curve; it follows also by differentiating $g_{ij}(\gamma)X^iY^j$. □

Lemma 1.4.14 (Realizing a variational field)**.** *Let V be a continuous, piecewise smooth field along a piecewise regular admissible $\gamma : [a,b] \to M$, with the same partition. Then V is the variational field of a variation that fixes every t_0 where $V(t_0) = 0$.*

Proof. Define $F(s,t) = \exp_{\gamma(t)}(sV(t))$. The preimage of the open exponential domain under $(s,t) \mapsto (\gamma(t), sV(t))$ contains $\{0\} \times [a,b]$. Compactness gives a common $\varepsilon > 0$ for which this formula is defined for $|s| < \varepsilon$ and all t. It is continuous and smooth on each strip. Since $\partial_t F(0,t) = \dot\gamma(t)$ is nonzero on each compact smooth piece, shrinking ε makes each varied curve regular there. Moreover,

$$\partial_s F(0,t) = (d\exp_{\gamma(t)})_0 V(t) = V(t).$$

If $V(t_0) = 0$, then $F(s,t_0) = \exp_{\gamma(t_0)}(0) = \gamma(t_0)$. □

Theorem 1.4.15 (First variation formula for length)**.** *Suppose γ is parametrized by arc length on each piece, and F is a variation with variational field V. Put*

$$\Delta_i\dot\gamma := \dot\gamma(a_i^+) - \dot\gamma(a_i^-), \qquad 1 \le i < m.$$

Then

$$\left.\frac{d}{ds}\right|_{s=0} L_g(\gamma_s) = -\sum_{i=1}^{m}\int_{a_{i-1}}^{a_i} g(V, D_t\dot\gamma)\,dt - \sum_{i=1}^{m-1} g(V(a_i), \Delta_i\dot\gamma) + g(V(b), \dot\gamma(b^-)) - g(V(a), \dot\gamma(a^+)). \tag{1.4}$$

Proof. On one smooth piece, differentiating gives

$$\begin{aligned}\left.\frac{d}{ds}\right|_{s=0}\int_{a_{i-1}}^{a_i} |\partial_t F|_g\,dt &= \left.\int_{a_{i-1}}^{a_i} \frac{g(D_s\partial_t F, \partial_t F)}{|\partial_t F|_g}\,dt\right|_{s=0}\\ &= \int_{a_{i-1}}^{a_i} g(D_t V, \dot\gamma)\,dt\\ &= -\int_{a_{i-1}}^{a_i} g(V, D_t\dot\gamma)\,dt + \big[g(V,\dot\gamma)\big]_{a_{i-1}}^{a_i}.\end{aligned}$$

We used $|\dot\gamma|_g = 1$, the differentiation identities, and integration by parts. Summing over the pieces yields the endpoint terms. At a breakpoint, the contribution is

$$g(V(a_i), \dot\gamma(a_i^-)) - g(V(a_i), \dot\gamma(a_i^+)) = -g(V(a_i), \Delta_i\dot\gamma),$$

which fixes the sign of the corner term. □

Remark 1.4.16. The unit-speed assumption is essential for this form. For a general regular parametrization, replace $\dot\gamma$ by $T = \dot\gamma/|\dot\gamma|_g$ in $g(V, D_t T)$ and in the boundary and corner terms. With the convention *right limit minus left limit* for $\Delta_i\dot\gamma$, the corner sum has a minus sign.

Definition 1.4.17 (Shortest curve)**.** An admissible curve joining p to q is shortest if its length is no greater than that of any other admissible curve with the same endpoints; equivalently, its length is $d_g(p, q)$.

Theorem 1.4.18 (Shortest curves are geodesics)**.** *A nonconstant piecewise regular shortest admissible curve becomes a smooth geodesic after parametrization by arc length. In particular, it cannot have a corner.*

Proof. Parametrize γ by arc length. For any variation fixing the endpoints, the length has a minimum at $s = 0$, so its derivative is zero. First, choose V supported inside a single smooth piece. The realization lemma and first variation give

$$\int g(V, D_t\dot\gamma)\,dt = 0.$$

Take $V = \chi D_t\dot\gamma$, where χ is any nonnegative smooth cutoff supported there. It follows that $D_t\dot\gamma = 0$ on each piece, so γ is a broken geodesic.

Next, fix a breakpoint a_i. Choose a continuous piecewise smooth field with $V(a) = V(b) = 0$, $V(a_j) = 0$ for $j \neq i$, and $V(a_i) = \Delta_i \dot{\gamma}$, using local frames and cutoffs. The formula gives $0 = -|\Delta_i \dot{\gamma}|_g^2$. The velocities agree at every breakpoint; uniqueness of the geodesic ODE joins the pieces into a single smooth geodesic. □

Example 1.4.19. A geodesic need not be globally shortest. On a round sphere, two nonantipodal points on a great circle are joined by a shorter arc and a longer arc. Both are geodesic segments, but the longer one is not length-minimizing.

1.4.3 Gauss Lemma and Local Minimality

Definition 1.4.20 (Normal neighborhood and geodesic ball)**.** If $\exp_p$ is a diffeomorphism from an open neighborhood $V \subset T_pM$ of 0 onto $N \subset M$, then N is a normal neighborhood of p. Write

$$B_r^{T_pM}(0) = \{v \in T_pM : |v|_{g_p} < r\}.$$

When $\exp_p$ is a diffeomorphism on this ball, its image $\mathcal{B}_r(p) = \exp_p(B_r^{T_pM}(0))$ is called a geodesic ball. We distinguish this from the metric ball $B_g(p, r)$ until their equality has been proved. For $0 < t < r$, put

$$S_t(p) = \exp_p\{v \in T_pM : |v|_{g_p} = t\}.$$

Lemma 1.4.21 (Gauss lemma)**.** *Let $\mathcal{B}_r(p)$ be a geodesic ball and $\gamma(t) = \exp_p(tu)$, where $|u|_{g_p} = 1$. Then $|\dot{\gamma}(t)|_g = 1$, and*

$$g(\dot{\gamma}(t), W) = 0 \qquad (W \in T_{\gamma(t)}S_t(p), \quad 0 < t < r).$$

Proof. Use local polar coordinates $(t, \theta^1, \ldots, \theta^{n-1})$ in $T_pM \setminus \{0\}$ and transport them by $\exp_p$. Write $R = \partial_t$, $A_\alpha = \partial_{\theta^\alpha}$. Radial curves are unit-speed geodesics, so $g(R, R) = 1$ and $\nabla_R R = 0$. Since $[R, A_\alpha] = 0$, torsion-freeness gives

$$\frac{d}{dt} g(R, A_\alpha) = g(R, \nabla_R A_\alpha) = g(R, \nabla_{A_\alpha} R) = \tfrac{1}{2} A_\alpha g(R, R) = 0.$$

To determine this constant, use normal coordinates $x = tu(\theta)$. There R has components u^i, and A_α has components $t\,\partial_{\theta^\alpha} u^j$. Thus

$$g(R, A_\alpha) = t\, g_{ij}(tu) u^i \partial_{\theta^\alpha} u^j \longrightarrow 0 \quad (t \downarrow 0).$$

The constant is therefore zero. The angular fields span $T_{\gamma(t)}S_t(p)$, proving the claim. In dimension one, the tangent space is zero, and the claim is immediate. □

Theorem 1.4.22 (Radial geodesics minimize length)**.** *If $q = \exp_p(v) \in \mathcal{B}_r(p)$, then $t \mapsto \exp_p(tv)$, $0 \le t \le 1$, is, up to orientation-preserving reparametrization, the unique shortest curve from p to q. Its length is $|v|_{g_p}$.*

Proof. The case $q = p$ is immediate, with the constant curve. Otherwise, put $\rho = |v|_{g_p} < r$. For a competitor σ contained in the geodesic ball, use polar coordinates wherever $\sigma \neq p$:

$$\sigma(s) = \exp_p(r(s)u(s)), \qquad |u(s)|_{g_p} = 1.$$

Its velocity splits into $r'R$ and an angular vector. Gauss's lemma gives

$$|\dot\sigma|_g^2 = |r'|^2 + |\dot\sigma_{\mathrm{ang}}|_g^2 \ge |r'|^2.$$

The radius $r(s) = |\exp_p^{-1}(\sigma(s))|$ is absolutely continuous on each smooth piece, including at its zeros. The estimate holds almost everywhere, so

$$L_g(\sigma) \ge \int |r'|\, ds \ge r(b) - r(a) = \rho.$$

Equality forces the angular velocity to vanish and the radius to be nondecreasing. Thus, the curve follows the same radial segment without backtracking.

If σ leaves $\mathcal{B}_r(p)$, choose R_0 with $\rho < R_0 < r$. Before leaving, it first meets $S_{R_0}(p)$: the image of the closed tangent ball of radius R_0 is compact and contained in the normal neighborhood. The preceding estimate up to that meeting gives $L_g(\sigma) \ge R_0 > \rho$. Such a curve cannot be minimizing. □

Corollary 1.4.23 (Geodesic and metric balls agree)**.** *With the preceding hypotheses,*

$$\mathcal{B}_r(p) = B_g(p, r), \qquad d_g(p, \exp_p(v)) = |v|_{g_p} \quad (|v|_{g_p} < r).$$

Proof. The distance identity and the first inclusion follow from the theorem. If $q \notin \mathcal{B}_r(p)$, every curve from p to q meets $S_R(p)$ for every $0 < R < r$, so its length is at least R. Taking the infimum and then $R \uparrow r$ yields $d_g(p, q) \ge r$. □

In geodesic polar coordinates, the same orthogonality says

$$g = dr^2 + \sum_{\alpha,\beta=1}^{n-1} h_{\alpha\beta}(r, \theta)\, d\theta^\alpha d\theta^\beta.$$

For fixed r, the angular block is the induced metric on $S_r(p)$. Polar coordinates are used away from their singularity at p.

Definition 1.4.24 (Local minimality and uniformly normal neighborhoods)**.** A curve is locally length-minimizing if each parameter value has a subinterval on which

every compact subsegment is shortest between its endpoints. An open set W is uniformly δ-normal if, for every $q \in W$, $\exp_q$ is a diffeomorphism on $B_\delta^{T_qM}(0)$ and

$$W \subset \mathcal{B}_\delta(q) = B_g(q, \delta).$$

Theorem 1.4.25 (Uniformly normal neighborhoods)**.** *Every point has a uniformly δ-normal neighborhood for some $\delta > 0$.*

Proof. In a local trivialization of TM near p, consider $\Phi(q, v) = (q, \exp_q(v))$. Since $\exp_q(0) = q$ and $(d \exp_q)_0 = \mathrm{id}$,

$$d\Phi_{(p,0)} = \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}$$

is invertible. The inverse function theorem gives an open neighborhood $\mathcal{O}$ of $(p, 0)$ on which Φ is a diffeomorphism onto its image. Choose a small relatively compact base neighborhood V and a coordinate vector radius $\eta > 0$ with $V \times B_\eta(0) \subset \mathcal{O}$. Uniform comparison of g_q with the Euclidean norm yields $\delta > 0$ such that

$$B_{2\delta}^{T_qM}(0) \subset B_\eta(0) \qquad (q \in V).$$

Restricting Φ to each fiber proves that $\exp_q$ is a diffeomorphism on the ball of radius 2δ, hence also on the ball of radius δ. Choose an open neighborhood $W \subset V \cap B_g(p, \delta/2)$. For $q, z \in W$, the triangle inequality gives $d_g(q, z) < \delta$, so $z \in \mathcal{B}_\delta(q)$. □

Theorem 1.4.26. *Every geodesic is locally length-minimizing.*

Proof. Choose a uniformly δ-normal neighborhood W of $\gamma(t_0)$. The speed c is constant. Choose an interval J about t_0 with $\gamma(J) \subset W$ and $c|b - a| < \delta$ for all $a, b \in J$. For $a < b$, uniqueness and rescaling give

$$\gamma(t) = \exp_{\gamma(a)}((t - a)\dot{\gamma}(a)), \qquad a \le t \le b.$$

The initial displacement has norm $c(b - a) < \delta$, so radial minimality applies. Constant geodesics are minimizing trivially. □

Definition 1.4.27 (Geodesic convexity)**.** An open set $W \subset M$ is geodesically convex if every two points in W can be joined by a shortest geodesic in M lying entirely in W.

Exercise from the lecture. For every $p \in M$, there is $r_0 > 0$ such that $B_g(p, r)$ is geodesically convex for every $0 < r \le r_0$. The uniformly normal property alone does not assert that the connecting segment stays inside W; that is the additional conclusion here.

1.4.4 Injectivity Radius and Completeness

Definition 1.4.28 (Injectivity radius)**.** Consider the radii $r > 0$ for which the restriction

$$\exp_p : B_r^{T_pM}(0) \longrightarrow M$$

is defined and is a diffeomorphism onto its image. Their supremum is the injectivity radius at p, denoted $\mathrm{inj}_p(M, g)$. The injectivity radius of M is

$$\mathrm{inj}(M, g) = \inf_{p \in M} \mathrm{inj}_p(M, g).$$

Each point has a positive injectivity radius, but the infimum over a noncompact manifold can be zero, as suggested by a narrowing end in the lecture sketch.

Proposition 1.4.29. *If M is compact, then* $\mathrm{inj}(M, g) > 0$.

Proof. Cover M by uniformly δ_p-normal neighborhoods W_p. Choose a finite subcover $W_{p_1}, \ldots, W_{p_N}$ and let $\delta = \min_j \delta_{p_j} > 0$. Every point lies in one of these neighborhoods and has an injectivity radius at least the corresponding δ_{p_j}, hence at least δ. □

1.4.5 Hopf–Rinow Theorem

A manifold is geodesically complete if all maximal geodesics have domain $\mathbb{R}$. It is metrically complete if every Cauchy sequence for d_g converges.

Theorem 1.4.30 (Hopf–Rinow)**.** *Let (M, g) be connected. The following conditions are equivalent:*

1. *The exponential map is defined on all of TM.*
2. *For some $p \in M$, $\exp_p$ is defined on all of T_pM.*
3. *(M, d_g) is a complete metric space.*

These are also equivalent to geodesic completeness. When they hold, every two points are joined by a length-minimizing geodesic.

Proof. Rescaling shows that (1) is equivalent to geodesic completeness, and (1) immediately implies (2).

(3) implies (1). Let $\gamma : (a, b) \to M$ be a maximal nonconstant geodesic. After rescaling, suppose it has unit speed. If $b < \infty$, then

$$d_g(\gamma(s), \gamma(t)) \leq |s - t|,$$

so completeness gives $q = \lim_{t\uparrow b}\gamma(t)$. Choose a uniformly δ-normal neighborhood W of q and t_j close enough to b that $\gamma(t_j) \in W$ and $b - t_j < \delta$. The curve

$$\sigma(t) = \exp_{\gamma(t_j)}((t-t_j)\dot\gamma(t_j)), \qquad |t-t_j| < \delta,$$

is a geodesic with the same position and velocity as γ at t_j. Uniqueness makes it an extension beyond b, a contradiction. Reverse time to treat the left endpoint. Constant geodesics already exist for all time.

From (2), first construct minimizing geodesics from p. Fix $q \neq p$ and set $\ell = d_g(p,q)$. Choose $0 < \varepsilon < \ell$ so small that $\mathcal{B}_{2\varepsilon}(p)$ is a geodesic ball. The sphere $S_\varepsilon(p)$ is compact, so $x \mapsto d_g(x,q)$ attains a minimum there, at a point x. Every curve from p to q crosses this sphere; splitting its length there and taking the infimum gives

$$\ell \geq \varepsilon + \min_{y\in S_\varepsilon(p)} d_g(y,q).$$

The triangle inequality gives the reverse inequality. Thus $\ell = \varepsilon + d_g(x,q)$. Write $x = \exp_p(\varepsilon u)$ with $|u| = 1$. Assumption (2) and rescaling define $\gamma(t) = \exp_p(tu)$ for all $t \in \mathbb{R}$. Consider the closed nonempty set

$$A = \{t \in [\varepsilon, \ell] : \ell = t + d_g(\gamma(t), q)\}.$$

For $t \in A$, the triangle inequality and the length of $\gamma|_{[0,t]}$ give $d_g(p,\gamma(t)) = t$. Let $T = \max A$. If $T < \ell$, put $z = \gamma(T)$ and choose $0 < \eta < \ell - T$ so small that $\mathcal{B}_{2\eta}(z)$ is a geodesic ball. The same sphere argument supplies $y \in S_\eta(z)$ with

$$d_g(z,q) = \eta + d_g(y,q).$$

Join z to y by its radial unit-speed geodesic σ. Then

$$\ell \leq d_g(p,y) + d_g(y,q) \leq T + \eta + d_g(y,q) = \ell.$$

All inequalities are equalities. The concatenation of $\gamma|_{[0,T]}$ and σ is therefore a shortest curve from p to y. By Theorem 1.4.18, it has no corner. Uniqueness forces σ to continue γ, so $y = \gamma(T+\eta)$ and $T + \eta \in A$, a contradiction. Hence $T = \ell$, $\gamma(\ell) = q$, and $L_g(\gamma|_{[0,\ell]}) = \ell$.

(2) implies (3). Let (q_j) be Cauchy. The distances $d_g(p,q_j)$ are bounded. The preceding construction provides $v_j \in T_pM$ with

$$\exp_p(v_j) = q_j, \qquad |v_j|_{g_p} = d_g(p,q_j).$$

A subsequence v_{j_k} converges in the finite-dimensional space T_pM, say to v. Continuity gives $q_{j_k} \to \exp_p(v)$. A Cauchy sequence with a convergent subsequence

converges, proving completeness.

The equivalences are now proved. Under (1), condition (2) holds at every base point, so the minimizing-geodesic construction applies to any pair of points. □

Corollary 1.4.31. *Every Riemannian metric on a compact manifold is complete.*

Proof. On each connected component, d_g induces the manifold topology. Compactness therefore implies metric completeness, and Hopf–Rinow gives geodesic completeness. This applies to every component. □

1.4.6 Riemannian Coverings and Homotopy Classes

Definition 1.4.32 (Riemannian covering)**.** A smooth covering map $\pi : (\widetilde{M}, \widetilde{g}) \to (M, g)$ is a Riemannian covering if it is a local isometry, that is, $\widetilde{g} = \pi^* g$:

$$\widetilde{g}_{\widetilde{p}}(X, Y) = g_{\pi(\widetilde{p})}(d\pi_{\widetilde{p}} X, d\pi_{\widetilde{p}} Y).$$

A local isometry preserves the Levi-Civita connection. Indeed, transport the source connection to a target neighborhood using a local inverse. The resulting connection has zero torsion and preserves the target metric, so uniqueness implies that it coincides with the target connection. Local isometries preserve lengths and send geodesics to geodesics. A lift of a geodesic through a local isometry is also a geodesic.

Proposition 1.4.33. *If $\pi : (\widetilde{M}, \widetilde{g}) \to (M, g)$ is a Riemannian covering and M is complete, then $\widetilde{M}$ is geodesically complete, and each of its connected components is metrically complete.*

Proof. For initial data $(\widetilde{p}, \widetilde{v})$, take the geodesic $\gamma : \mathbb{R} \to M$ with initial data $(\pi(\widetilde{p}), d\pi\widetilde{v})$. The path-lifting property gives a unique lift through $\widetilde{p}$ on all of $\mathbb{R}$. It is geodesic and has initial velocity $\widetilde{v}$, since $d\pi$ is an isomorphism. □

Theorem 1.4.34 (Shortest geodesics in fixed-endpoint homotopy classes)**.** *Let M be connected and complete, and let $p, q \in M$. Every homotopy class of paths from p to q, with endpoints fixed during the homotopy, contains a geodesic of minimal length among admissible paths in that class.*

Proof. Take the universal covering $\pi : \widetilde{M} \to M$ with $\widetilde{g} = \pi^* g$. It is a complete Riemannian covering. Fix a lift $\widetilde{p}$ of p, choose a representative path α, and let its lift starting at $\widetilde{p}$ end at $\widetilde{q}$.

Hopf–Rinow gives a shortest geodesic $\widetilde{\gamma}$ from $\widetilde{p}$ to $\widetilde{q}$. Its projection γ is geodesic and belongs to the class of α: paths in the simply connected cover with the same

endpoints are homotopic relative to those endpoints. Every other admissible path β in that class lifts from $\widetilde{p}$ to the same $\widetilde{q}$. Hence

$$L_g(\beta) = L_{\widetilde{g}}(\widetilde{\beta}) \geq L_{\widetilde{g}}(\widetilde{\gamma}) = L_g(\gamma).$$

This is minimality within the chosen class, not necessarily among all paths from p to q. □

Definition 1.4.35 (Closed geodesics and free homotopy)**.** A geodesic segment $\gamma : [a, b] \to M$ is closed if

$$\gamma(a) = \gamma(b), \qquad \dot{\gamma}(a) = \dot{\gamma}(b).$$

These conditions give a smooth periodic geodesic by uniqueness. Loops $\alpha_0, \alpha_1 : [a, b] \to M$ are freely homotopic if there is a continuous $H : [0, 1] \times [a, b] \to M$ with $H(0, t) = \alpha_0(t)$, $H(1, t) = \alpha_1(t)$, and $H(s, a) = H(s, b)$ for all s. The common endpoint may move with s.

Exercise from the lecture. If M is compact, every nontrivial free homotopy class contains a nonconstant closed geodesic minimizing length in that class. In the trivial class, a constant loop is the length-minimizing representative. If closed geodesics are required to be nonconstant, the nontrivial-class qualification is necessary.

Theorem 1.4.36 (A complete local isometry is a covering)**.** *Let $\pi : (\widetilde{M}, \widetilde{g}) \to (M, g)$ be a local isometry between nonempty connected Riemannian manifolds. If $\widetilde{M}$ is complete, then M is complete and π is a surjective Riemannian covering.*

Proof. completeness and surjectivity. Choose $\widetilde{p} \in \widetilde{M}$ and put $p = \pi(\widetilde{p})$. For any $v \in T_pM$, set $\widetilde{v} = (d\pi_{\widetilde{p}})^{-1}v$. The geodesic with initial data $(\widetilde{p}, \widetilde{v})$ exists for all time. Its projection is the geodesic with initial data (p, v), so $\exp_p$ is defined on all of T_pM. Hopf–Rinow makes M complete. Every $q \in M$ can now be joined to p by a geodesic. Choose its lifted initial velocity at $\widetilde{p}$ and use completeness of $\widetilde{M}$ to obtain the corresponding geodesic on the whole parameter interval. By geodesic uniqueness, its projection is the chosen geodesic from p to q. Its endpoint therefore maps to q, proving surjectivity without assuming the covering property in advance.

Evenly covered neighborhoods. Choose a normal ball $U = B_g(p, r)$ and write $\pi^{-1}(p) = \{\widetilde{p}_\alpha\}_\alpha$. Geodesic uniqueness gives

$$\pi \circ \exp_{\widetilde{p}_\alpha} = \exp_p \circ d\pi_{\widetilde{p}_\alpha}.$$

For each α, define a smooth inverse branch on U:

$$s_\alpha(q) = \exp_{\widetilde{p}_\alpha}\big((d\pi_{\widetilde{p}_\alpha})^{-1} \exp_p^{-1}(q)\big).$$

Then $\pi \circ s_\alpha = \mathrm{id}_U$. Since $d\pi$ is invertible, ds_α is invertible too. Thus $\widetilde{U}_\alpha = s_\alpha(U)$ is open, and $\pi : \widetilde{U}_\alpha \to U$ is a diffeomorphism.

If $s_\alpha(q) = s_\beta(q)$, reverse the radial geodesic from p to q. Its two lifts from this common point have the same initial velocity, since their images under $d\pi$ agree. Uniqueness makes the lifts identical, so $\widetilde{p}_\alpha = \widetilde{p}_\beta$. Thus, the sheets are pairwise disjoint.

Given $\widetilde{q} \in \pi^{-1}(U)$, lift the reversed radial segment from $q = \pi(\widetilde{q})$ to p by its initial velocity at $\widetilde{q}$. Completeness ensures this geodesic exists along the whole segment. Its endpoint is some $\widetilde{p}_\alpha$; reversing again gives $\widetilde{q} = s_\alpha(q)$. Hence

$$\pi^{-1}(U) = \coprod_\alpha \widetilde{U}_\alpha,$$

proving the covering property. These sheets also equal $B_{\widetilde{g}}(\widetilde{p}_\alpha, r)$, as in the lecture: one inclusion follows from lifted radial lengths; conversely, a minimizing segment of length less than r projects to the unique radial segment in U. □

Remark 1.4.37. Under the usual second-countability assumption for manifolds, the sheets are at most countable. The fiber is discrete, and its distinct points can be assigned distinct members of a countable basis meeting the fiber in only that point.

Chapter 2

Curvature and Topology

2.1 Curvature Tensors

Curvature measures the failure of covariant derivatives to commute. We follow the lecture from tensor algebra to the curvature tensor, parallel transport and flatness, curvature identities, and finally the second variation and its global applications.

2.1.1 Tensor Bundles

Let V, W be finite-dimensional real vector spaces of dimensions m, n. The tensor product $V \otimes W$ is generated by symbols $v \otimes w$ subject to bilinearity. If $v_1, \dots, v_m$ and $w_1, \dots, w_n$ are bases, then $v_i \otimes w_a$ form a basis, so $\dim(V \otimes W) = mn$. Linear maps $A : V_1 \to V_2$ and $B : W_1 \to W_2$ induce

$$A \otimes B : V_1 \otimes W_1 \longrightarrow V_2 \otimes W_2, \qquad v \otimes w \longmapsto Av \otimes Bw.$$

Proposition 2.1.1. *There is a natural isomorphism*

$$V^* \otimes W^* \cong \mathrm{Bil}(V \times W, \mathbb{R}).$$

Proof. Send $\alpha \otimes \beta$ to $(v, w) \mapsto \alpha(v)\beta(w)$ and extend linearly. In dual bases v^i, w^a, a bilinear form B has the unique expansion

$$B = \sum_{i,a} B(v_i, w_a) v^i \otimes w^a.$$

This supplies an inverse to the stated map. □

If $E, F \to M$ are smooth vector bundles, their tensor product has fiber $(E \otimes F)_p = E_p \otimes F_p$. Local frames e_i, f_a give the frame $e_i \otimes f_a$. Its transition matrices are tensor products of the original smooth transition matrices, which equips the fiberwise construction with a smooth bundle structure. Duals and iterated tensor products are constructed similarly.

Definition 2.1.2 (Tensor fields)**.** The bundle of tensors of type (r, s) is

$$T^r_s M = (TM)^{\otimes r} \otimes (T^*M)^{\otimes s}.$$

A smooth (r, s)-tensor field is a smooth section of this bundle. It can be evaluated on r one-forms and s vector fields to give a smooth function.

Proposition 2.1.3 (Tensoriality criterion)**.** *Smooth (r, s)-tensor fields are equivalent to $C^\infty(M)$-multilinear maps*

$$\underbrace{\Omega^1(M) \times \cdots \times \Omega^1(M)}_{r} \times \underbrace{\mathscr{X}(M) \times \cdots \times \mathscr{X}(M)}_{s} \longrightarrow C^\infty(M).$$

Proof. A tensor field gives such a map by pointwise evaluation. Conversely, multilinearity over smooth functions implies locality in each slot: if an argument vanishes near p, multiply it by a cutoff equal to 1 near p to see that its contribution at p is zero. In a local frame, expand each argument and use cutoffs to extend the frame near p. Multilinearity then shows that the value at p depends only on the arguments at p. Evaluating on frame elements gives smooth coefficients, hence a smooth tensor field. The construction is independent of the frame because it reproduces the given map. □

Example 2.1.4. The metric is a $(0,2)$-tensor. For a connection on TM, its torsion

$$\mathcal{T}(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]$$

is a $(1,2)$-tensor: the derivative terms cancel when either argument is multiplied by a function. Lowering its output index gives the $(0,3)$-tensor $g(\mathcal{T}(X,Y),Z)$. For the Levi-Civita connection, $\mathcal{T} = 0$.

2.1.2 The Curvature Tensor and Flatness

Definition 2.1.5 (Riemann curvature tensor)**.** For the Levi-Civita connection, set

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z. \tag{2.1}$$

Its metric version is $R(X,Y,Z,W) = g(R(X,Y)Z,W)$.

Proposition 2.1.6. *$R(X,Y)Z$ is $C^\infty(M)$-linear in all three arguments. Thus it defines a $(1,3)$-tensor, and its metric version is a $(0,4)$-tensor.*

Proof. Using $[fX,Y] = f[X,Y] - Y(f)X$, we obtain

$$\begin{aligned} R(fX,Y)Z &= f\nabla_X \nabla_Y Z - \nabla_Y(f\nabla_X Z) - f\nabla_{[X,Y]}Z + Y(f)\nabla_X Z \\ &= fR(X,Y)Z. \end{aligned}$$

Skew-symmetry in X,Y gives linearity in the second argument. Expanding in the third argument gives

$$R(X,Y)(fZ) = fR(X,Y)Z + \big(XY(f) - YX(f) - [X,Y](f)\big)Z = fR(X,Y)Z.$$

The mixed first-derivative terms also cancel by the Leibniz rule. Pairing with g proves the final assertion. □

Coordinate formula. Write $\partial_i = \partial/\partial x^i$ and define

$$R(\partial_i,\partial_j)\partial_k = R^{\ell}{}_{ijk}\partial_\ell, \qquad R_{ijk\ell} = g_{a\ell}R^{a}{}_{ijk}.$$

Since coordinate fields commute,

$$\nabla_{\partial_i}\nabla_{\partial_j}\partial_k = \big(\partial_i\Gamma^\ell_{jk} + \Gamma^a_{jk}\Gamma^\ell_{ia}\big)\partial_\ell.$$

Subtracting the expression with i, j interchanged yields

$$R^\ell{}_{ijk} = \partial_i\Gamma^\ell_{jk} - \partial_j\Gamma^\ell_{ik} + \Gamma^a_{jk}\Gamma^\ell_{ia} - \Gamma^a_{ik}\Gamma^\ell_{ja}. \tag{2.2}$$

Thus torsion-freeness does not imply that second covariant derivatives commute: even though $\nabla_{\partial_i}\partial_j = \nabla_{\partial_j}\partial_i$, the expression above need not vanish.

Definition 2.1.7 (Flatness)**.** A Riemannian manifold is flat if each point has a coordinate neighborhood on which all metric coefficients are constant. A linear change of coordinates makes this constant positive-definite matrix the identity, so this means local isometry to Euclidean space.

Lemma 2.1.8. *A flat Riemannian manifold has $R = 0$.*

Proof. In the stated coordinates, the Christoffel symbols vanish throughout the neighborhood. Formula (2.2) gives $R = 0$. □

To prove the converse, the lecture introduces parallel transport.

Definition 2.1.9 (Parallel fields and parallel transport)**.** A field V along a smooth curve $\gamma : I \to M$ is parallel if $D_tV = 0$. For $t_0, t_1 \in I$, parallel transport sends $v \in T_{\gamma(t_0)}M$ to $V(t_1)$, where V is parallel and $V(t_0) = v$.

Proposition 2.1.10. *Every initial vector determines a unique parallel field along all of γ. Parallel transport is a linear isometry, and reversal of the curve gives its inverse.*

Proof. In coordinates, the equation is the linear system

$$\dot V^k + \Gamma^k_{ij}(\gamma(t))\dot\gamma^i V^j = 0.$$

Existence and uniqueness for linear ODEs, patched over coordinate subintervals, give a solution on I. Solutions depend linearly on the initial vector. If V, W are parallel, metric compatibility gives

$$\frac{d}{dt}g(V, W) = g(D_tV, W) + g(V, D_tW) = 0.$$

Thus inner products are preserved. Uniqueness gives the inverse under reversal. The same construction works along piecewise smooth curves by composition. □

Theorem 2.1.11 (Vanishing curvature and local flatness)**.** *A Riemannian manifold is flat if and only if $R = 0$.*

Proof. Only the converse remains. Assume $R = 0$ and choose a coordinate cube about p, with $x(p) = 0$. For $v \in T_pM$, we construct a parallel field on a smaller cube. Set

$$M_k = \{x^{k+1} = \cdots = x^n = 0\}, \qquad M_0 = \{p\}.$$

Start with v at p. After defining V on M_k, extend it to M_{k+1} by parallel transport along the x^{k+1}-coordinate lines. Smooth dependence for linear ODEs makes each extension smooth.

Inductively, $\nabla_{\partial_i} V = 0$ on M_k for $i \le k$. By construction, $\nabla_{\partial_{k+1}} V = 0$ on M_{k+1}. For $i \le k$, zero curvature and commuting coordinate fields give

$$\nabla_{\partial_{k+1}}(\nabla_{\partial_i} V) = \nabla_{\partial_i}(\nabla_{\partial_{k+1}} V) = 0.$$

The field $\nabla_{\partial_i} V$ is therefore parallel along each new coordinate line and has zero initial value on M_k. It is identically zero. At the last step, $\nabla_X V = 0$ for every X.

Apply the construction to an orthonormal basis at p. This gives parallel fields $E_1, \ldots, E_n$. Since $Xg(E_i, E_j) = 0$, they remain orthonormal. Torsion-freeness gives $[E_i, E_j] = 0$. Compose their commuting local flows and apply the inverse function theorem to obtain local coordinates y^i satisfying

$$E_i = \frac{\partial}{\partial y^i}, \qquad g_{ij} = \delta_{ij}.$$

□

Remark 2.1.12. Every Riemannian manifold has local orthonormal frames, but these need not be parallel or commute. Likewise, normal coordinates make Γ_{ij}^k vanish at one point, not generally on a whole neighborhood. The theorem concerns local flatness, not a global isometry to $\mathbb{R}^n$.

2.1.3 Model Spaces and Curvature Symmetries

Example 2.1.13 (Euclidean space)**.** For the standard Euclidean metric, $\Gamma_{ij}^k = 0$ in Cartesian coordinates and $R = 0$.

Example 2.1.14 (The hyperbolic ball)**.** On $B_a(0) \subset \mathbb{R}^n$, consider the Poincare metric

$$g_a = \frac{4a^4}{(a^2 - |x|^2)^2} \sum_i (dx^i)^2 = e^{2h} \sum_i (dx^i)^2, \qquad h = \log \frac{2a^2}{a^2 - |x|^2}.$$

The conformal factor is e^{2h}, so the Christoffel formula gives

$$\Gamma_{ij}^k = h_i \delta_j^k + h_j \delta_i^k - h_k \delta_{ij}.$$

For clarity, set $A_{ij} = h_{ij} - h_i h_j + \frac{1}{2}|dh|_0^2 \delta_{ij}$, where the norm is Euclidean. Substitution into (2.2) gives

$$R_{ijk\ell} = e^{2h}(A_{ik}\delta_{j\ell} + A_{j\ell}\delta_{ik} - A_{i\ell}\delta_{jk} - A_{jk}\delta_{i\ell}).$$

Here, putting $D = a^2 - |x|^2$, we have

$$h_i = \frac{2x^i}{D}, \qquad h_{ij} = \frac{2\delta_{ij}}{D} + \frac{4x^i x^j}{D^2}, \qquad A_{ij} = \frac{2a^2}{D^2}\delta_{ij} = \frac{e^{2h}}{2a^2}\delta_{ij}.$$

Consequently,

$$R(X,Y)Z = -\frac{1}{a^2}\big(g_a(Y,Z)X - g_a(X,Z)Y\big).$$

Example 2.1.15 (The round sphere)**.** Let $\mathbb{S}^n(a) \subset \mathbb{R}^{n+1}$ have its induced metric, and write D for the ambient Euclidean connection. The unit normal is $N(p) = p/a$, so $D_X N = X/a$. For tangent X, Y,

$$g(D_X Y, N) = -g(Y, D_X N) = -\frac{1}{a}g(X,Y),$$
$$\nabla_X Y = D_X Y + \frac{1}{a^2}g(X,Y)p.$$

Taking the tangential component gives

$$\nabla_X \nabla_Y Z = (D_X D_Y Z)^\top + a^{-2} g(Y,Z)X.$$

Subtract the expression with X, Y interchanged and the bracket term. The ambient curvature is zero, so the result is

$$R(X,Y)Z = \frac{1}{a^2}\big(g(Y,Z)X - g(X,Z)Y\big).$$

These three examples will have sectional curvatures $0, -a^{-2}, a^{-2}$, respectively, with the convention fixed below.

Proposition 2.1.16 (Curvature symmetries)**.** *The metric curvature tensor satisfies*

$$R(X,Y,Z,W) = -R(Y,X,Z,W),$$
$$R(X,Y,Z,W) = -R(X,Y,W,Z),$$
$$R(X,Y,Z,W) = R(Z,W,X,Y),$$
$$R(X,Y)Z + R(Y,Z)X + R(Z,X)Y = 0.$$

The last identity is the first Bianchi identity.

Proof. The first identity follows from the definition. For the second, metric com-

patibility gives

$$R(X,Y,Z,Z) = \tfrac{1}{2}\big(XY - YX - [X,Y]\big)g(Z,Z) = 0.$$

Polarizing in Z proves skew-symmetry in the last two entries.

For the first Bianchi identity, use torsion-freeness in the cyclic sum:

$$\begin{aligned}\sum_{\text{cyc}} R(X,Y)Z &= \sum_{\text{cyc}} \big(\nabla_X[Y,Z] - \nabla_{[Y,Z]}X\big) \\ &= [X,[Y,Z]] + [Y,[Z,X]] + [Z,[X,Y]] = 0.\end{aligned}$$

The last step is the Jacobi identity.

For completeness, pair interchange can be checked at p in normal coordinates. The Christoffel formula and $\partial g(p) = 0$ give

$$R_{ijk\ell}(p) = \tfrac{1}{2}\big(\partial_i\partial_k g_{j\ell} - \partial_i\partial_\ell g_{jk} - \partial_j\partial_k g_{i\ell} + \partial_j\partial_\ell g_{ik}\big)(p).$$

Symmetry of g and commuting partial derivatives show this equals $R_{k\ell ij}(p)$. Tensoriality then proves the identity for arbitrary vectors. $\square$

2.1.4 Covariant Derivatives of Tensors and the Ricci Identity

The connection on TM extends to one-forms by requiring compatibility with evaluation:

$$(\nabla_X \alpha)(Y) = X(\alpha(Y)) - \alpha(\nabla_X Y).$$

For $\alpha = \alpha_j dx^j$, this reads

$$(\nabla_i \alpha)_j = \partial_i \alpha_j - \Gamma_{ij}^k \alpha_k.$$

Extend to tensor products by the product rule

$$\nabla_X(A \otimes B) = (\nabla_X A) \otimes B + A \otimes (\nabla_X B).$$

Equivalently, for a tensor T of type (r,s),

$$\begin{aligned}(\nabla_X T)(\alpha_1,\dots,\alpha_r,Y_1,\dots,Y_s) &= X\big(T(\alpha_1,\dots,\alpha_r,Y_1,\dots,Y_s)\big) \\ &\quad - \sum_{a=1}^{r} T(\alpha_1,\dots,\nabla_X\alpha_a,\dots,\alpha_r,Y_1,\dots,Y_s) \\ &\quad - \sum_{b=1}^{s} T(\alpha_1,\dots,\alpha_r,Y_1,\dots,\nabla_X Y_b,\dots,Y_s).\end{aligned}$$

This formula is tensorial in the displayed arguments. It gives a connection on every tensor bundle and commutes with contractions. In particular, metric compatibility is exactly $\nabla g = 0$.

We put the differentiation slot last:

$$(\nabla T)(\alpha_1, \dots, \alpha_r, Y_1, \dots, Y_s, X) = (\nabla_X T)(\alpha_1, \dots, \alpha_r, Y_1, \dots, Y_s).$$

Thus ∇T has type $(r, s+1)$. In coordinates,

$$\begin{aligned}(\nabla_m T)^{i_1 \cdots i_r}{}_{j_1 \cdots j_s} &= \partial_m T^{i_1 \cdots i_r}{}_{j_1 \cdots j_s} \\ &\quad + \sum_{a=1}^{r} \Gamma^{i_a}_{mk} T^{i_1 \cdots k \cdots i_r}{}_{j_1 \cdots j_s} - \sum_{b=1}^{s} \Gamma^{k}_{m j_b} T^{i_1 \cdots i_r}{}_{j_1 \cdots k \cdots j_s}.\end{aligned}$$

At the center of normal coordinates the connection terms vanish. Their derivatives need not vanish; this distinction matters in second derivatives.

Theorem 2.1.17 (Second Bianchi identity)**.** *With the differentiation slot last,*

$$\begin{aligned}(\nabla R)(X, Y, Z, W, U) + (\nabla R)(X, Y, W, U, Z) \\ + (\nabla R)(X, Y, U, Z, W) = 0.\end{aligned}$$

Proof. Choose normal coordinates at p. In differentiating (2.2) at p, the derivatives of the quadratic Christoffel terms vanish because $\Gamma(p) = 0$. Therefore

$$(\nabla_m R)^{\ell}{}_{ijk}(p) = \partial_m \partial_i \Gamma^{\ell}_{jk}(p) - \partial_m \partial_j \Gamma^{\ell}_{ik}(p).$$

The cyclic sum in m, i, j cancels term by term. Lowering the output index is compatible with ∇, so

$$(\nabla_m R)_{ijk\ell} + (\nabla_i R)_{jmk\ell} + (\nabla_j R)_{mik\ell} = 0.$$

Pair interchange converts this to the stated form, cyclic in the last two curvature entries and the differentiation entry. All expressions are tensorial, so the identity holds everywhere. □

Define the second covariant derivative in two directions by

$$\nabla^2_{X,Y} T = \nabla_X (\nabla_Y T) - \nabla_{\nabla_X Y} T.$$

Under the last-slot convention, this is $\nabla^2 T$ evaluated in the final two slots as (Y, X), not (X, Y).

Proposition 2.1.18 (Ricci identity)**.** *For a vector field Z and a one-form α,*

$$\nabla^2_{X,Y} Z - \nabla^2_{Y,X} Z = R(X,Y)Z,$$
$$(\nabla^2_{X,Y}\alpha - \nabla^2_{Y,X}\alpha)(Z) = -\alpha(R(X,Y)Z).$$

More generally, write $A = R(X,Y)$ and $A^*\alpha = \alpha \circ A$. For a tensor T of type (r,s),

$$\begin{aligned}&\big(\nabla^2_{X,Y}T - \nabla^2_{Y,X}T\big)(\alpha_1,\dots,\alpha_r,Z_1,\dots,Z_s)\\ &= \sum_{a=1}^{r} T(\alpha_1,\dots,A^*\alpha_a,\dots,\alpha_r,Z_1,\dots,Z_s)\\ &\quad - \sum_{b=1}^{s} T(\alpha_1,\dots,\alpha_r,Z_1,\dots,AZ_b,\dots,Z_s).\end{aligned}$$

Proof. Torsion-freeness identifies the operator on the left with

$$\mathcal{R}_{X,Y} = \nabla_X\nabla_Y - \nabla_Y\nabla_X - \nabla_{[X,Y]}.$$

It equals $R(X,Y)$ on vector fields and vanishes on functions. Applying it to the contraction $\alpha(Z)$ yields

$$0 = (\mathcal{R}_{X,Y}\alpha)(Z) + \alpha(R(X,Y)Z),$$

which proves the one-form identity. The product rule on tensor products gives the general formula: a plus sign on each vector factor, a minus sign on each covector factor. In components, the same formula follows by differentiating the preceding coordinate expression and antisymmetrizing. The surviving derivatives of Γ give ${R^\ell}_{ijk}$. □

2.1.5 Sectional, Ricci, and Scalar Curvature

Definition 2.1.19 (Sectional curvature)**.** For a two-plane $\Pi \subset T_pM$ and an orthonormal basis u, v of Π, set

$$K(\Pi) = R(u,v,v,u).$$

Remark 2.1.20 (Sign convention). Throughout, we use (2.1) and lower the output index by

$$R(X,Y,Z,W) = g(R(X,Y)Z,W).$$

Thus the round sphere has positive sectional curvature. For orthonormal u, v, this convention gives $R(u,v,u,v) = -K(\Pi)$. Interchanging the last two entries requires a minus sign.

Proposition 2.1.21. *Sectional curvature is independent of the orthonormal basis. For any linearly independent u, v,*

$$K(\mathrm{span}\{u,v\}) = \frac{R(u,v,v,u)}{g(u,u)g(v,v) - g(u,v)^2}.$$

Proof. Under a basis change $(\widetilde{u}, \widetilde{v}) = (au+bv, cu+dv)$, skew-symmetry in both pairs multiplies the numerator by $(ad-bc)^2$. The Gram determinant in the denominator transforms by the same factor. For an orthonormal basis it equals 1. □

Proposition 2.1.22. *The sectional curvatures at a point determine the entire Riemann curvature tensor there.*

Proof. Let B be the difference of two algebraic curvature tensors with the same sectional curvatures. Then $B(x,y,y,x) = 0$ for all x, y, including dependent pairs. Polarizing in y gives $B(x,y,z,x) = 0$. Polarizing this in x gives

$$B(x,y,z,w) + B(w,y,z,x) = 0.$$

Using pair interchange and the two skew-symmetries, this implies $B(x,y,z,w) = -B(x,z,y,w)$. Hence B is alternating in its first three entries. The first Bianchi identity is then $3B(x,y,z,w) = 0$. □

Remark 2.1.23 (Curvature on bivectors). The blackboard also views curvature as a symmetric bilinear form on $\Lambda^2 T_pM$. With our sectional-curvature convention, define

$$\mathcal{B}(u \wedge v, z \wedge w) = R(u,v,w,z).$$

Skew-symmetry in each pair makes this well-defined, and pair interchange makes it symmetric. If u, v are orthonormal, then $\mathcal{B}(u \wedge v, u \wedge v) = K(\mathrm{span}\{u,v\})$. For arbitrary bivectors ξ, η, polarization reads

$$2\mathcal{B}(\xi,\eta) = \mathcal{B}(\xi+\eta, \xi+\eta) - \mathcal{B}(\xi,\xi) - \mathcal{B}(\eta,\eta).$$

The sum of two simple bivectors need not be simple. This is why the preceding determination argument also uses the first Bianchi identity.

Definition 2.1.24 (Ricci and scalar curvature)**.** For an orthonormal basis $e_1, \ldots, e_n$ of T_pM, define

$$\mathrm{Ric}(u,v) = \sum_i R(e_i,u,v,e_i), \qquad S = \sum_j \mathrm{Ric}(e_j,e_j).$$

These are traces, so they do not depend on the orthonormal basis. In arbitrary

coordinates,

$$\mathrm{Ric}_{jk} = g^{i\ell} R_{ijk\ell}, \qquad S = g^{jk}\,\mathrm{Ric}_{jk}\,.$$

The first Bianchi identity, paired with e_i, gives $R(e_i, u, v, e_i) = R(e_i, v, u, e_i)$; thus Ricci is symmetric. For a unit vector $u = e_1$,

$$\mathrm{Ric}(u,u) = \sum_{i=2}^{n} K(\mathrm{span}\{u, e_i\}), \qquad S = 2\sum_{i<j} K(\mathrm{span}\{e_i, e_j\}).$$

These are sums of sectional curvatures, rather than normalized averages. Constant sectional curvature c gives $\mathrm{Ric} = (n-1)cg$ and $S = n(n-1)c$.

Low-dimensional consequences. On a surface, the only tangent two-plane is T_pM, so one function K determines R; moreover $\mathrm{Ric} = Kg$ and $S = 2K$. In dimension three, Ricci determines R. Indeed, for any orthonormal triple u, v, w,

$$K(u,v) = \mathrm{Ric}(u,u) + \mathrm{Ric}(v,v) - \frac{S}{2}.$$

This follows by solving the three equations expressing Ricci as sums of two sectional curvatures. Since every two-plane admits such a triple, the preceding proposition determines R.

2.1.6 Second Variation of Length

The manuscript derives the length formula, including its endpoint terms. We state that formula first and then relate it to energy. Let $F(s,t)$ be a variation, and put $S_F = \partial_s F$ and $T_F = \partial_t F$; the notation S_F is distinct from scalar curvature.

Proposition 2.1.25 (Commuting derivatives along a variation)**.** *For any field W along F,*

$$D_s D_t W - D_t D_s W = R(S_F, T_F)W.$$

Proof. The left-hand side is linear over smooth functions of (s,t): the mixed derivatives of a coefficient cancel. It is therefore enough to check the fields $\partial_k|_F$. Expanding the two covariant derivatives gives

$$(D_s D_t - D_t D_s)\partial_k|_F = \partial_s F^i \partial_t F^j R^{\ell}{}_{ijk}(F)\partial_\ell|_F.$$

Terms containing $\partial_s\partial_t F^i - \partial_t\partial_s F^i$ cancel. The remaining coefficient is precisely (2.2), proving the claim. □

Theorem 2.1.26 (Second variation of length)**.** *Let $\gamma : [a,b] \to M$ be a unit-speed geodesic and F a smooth variation of γ through regular curves. Set*

$$V(t) = \partial_s F(0,t), \qquad V^{\perp} = V - g(V,\dot\gamma)\dot\gamma, \qquad A(t) = (D_s S_F)(0,t).$$

Then

$$\frac{d^2}{ds^2}\Big|_{s=0} L_g(F(s,\cdot)) = \int_a^b \Big(|D_t V^\perp|^2 - R(V^\perp, \dot\gamma, \dot\gamma, V^\perp)\Big)\, dt + \big[g(A, \dot\gamma)\big]_a^b. \tag{2.3}$$

For fixed endpoints the boundary term is zero.

Proof. Differentiate the speed twice. Since $|T_F| = 1$ at $s = 0$,

$$\partial_s^2 |T_F|\big|_{s=0} = |D_s T_F|^2 - g(D_s T_F, T_F)^2 + g(D_s D_s T_F, T_F)\big|_{s=0}.$$

Torsion-freeness gives $D_s T_F = D_t S_F$, and the commutation formula gives

$$D_s D_s T_F = D_s D_t S_F = D_t D_s S_F + R(S_F, T_F) S_F.$$

Thus the integrand at $s = 0$ is

$$|D_t V|^2 - g(D_t V, \dot\gamma)^2 - R(V, \dot\gamma, \dot\gamma, V) + \frac{d}{dt} g(A, \dot\gamma),$$

where $D_t \dot\gamma = 0$ was used. If $f = g(V, \dot\gamma)$, then $D_t V = D_t V^\perp + f' \dot\gamma$, with $D_t V^\perp \perp \dot\gamma$. The first two terms therefore reduce to $|D_t V^\perp|^2$. Curvature skew-symmetries remove the tangential part of V from the curvature term. Integration proves the formula. When endpoints are fixed, their velocity S_F and acceleration $D_s S_F$ vanish there. □

The same formula holds for a continuous variation smooth on a fixed finite set of strips, with γ a smooth geodesic and the usual smooth endpoint traces. Apply the computation on each strip; the acceleration terms at internal breakpoints cancel, since the endpoint paths agree and $\dot\gamma$ has no jump.

Definition 2.1.27 (Index form)**.** Along a geodesic, the index form is

$$I(V, W) = \int_a^b \left(g(D_t V, D_t W) - R(V, \dot\gamma, \dot\gamma, W)\right) dt.$$

It is symmetric by the curvature identities. A proper field is a continuous piecewise smooth field with $V(a) = V(b) = 0$.

Corollary 2.1.28. *If a unit-speed geodesic minimizes length between its endpoints, then $I(V, V) \geq 0$ for every proper field V.*

Proof. First take V normal and realize it by $F(s, t) = \exp_{\gamma(t)}(sV(t))$. The endpoints are fixed, and length has a minimum at $s = 0$, so (2.3) gives $I(V, V) \geq 0$. For an

arbitrary proper field $V = V^\perp + f\dot{\gamma}$,

$$I(V,V) = I(V^\perp, V^\perp) + \int_a^b (f')^2\,dt \geq 0.$$

□

Remark 2.1.29 (Energy normalization). If energy is defined without a factor $1/2$,

$$E(\sigma) = \int_a^b |\dot{\sigma}|^2\,dt,$$

then a fixed-endpoint variation of a geodesic satisfies

$$\frac{1}{2}E''(0) = I(V,V).$$

Indeed, differentiating $g(T_F, T_F)$ twice and using the same commutation calculation gives the integral above; the endpoint acceleration term vanishes. For length, the corresponding expression is $I(V^\perp, V^\perp)$ at a unit-speed geodesic. In particular, for normal fields it is $I(V,V)$; the varied curves need not themselves have constant speed. Geodesics are critical points for fixed-endpoint variations, but criticality alone does not imply minimality.

2.1.7 Applications to Topology and Diameter

Theorem 2.1.30 (Synge's theorem)**.** *Let (M,g) be connected and compact with positive sectional curvature.*

- *If $\dim M$ is even and M is orientable, then M is simply connected.*
- *If $\dim M$ is odd, then M is orientable.*

Proof. We first give the even-dimensional argument from the lecture. If M is not simply connected, choose a nontrivial free homotopy class and a shortest closed geodesic in that class, as in the preceding section. Parametrize it by unit speed on $[0, L]$, with matching endpoint velocities. Parallel transport around γ is an orthogonal map $P : T_{\gamma(0)}M \to T_{\gamma(0)}M$ fixing $\dot{\gamma}(0)$. Orientability implies $\det P = 1$: a continuously transported oriented frame cannot change its orientation.

The restriction of P to $\dot{\gamma}(0)^\perp$ is therefore an orientation-preserving orthogonal map of an odd-dimensional space. It has an eigenvector $e \neq 0$ with eigenvalue 1. To see this, nonreal eigenvalues occur in conjugate pairs, real eigenvalues are ± 1, and determinant 1 forces an even number of -1 eigenvalues. Normalize e and parallel transport it to a periodic unit normal field E along γ.

The variation $F(s,t) = \exp_{\gamma(t)}(sE(t))$ consists of closed loops in the same free homotopy class. Its s-curves are geodesics, so $D_s S_F = 0$. By the second variation

formula,

$$L''(0) = I(E,E) = -\int_0^L R(E,\dot\gamma,\dot\gamma,E)\,dt < 0,$$

contradicting minimality in that class. Notice that this is a variation of closed loops, not a fixed-endpoint variation; the boundary term vanishes for the stated reason.

For the odd-dimensional assertion, suppose that M is nonorientable. Choose a loop that reverses orientation and minimize length in its free homotopy class. The resulting nonconstant closed geodesic has $\det P = -1$. Its normal space has even dimension, and P restricted to that space has determinant -1. Such an orthogonal map also has eigenvalue 1: conjugate pairs contribute positive determinant, and an odd number of -1 eigenvalues leaves an odd number of $+1$ eigenvalues in an even-dimensional space. The same parallel-field variation gives the contradiction. In dimension one, every connected compact manifold without boundary is a circle and is already orientable. □

Corollary 2.1.31. *If M is connected, compact, even-dimensional and nonorientable with positive sectional curvature, then*

$$\pi_1(M) \cong \mathbb{Z}/2\mathbb{Z}.$$

Proof. The orientation double cover is connected, compact, orientable and has the lifted positive-curvature metric. Synge's theorem makes it simply connected. It is therefore the universal cover, with two sheets, so its deck group and $\pi_1(M)$ have order two. □

Example 2.1.32 (Spherical quotients)**.** The orientability assumption is essential: $\mathbb{RP}^{2m}$, $m \geq 1$, has positive sectional curvature and fundamental group $\mathbb{Z}/2\mathbb{Z}$. More generally, a finite group acting freely by isometries on $\mathbb{S}^{2m}$ gives a quotient with positive sectional curvature. Synge's theorem and its corollary imply that the group is trivial or has order two. The qualifications "finite" and "isometric" ensure that this is a Riemannian covering argument; an arbitrary group action need not have those properties.

Odd-dimensional spheres behave differently. For $k \geq 2$, scalar multiplication by $e^{2\pi i/k}$ acts freely and isometrically on

$$\mathbb{S}^{2m+1} \subset \mathbb{C}^{m+1}.$$

Its quotient is a lens space (the equal-weight case), and for $m \geq 1$ its fundamental group is $\mathbb{Z}/k\mathbb{Z}$.

Theorem 2.1.33 (Bonnet–Myers)**.** *Let (M^n, g) be connected and complete, with*

$n \geq 2$. *Suppose for a constant $c > 0$ that*

$$\mathrm{Ric} \geq (n-1)cg, \quad \textit{meaning} \quad \mathrm{Ric}(v,v) \geq (n-1)c\, g(v,v) \quad \textit{for all } v.$$

Then M is compact and

$$\mathrm{diam}(M) \leq \frac{\pi}{\sqrt{c}}.$$

Proof. By Hopf–Rinow, two distinct points are joined by a unit-speed minimizing geodesic $\gamma : [0, L] \to M$, where L is their distance. Choose parallel orthonormal normal fields $E_1, \ldots, E_{n-1}$ and set

$$V_i(t) = \sin\left(\frac{\pi t}{L}\right) E_i(t), \qquad D_t V_i = \frac{\pi}{L} \cos\left(\frac{\pi t}{L}\right) E_i(t).$$

Each field vanishes at both endpoints. The index inequality and the Ricci bound give

$$\begin{aligned} 0 \leq \sum_{i=1}^{n-1} I(V_i, V_i) &= \int_0^L \left((n-1)\frac{\pi^2}{L^2} \cos^2 \frac{\pi t}{L} - \sin^2 \frac{\pi t}{L}\, \mathrm{Ric}(\dot\gamma, \dot\gamma) \right) dt \\ &\leq \frac{(n-1)L}{2} \left(\frac{\pi^2}{L^2} - c \right). \end{aligned}$$

Hence $L \leq \pi/\sqrt{c}$. Every point is reached by a minimizing geodesic from a fixed p, so

$$M = \exp_p\left(\overline{B}^{T_pM}_{\pi/\sqrt{c}}(0) \right).$$

Completeness makes $\exp_p$ defined on the whole tangent space. The displayed closed ball is compact, and its continuous image is therefore compact. □

Corollary 2.1.34. *Under the hypotheses of Bonnet–Myers, $\pi_1(M)$ is finite.*

Proof. The universal cover with the lifted metric is complete and has the same Ricci lower bound. Bonnet–Myers makes it compact. A fiber of the covering is closed and discrete, hence finite in this compact space. Its cardinality is the number of sheets, equal to the order of $\pi_1(M)$. □

Remark 2.1.35. The constant $c > 0$ is a uniform lower bound. Pointwise strict positivity of Ricci curvature on a noncompact complete manifold is not by itself the hypothesis of Bonnet–Myers. The round sphere of radius $1/\sqrt{c}$ attains the diameter bound.

Further remarks from the lecture. Nonpositive sectional curvature has a different global effect: the Cartan–Hadamard theorem identifies the universal cover of a complete connected manifold of nonpositive sectional curvature with $\mathbb{R}^n$. Negative Ricci curvature alone is much less restrictive.

Theorem 2.1.36 (Lohkamp, 1994 [5])**.** *Every smooth manifold M^n, $n \geq 3$, admits a complete metric of strictly negative Ricci curvature. More precisely, there are constants $a(n) > b(n) > 0$, depending only on n, for which one can arrange*

$$-a(n)g < \mathrm{Ric}_g < -b(n)g.$$

The inequalities are inequalities of quadratic forms. In particular, every compact manifold of dimension at least three admits such a metric, as stated in the lecture.

Remark 2.1.37 (The Ricci-pinching aside). The closing blackboard note mentions Hamilton's pinching conjecture and Lee–Topping. The precise three-dimensional statement is: a complete connected three-manifold satisfying

$$\mathrm{Ric} \geq \varepsilon S g \geq 0 \qquad \text{for some constant } \varepsilon > 0$$

is flat or compact. In particular, if $S > 0$ everywhere, it is compact. This is a research-level aside, not an application of Bonnet–Myers with an assumed uniform positive lower bound on S. Lee and Topping's 2022 work, combined with work of Lott and Deruelle–Schulze–Simon, establishes this statement.[1] The dimension and pinching constant are stated explicitly here to avoid reading the abbreviated blackboard inequality as a theorem in all dimensions.

[1] See M.-C. Lee and P. M. Topping, *Three-manifolds with non-negatively pinched Ricci curvature*, arXiv:2204.00504, Theorem 1.1.

2.2 Riemannian Submanifolds

Let $\iota : M^m \hookrightarrow (\widetilde{M}, \widetilde{g})$ be a smooth embedded submanifold, with induced metric $g = \iota^* \widetilde{g}$. We identify tangent vectors to M with their images under $d\iota$. Along M there is an orthogonal splitting

$$T\widetilde{M}|_M = TM \oplus NM, \qquad N_pM = (T_pM)^\perp.$$

The normal spaces form a smooth vector bundle: locally, extend a frame of TM to an ambient frame and apply the smooth Gram–Schmidt process. We write $\top$ and $\perp$ for the corresponding projections. The ambient and induced Levi-Civita connections are denoted by $\widetilde{\nabla}$ and ∇.

2.2.1 The Second Fundamental Form

As established for induced connections, if X, Y are tangent fields and $\widetilde{X}, \widetilde{Y}$ are local ambient extensions, then

$$\nabla_X Y = \big(\widetilde{\nabla}_{\widetilde{X}} \widetilde{Y}\big)^\top\big|_M.$$

The normal part records the extrinsic bending of M.

Definition 2.2.1 (Second fundamental form)**.** The second fundamental form is the normal-valued tensor

$$\Pi \in \Gamma\big(\mathrm{Sym}^2 T^*M \otimes NM\big), \qquad \Pi(X,Y) = \big(\widetilde{\nabla}_{\widetilde{X}} \widetilde{Y}\big)^\perp\big|_M.$$

Equivalently, each $\Pi_p : T_pM \times T_pM \to N_pM$ is a symmetric bilinear map. The assertions implicit in this definition are proved next.

Proposition 2.2.2. *The definition is independent of both ambient extensions. It is smooth and satisfies*

$$\Pi(fX, Y) = \Pi(X, fY) = f\Pi(X,Y), \qquad \Pi(X,Y) = \Pi(Y,X).$$

In particular, $\Pi(X,Y)(p)$ *depends only on* X_p, Y_p.

Proof. Dependence on $\widetilde{X}$ at p is only through its value X_p. To check independence of $\widetilde{Y}$, let W be the difference of two extensions, so $W|_M = 0$. Choose a curve γ in M with $\gamma(0) = p$ and $\dot\gamma(0) = X_p$. Then

$$\widetilde{\nabla}_{X_p} W = \widetilde{D}_t(W \circ \gamma)\Big|_{t=0} = 0.$$

Thus even the full ambient derivative along a tangent direction is independent of the extension. Its normal projection is smooth.

Linearity over functions in the first entry follows from the connection axiom. In the second entry, extend f locally off M and use the Leibniz rule:

$$\big(\widetilde{\nabla}_X(fY)\big)^\perp = \big(X(f)Y + f\widetilde{\nabla}_X Y\big)^\perp = f\Pi(X,Y).$$

The notation here and below suppresses ambient extensions and restriction to M. Since the ambient connection has zero torsion,

$$\Pi(X,Y) - \Pi(Y,X) = \big[\widetilde{X},\widetilde{Y}\big]^\perp\big|_M = 0.$$

Indeed, the bracket restricts to the tangent field $[X,Y]$. Finally, expanding X, Y in a local tangent frame shows that their coefficients enter pointwise, with no derivatives. This proves the last assertion. □

Proposition 2.2.3 (Gauss formula)**.** *For tangent fields X, Y,*

$$\widetilde{\nabla}_X Y = \nabla_X Y + \Pi(X,Y).$$

This is the orthogonal decomposition of the ambient derivative, with all terms evaluated along M.

Example 2.2.4 (A parametrized surface)**.** For a regular surface $F(u,v) \subset \mathbb{R}^3$ and a local unit normal N, the formula recovers

$$F_{uu} = \Gamma^1_{11} F_u + \Gamma^2_{11} F_v + eN, \qquad e = \langle F_{uu}, N\rangle.$$

The analogous formulas for F_{uv} and F_{vv} give the other coefficients of the classical second fundamental form.

2.2.2 The Gauss Equation

We retain the curvature convention of the preceding section:

$$R(W,X)Y = \nabla_W\nabla_X Y - \nabla_X\nabla_W Y - \nabla_{[W,X]}Y, \qquad R(W,X,Y,Z) = g(R(W,X)Y,Z).$$

The ambient tensor $\widetilde{R}$ uses the same convention.

Theorem 2.2.5 (Gauss equation)**.** *For tangent fields W, X, Y, Z,*

$$\begin{aligned}\widetilde{R}(W,X,Y,Z) = {} & R(W,X,Y,Z) \\ & - \langle \Pi(W,Z), \Pi(X,Y)\rangle + \langle \Pi(W,Y), \Pi(X,Z)\rangle.\end{aligned}$$

Here the inner products between normal vectors are taken with $\widetilde{g}$.

Proof. For a normal field η and tangent Z, differentiation of $\langle \eta, Z\rangle = 0$ gives

$$\langle \widetilde{\nabla}_W \eta, Z\rangle = -\langle \eta, \widetilde{\nabla}_W Z\rangle = -\langle \eta, \Pi(W, Z)\rangle.$$

Use the Gauss formula first on $\widetilde{\nabla}_X Y$ and then on the derivative of its tangent part. Pairing with Z yields

$$\begin{aligned}\langle \widetilde{\nabla}_W \widetilde{\nabla}_X Y, Z\rangle &= \langle \widetilde{\nabla}_W(\nabla_X Y), Z\rangle + \langle \widetilde{\nabla}_W \Pi(X, Y), Z\rangle \\ &= g(\nabla_W \nabla_X Y, Z) - \langle \Pi(X, Y), \Pi(W, Z)\rangle.\end{aligned}$$

Interchange W, X and subtract. The bracket $[W, X]$ is tangent, so

$$\langle \widetilde{\nabla}_{[W,X]} Y, Z\rangle = g(\nabla_{[W,X]} Y, Z).$$

Combining these three terms gives the claimed equation. □

Corollary 2.2.6. *For orthonormal $u, v \in T_pM$,*

$$K_M(u, v) = K_{\widetilde{M}}(u, v) + \langle \Pi(u, u), \Pi(v, v)\rangle - |\Pi(u, v)|^2.$$

Proof. Put $(W, X, Y, Z) = (u, v, v, u)$ in the Gauss equation and use symmetry of Π. Recall that $K(u, v) = R(u, v, v, u)$. □

2.2.3 Totally Geodesic Submanifolds

Definition 2.2.7. A submanifold is totally geodesic if $\Pi \equiv 0$.

Example 2.2.8. Every affine linear subspace of Euclidean space is totally geodesic: differentiation of a tangent field stays in its fixed tangent subspace. A great sphere

$$\mathbb{S}^k = L \cap \mathbb{S}^n, \qquad L \subset \mathbb{R}^{n+1}, \quad \dim L = k + 1,$$

is totally geodesic in $\mathbb{S}^n$. Indeed, the spherical connection is the Euclidean derivative plus a multiple of the position vector. For fields tangent to the great sphere both terms lie in L, and their sum is tangent to $\mathbb{S}^n$, hence to $L \cap \mathbb{S}^n$.

In general, an intrinsic geodesic and an ambient geodesic with the same initial data are different. The following along-curve version of the Gauss formula makes the distinction precise.

Proposition 2.2.9. *If $\gamma : I \to M$ is smooth and $V(t) \in T_{\gamma(t)}M$ is a smooth tangent field along it, then*

$$\widetilde{D}_t V = D_t V + \Pi(\dot\gamma, V).$$

Proof. Locally write $V = v^i(t)E_i(\gamma(t))$ in a tangent frame. Differentiate by the Leibniz rule and apply the Gauss formula to $\widetilde{\nabla}_{\dot\gamma} E_i$. The tangent terms give $D_t V$, and the normal terms give $v^i \Pi(\dot\gamma, E_i) = \Pi(\dot\gamma, V)$. □

Theorem 2.2.10 (Geodesic characterizations)**.** *For an embedded Riemannian submanifold $M \subset \widetilde{M}$, the following are equivalent:*

1. *M is totally geodesic.*
2. *Every geodesic in M is also an ambient geodesic on its domain.*
3. *If an ambient geodesic is tangent to M at time t_0, it lies in M on some interval about t_0.*

Proof. For an intrinsic geodesic γ, the preceding proposition gives

$$\widetilde{D}_t \dot\gamma = \Pi(\dot\gamma, \dot\gamma).$$

Thus (1) implies (2). If (2) holds, take the intrinsic geodesic with arbitrary initial data $(p, v) \in TM$. At its initial point the equation gives $\Pi_p(v, v) = 0$. Polarization of the symmetric bilinear form gives $\Pi_p(u, v) = 0$ for all u, v, proving (1).

Assume (2) and let an ambient geodesic have tangent initial data at t_0. The intrinsic geodesic with those data is also an ambient geodesic. Uniqueness identifies the two on their common interval, proving (3).

Conversely, under (3), the ambient geodesic with initial data $(p, v) \in TM$ lies locally in M. Its zero acceleration splits as

$$0 = D_t \dot\gamma + \Pi(\dot\gamma, \dot\gamma).$$

The two terms are respectively tangent and normal, so each vanishes. Evaluating at t_0 and polarizing again yields (1). □

Remark 2.2.11 (Local versus global containment)*.* The third condition is local. An open ball in an affine plane is totally geodesic, yet an ambient straight line can leave that ball. If the induced metric on M is complete, its intrinsic geodesics exist for all time; uniqueness then gives global containment of the corresponding ambient geodesics. No completeness assumption is needed for the local equivalence above.

2.2.4 Mean Curvature and Minimal Submanifolds

Definition 2.2.12 (Mean curvature vector)**.** For $m = \dim M \geq 1$, the mean curvature vector is the normalized trace

$$\mathbf{H} = \frac{1}{m} \operatorname{tr}_g \Pi = \frac{1}{m} \sum_{i=1}^{m} \Pi(e_i, e_i),$$

where $e_1, \dots, e_m$ is any local orthonormal tangent frame. This trace is independent of the frame and is a smooth normal field. A submanifold is minimal if $\mathbf{H} \equiv 0$.

In a coordinate frame the same definition is

$$\mathbf{H} = \frac{1}{m} g^{ij} \Pi(\partial_i, \partial_j).$$

Some authors omit $1/m$; we keep the normalization in the manuscript. Accordingly, a factor m appears in the first variation of volume. Totally geodesic submanifolds are minimal, but the converse fails: zero trace does not force the whole second fundamental form to vanish.

Example 2.2.13 (Catenoid and helicoid)**.** The two examples named in the lecture are minimal surfaces in $\mathbb{R}^3$. For $a > 0$, a catenoid is parametrized by

$$C(u, v) = (a \cosh v \cos u,\ a \cosh v \sin u,\ av), \qquad u \in \mathbb{R}/2\pi\mathbb{Z},\ v \in \mathbb{R}.$$

Its induced metric is $a^2 \cosh^2 v\,(du^2 + dv^2)$. Since $C_{uu} + C_{vv} = 0$, the coordinate trace formula gives

$$2\mathbf{H} = \frac{(C_{uu} + C_{vv})^{\perp}}{a^2 \cosh^2 v} = 0.$$

A helicoid is parametrized by

$$Q(u, v) = (u \cos v,\ u \sin v,\ av), \qquad (u, v) \in \mathbb{R}^2.$$

Its metric is $du^2 + (u^2 + a^2)dv^2$. Here $Q_{uu} = 0$ and $Q_{vv} = -uQ_u$ is tangent, so both diagonal normal components vanish and again $\mathbf{H} = 0$. Neither example is totally geodesic; their second fundamental forms are nonzero.

2.2.5 First Variation of Volume

The lecture now characterizes minimal submanifolds as critical points of volume. The argument also applies to immersions, so let $F_0 : M^m \to \widetilde{M}$ be a smooth immersion and $g = F_0^* \widetilde{g}$. Tangent and normal components are understood along F_0, rather than as fields on a possibly self-intersecting image.

Volume density and variations. The Riemannian volume density has the local expression

$$d\mu_g = \sqrt{\det(g_{ij})}\, |dx^1 \cdots dx^m|.$$

Using a density avoids any orientability assumption; on an oriented chart this is the usual volume form. A variation of F_0 is a smooth map

$$F : (-\varepsilon, \varepsilon) \times M \to \widetilde{M}, \qquad F_s = F(s, \cdot), \qquad F_{s=0} = F_0,$$

such that every F_s is an immersion. Its variational field is $V = \partial_s F|_{s=0}$. We consider compactly supported variations: there is a fixed compact set $K \subset M$ such that $F_s = F_0$ outside K for all s.

Choose a relatively compact domain U containing K in its interior and define

$$\mathcal{V}(s) = \int_U d\mu_{g_s}, \qquad g_s = F_s^* \widetilde{g}.$$

This is the volume of the parametrized immersion, counted with multiplicity if necessary. The variation is fixed near ∂U, so enlarging U changes $\mathcal{V}(s)$ only by a constant. This formulation makes sense even when M has infinite total volume.

Lemma 2.2.14 (Differentiating a determinant)**.** *For a differentiable family of invertible matrices $A(s)$,*

$$\frac{d}{ds} \det A = (\det A)\, \mathrm{tr}(A^{-1} A').$$

Consequently, for a positive-definite metric matrix,

$$\partial_s \sqrt{\det g_s}\Big|_0 = \tfrac{1}{2} \sqrt{\det g}\, g^{ij} \dot{g}_{ij}, \qquad \dot{g}_{ij} = \partial_s (g_s)_{ij}|_0 \,.$$

Proof. Differentiating the determinant polynomial gives the sum of each cofactor times the derivative of its matrix entry. The identity

$$\mathrm{cof}_{ij}(A) = (\det A)(A^{-1})_{ji}$$

then gives the first formula. Apply the ordinary derivative of the square root to obtain the second. In particular, the factor is $\det A$, not its reciprocal. □

Theorem 2.2.15 (First variation of volume)**.** *For a compactly supported variation as above,*

$$\mathcal{V}'(0) = -m \int_U \langle V, \mathbf{H} \rangle \, d\mu_g.$$

More generally, the pointwise variation of the volume density is

$$\partial_s d\mu_{g_s}|_0 = \big(\mathrm{div}_M V^\top - m\langle V, \mathbf{H}\rangle\big) d\mu_g.$$

Proof. Write $F_i = \partial_i F$ and let $\widetilde{D}_i, \widetilde{D}_s$ denote the ambient covariant derivatives along F. Metric compatibility and torsion-freeness give, at $s = 0$,

$$\begin{aligned} \dot{g}_{ij} &= \langle \widetilde{D}_s F_i, F_j \rangle + \langle F_i, \widetilde{D}_s F_j \rangle \\ &= \langle \widetilde{D}_i V, F_j \rangle + \langle F_i, \widetilde{D}_j V \rangle. \end{aligned}$$

Here $\widetilde{D}_s F_i = \widetilde{D}_i \partial_s F$ because the parameter coordinate fields commute. Contracting

with the symmetric matrix g^{ij} yields

$$\tfrac{1}{2} g^{ij} \dot{g}_{ij} = g^{ij} \langle \widetilde{D}_i V, F_j \rangle.$$

Split $V = V^\top + V^\perp$. For its tangent part, the Gauss formula gives

$$g^{ij} \langle \widetilde{D}_i V^\top, F_j \rangle = g^{ij} g(\nabla_{\partial_i} V^\top, \partial_j) = \operatorname{div}_M V^\top.$$

For the normal part, differentiate $\langle V^\perp, F_j \rangle = 0$:

$$\begin{aligned} g^{ij} \langle \widetilde{D}_i V^\perp, F_j \rangle &= -g^{ij} \langle V^\perp, \widetilde{D}_i F_j \rangle \\ &= -\langle V^\perp, g^{ij} \mathrm{II}(\partial_i, \partial_j) \rangle = -m \langle V, \mathbf{H} \rangle. \end{aligned}$$

The determinant lemma now proves the pointwise formula. Differentiation under the integral is justified on the compact closure of U. Finally, $V^\top$ is compactly supported in U, so its divergence integrates to zero. This proves the integral formula. □

The divergence term. For clarity, in local coordinates a tangent field $Y = Y^i \partial_i$ satisfies

$$\operatorname{div}_M Y = \partial_i Y^i + \Gamma^i_{ik} Y^k = \frac{1}{\sqrt{\det g}} \partial_i \big(\sqrt{\det g}\, Y^i \big).$$

The last equality follows from the Christoffel formula and the determinant lemma, which give $\Gamma^i_{ik} = \partial_k \log \sqrt{\det g}$. Thus the zero integral for compact support is ordinary integration of a total derivative in a chart. A partition of unity gives the same result globally. Equivalently, on a smooth relatively compact domain,

$$\int_U \operatorname{div}_M Y \, d\mu_g = \int_{\partial U} g(Y, \nu) \, d\sigma_g,$$

where ν is the outward unit conormal, identified with a tangent vector to M. For variations not fixed near the boundary, this boundary integral must be retained.

Corollary 2.2.16 (Variational characterization of minimality)**.** *An immersion is minimal if and only if $\mathcal{V}'(0) = 0$ for every compactly supported variation.*

Proof. If $\mathbf{H} = 0$, the first variation formula gives stationarity. Conversely, every compactly supported smooth field V along F_0 is realized, for sufficiently small s, by

$$F(s, p) = \exp^{\widetilde{M}}_{F_0(p)}(sV(p)).$$

Compact support gives a common interval of definition. The immersion condition persists for small s on that compact set, and the map is unchanged outside it.

Take $V = \varphi \mathbf{H}$ for any nonnegative smooth compactly supported function φ. Stationarity gives

$$0 = -m \int_U \varphi |\mathbf{H}|^2 \, d\mu_g.$$

If $\mathbf{H}$ were nonzero somewhere, a nonnegative cutoff supported near that point would make the integral positive, a contradiction. □

Remark 2.2.17. Minimality means stationarity of volume; it need not give a local minimum. Only the normal part of V appears in the compactly supported first variation, since the tangent part contributes a divergence. If M has boundary, use variations supported in its interior or retain the appropriate boundary term.

2.2.6 Hypersurfaces and Principal Curvatures

A hypersurface has codimension one, so its normal bundle is a line bundle. Locally it always has a smooth unit normal field. A global choice N is an additional assumption, called two-sidedness; a normal line bundle can be nontrivial. For example, the central circle in the total space of a Möbius line bundle has that nontrivial line bundle as its normal bundle and admits no global unit normal.

Definition 2.2.18 (Scalar second fundamental form and shape operator)**.** After choosing a unit normal N, put

$$h(X, Y) = \langle \Pi(X, Y), N \rangle, \qquad \Pi(X, Y) = h(X, Y)N.$$

The shape operator $A : TM \to TM$ is defined by

$$g(AX, Y) = h(X, Y).$$

It is self-adjoint, so its eigenvalues $\kappa_1, \ldots, \kappa_m$ are real. These are the principal curvatures.

The metric is essential here: a bilinear form does not have basis-independent eigenvalues without identifying it with an endomorphism. In coordinates,

$$A^i{}_j = g^{ik} h_{kj}.$$

Moreover, $\widetilde{\nabla}_X N$ is tangent because N has unit length and the normal space is one-dimensional. Differentiating $\langle N, Y \rangle = 0$ shows that

$$\widetilde{\nabla}_X N = -AX.$$

This is the Weingarten formula, with the sign fixed by our definition of Π.

Proposition 2.2.19. *For a hypersurface with chosen unit normal,*

$$\mathbf{H} = HN, \qquad H = \frac{1}{m}\operatorname{tr} A = \frac{1}{m}\sum_i \kappa_i.$$

For orthonormal principal directions e_i, e_j, $i \neq j$,

$$K_M(e_i, e_j) = K_{\widetilde{M}}(e_i, e_j) + \kappa_i \kappa_j.$$

Proof. In an orthonormal eigenbasis, $\mathrm{II}(e_i, e_j) = \kappa_i \delta_{ij} N$. Take its trace for the first formula and substitute into the Gauss equation for the second. □

Definition 2.2.20 (Gauss–Kronecker curvature)**.** The product of all principal curvatures is

$$K_{\mathrm{GK}} = \det A = \kappa_1 \cdots \kappa_m.$$

For a surface in Euclidean three-space, the Gauss equation identifies this with its intrinsic Gaussian curvature: $K = \kappa_1 \kappa_2$.

Remark 2.2.21 (Terminology and normal orientation). The manuscript calls this product Gaussian curvature. The name used above distinguishes it from the intrinsic curvature of a surface. In a curved ambient three-manifold the Gauss equation reads

$$K_M = K_{\widetilde{M}}(TM) + \kappa_1 \kappa_2.$$

Changing N to $-N$ reverses h, A, H and each principal curvature. The vector $\mathbf{H} = HN$ is unchanged, whereas K_{GK} is multiplied by $(-1)^m$.

Example 2.2.22 (Sign check on a round sphere)**.** For $\mathbb{S}^m(a) \subset \mathbb{R}^{m+1}$ with outward normal $N(p) = p/a$,

$$\widetilde{\nabla}_X N = \frac{1}{a} X, \qquad A = -\frac{1}{a}\,\mathrm{id}, \qquad \mathbf{H} = -\frac{1}{a} N.$$

Thus, for $m \geq 2$, the intrinsic sectional curvature is a^{-2}, in agreement with the preceding section. The sphere is not minimal; its mean curvature vector points inward with this convention.

2.3 Jacobi Fields

Jacobi fields describe the infinitesimal separation of geodesics. We follow the lecture from the linearized geodesic equation to the exponential map, constant-curvature models, conjugate points, the index lemma, and global classification results.

2.3.1 The Jacobi Equation and Geodesic Variations

Definition 2.3.1 (Jacobi field)**.** A smooth field J along a geodesic $\gamma : I \to M$ is a Jacobi field if

$$D_t^2 J + R(J, \dot\gamma)\dot\gamma = 0.$$

The curvature convention is the one fixed in Section 2.1.

Lemma 2.3.2. *If $F : (-\varepsilon, \varepsilon) \times I \to M$ is a geodesic variation of γ, then $J = \partial_s F|_{s=0}$ is a Jacobi field. Here each curve $t \mapsto F(s,t)$ is required to be an affinely parametrized geodesic.*

Proof. Put $S = \partial_s F$ and $T = \partial_t F$. Since $D_t T = 0$, the commutation formula and zero torsion give

$$0 = D_s D_t T = D_t D_s T + R(S,T)T = D_t^2 S + R(S,T)T.$$

Restricting to $s = 0$ proves the assertion. □

Definition 2.3.3 (Normal Jacobi field)**.** A Jacobi field is normal if $\langle J, \dot\gamma\rangle = 0$ everywhere. Equivalently, at one time t_0 it satisfies

$$\langle J(t_0), \dot\gamma(t_0)\rangle = 0, \qquad \langle D_t J(t_0), \dot\gamma(t_0)\rangle = 0.$$

Proof of the equivalence. For $f = \langle J, \dot\gamma\rangle$, the geodesic equation gives

$$f' = \langle D_t J, \dot\gamma\rangle, \qquad f'' = -R(J, \dot\gamma, \dot\gamma, \dot\gamma) = 0.$$

Thus f is affine; its value and derivative vanish at t_0 exactly when it vanishes identically. In particular, a Jacobi field vanishing at two distinct times is normal. □

Theorem 2.3.4 (Existence and uniqueness)**.** *For any $t_0 \in I$ and $U, W \in T_{\gamma(t_0)}M$, there is a unique Jacobi field on all of I with*

$$J(t_0) = U, \qquad D_t J(t_0) = W.$$

Proof. Parallel transport an orthonormal basis to obtain a parallel frame $E_i(t)$ and

write $J = j^i E_i$. The equation becomes

$$(j^k)'' + \sum_i R(E_i, \dot\gamma, \dot\gamma, E_k) j^i = 0, \qquad k = 1, \dots, n.$$

This is a linear second-order system with smooth coefficients. Linear ODE existence and uniqueness gives a solution throughout I for arbitrary position and derivative data. The correspondence with those data is linear, so the space of Jacobi fields has dimension $2n$. □

Theorem 2.3.5 (Realization by geodesic variations)**.** *Every Jacobi field is locally the variational field of a geodesic variation. The variation can be defined on the whole interval I if M is complete or I is compact.*

Proof. Choose $t_0 \in I$, set $p = \gamma(t_0)$, and take a curve σ with

$$\sigma(0) = p, \qquad \sigma'(0) = J(t_0).$$

Let $P_s : T_pM \to T_{\sigma(s)}M$ be parallel transport along σ and put

$$v(s) = P_s\big(\dot\gamma(t_0) + s D_t J(t_0)\big).$$

Then $v(0) = \dot\gamma(t_0)$ and $D_s v(0) = D_t J(t_0)$. Define

$$F(s,t) = \exp_{\sigma(s)}\big((t - t_0) v(s)\big).$$

Its central curve is γ by uniqueness of geodesics. If M is complete this formula is defined for every $t \in I$. If I is compact, openness of the geodesic-flow domain and compactness of the central segment give a common small interval in s; a geodesic on a closed interval is understood to extend slightly past its endpoints.

The variational field $\widehat{J}$ is Jacobi and satisfies

$$\widehat{J}(t_0) = \sigma'(0) = J(t_0), \qquad D_t\widehat{J}(t_0) = D_s v(0) = D_t J(t_0).$$

The second identity uses $D_t \partial_s F = D_s \partial_t F$. Uniqueness now gives $\widehat{J} = J$. Without either global hypothesis the same construction works on a sufficiently small interval about t_0. □

Corollary 2.3.6 (Differential of the exponential map)**.** *Let $\gamma(t) = \exp_p(tv)$. The Jacobi field with $J(0) = 0$ and $D_t J(0) = w$ is*

$$J(t) = (d\exp_p)_{tv}(tw).$$

Proof. Use $F(s,t) = \exp_p(t(v + sw))$. Its variational field has the required initial

data, and differentiating in s gives the formula. The tangent space of the vector space T_pM at tv is canonically identified with T_pM. □

2.3.2 Constant Curvature and Geodesic Polar Coordinates

Define the model function

$$s_c(t) = \begin{cases} \sin(\sqrt{c}\,t)/\sqrt{c}, & c > 0, \\ t, & c = 0, \\ \sinh(\sqrt{-c}\,t)/\sqrt{-c}, & c < 0. \end{cases} \qquad s_c(0) = 0, \quad s_c'(0) = 1.$$

It solves $s_c'' + cs_c = 0$. For $c > 0$ its first positive zero is $\pi/\sqrt{c}$; for $c \le 0$ it is positive on $(0, \infty)$.

Proposition 2.3.7. *Suppose M has constant sectional curvature c and γ has unit speed. If $E(t)$ is the parallel transport of $e \perp \dot\gamma(0)$, then*

$$J(t) = s_c(t)E(t)$$

is the unique Jacobi field with $J(0) = 0$ and $D_tJ(0) = e$.

Proof. The algebraic curvature tensor is

$$R(X, Y)Z = c\big(\langle Y, Z\rangle X - \langle X, Z\rangle Y\big).$$

Indeed, this tensor has the prescribed sectional curvatures, which determine R. Consequently $R(E, \dot\gamma)\dot\gamma = cE$. Since E is parallel, the Jacobi equation reduces to $s_c'' + cs_c = 0$. The initial conditions and uniqueness finish the proof. □

Theorem 2.3.8 (Polar form of a constant-curvature metric)**.** *On a punctured normal ball about p, write $\Phi(r, u) = \exp_p(ru)$ with $u \in \mathbb{S}_p^{n-1}$ a unit tangent vector. Then*

$$\Phi^* g = dr^2 + s_c(r)^2 g_{\mathbb{S}_p^{n-1}}.$$

Proof. The Gauss lemma gives unit radial length and orthogonality of the radial and angular directions. If $e \in T_u\mathbb{S}_p^{n-1} = u^\perp$, varying u with initial derivative e produces the Jacobi field

$$d\Phi_{(r,u)}(0, e) = (d\exp_p)_{ru}(re) = s_c(r)P_r e,$$

where P_r is parallel transport along the radial geodesic. Thus the inner product of two angular vectors e, f is $s_c(r)^2\langle e, f\rangle$. This proves the formula. □

Remark 2.3.9. For $c = 0$ this is the Euclidean expression $dr^2 + r^2 g_{\mathbb{S}^{n-1}}$. The pullback of the round metric by $x \mapsto x/|x|$ alone is degenerate in the radial direction; it is

not a Riemannian metric on the punctured tangent space. Polar coordinates also exclude the center, although the full metric extends smoothly there.

Corollary 2.3.10 (Local isometry of constant-curvature spaces)**.** *Two manifolds of the same dimension and the same constant sectional curvature are locally isometric. More precisely, for a linear isometry $L : T_pM_1 \to T_qM_2$,*

$$F = \exp_q \circ L \circ \exp_p^{-1}$$

is an isometry between sufficiently small normal balls.

Proof. The polar metric formula agrees on the two balls because L preserves the round metric on unit tangent spheres. The map is smooth at the center by normal coordinates, and its differential there is L. □

2.3.3 Conjugate Points and the Index Form

Definition 2.3.11 (Conjugate points)**.** Along a nonconstant geodesic $\gamma : [a, b] \to M$, the endpoint $\gamma(b)$ is conjugate to $\gamma(a)$ if a nonzero Jacobi field satisfies $J(a) = J(b) = 0$. The dimension of the space of such fields is the multiplicity. Conjugacy is a property along the specified geodesic, not merely of an unordered pair of points.

Proposition 2.3.12. *For $\gamma(t) = \exp_p(tv)$ and $b > 0$, the point $\gamma(b)$ is conjugate to p if and only if $(d\exp_p)_{bv}$ is singular. The multiplicity equals the dimension of its kernel.*

Proof. Corollary 2.3.6 identifies Jacobi fields with $J(0) = 0$ with their initial derivatives w, and $J(b) = (d\exp_p)_{bv}(bw)$. This gives the kernel correspondence. Such a field is normal because it vanishes at two times. The radial derivative of $\exp_p$ is nonzero, so the kernel is transverse to the radial direction. □

Example 2.3.13. In constant curvature $c \leq 0$ there are no conjugate points. In constant curvature $c > 0$, along a unit-speed geodesic the conjugate times are $k\pi/\sqrt{c}$ for positive integers k within its domain, each with multiplicity $n - 1$. This follows from the model Jacobi fields and from the affine tangential component. On a round sphere, the first such point is the antipode.

We now take $\gamma : [a, b] \to M$ to have unit speed. Recall the index form from the second variation:

$$I(X, Y) = \int_a^b \big(\langle D_tX, D_tY\rangle - R(X, \dot\gamma, \dot\gamma, Y)\big)\, dt.$$

Fields here are continuous and smooth on the closed pieces of a finite partition, with one-sided derivatives at the breakpoints. A proper field vanishes at both endpoints. For length variations we use normal fields; I itself is defined without that restriction.

Proposition 2.3.14 (Piecewise integration by parts)**.** *Let $a = t_0 < \cdots < t_N = b$ be a common smooth partition for X, Y. Set*

$$[D_tX]_{t_i} = D_tX(t_i^+) - D_tX(t_i^-).$$

Then

$$I(X,Y) = -\sum_{i=1}^{N} \int_{t_{i-1}}^{t_i} \langle D_t^2 X + R(X,\dot\gamma)\dot\gamma, Y\rangle \, dt \\ + [\langle D_tX, Y\rangle]_a^b - \sum_{i=1}^{N-1} \langle [D_tX]_{t_i}, Y(t_i)\rangle.$$

Proof. Integrate the derivative term on each smooth piece. At an internal endpoint the left interval contributes $\langle D_tX(t_i^-), Y(t_i)\rangle$ and the right interval contributes its negative with t_i^+. Their sum is minus the stated jump. □

Proposition 2.3.15 (Kernel of the index form)**.** *A continuous piecewise smooth normal field X is a Jacobi field if and only if $I(X,Y) = 0$ for every proper normal field Y.*

Proof. A smooth Jacobi field has zero interior term and no derivative jumps, so the formula proves one direction. Conversely, test with fields supported inside a smooth piece to obtain the Jacobi equation there. The remaining jump sum vanishes for every choice of $Y(t_i)$, so all jumps of D_tX are zero. The jumps are normal because X is normal. Position and derivative therefore match across the partition. Uniqueness for the Jacobi ODE identifies the adjoining pieces with a single smooth solution. □

Theorem 2.3.16 (An interior conjugate point prevents minimality)**.** *If $\gamma(c)$ is conjugate to $\gamma(a)$ for some $a < c < b$, then γ does not minimize length between its endpoints.*

Proof. Choose a nonzero Jacobi field J on $[a,c]$ vanishing at a, c. It is normal and $D_tJ(c) \neq 0$, since zero position and derivative would force $J = 0$. Extend J by zero to a continuous piecewise smooth field V on $[a,b]$. Integration by parts gives $I(V,V) = 0$.

Choose a smooth proper normal field W with $W(c) = D_tJ(c)$; a bump function times the parallel transport of that vector suffices. The same formula gives

$$I(V,W) = \langle D_tJ(c), W(c)\rangle = |D_tJ(c)|^2 > 0.$$

Hence, for small $\varepsilon > 0$,

$$I(V - \varepsilon W, V - \varepsilon W) = -2\varepsilon |D_tJ(c)|^2 + \varepsilon^2 I(W,W) < 0.$$

The second variation for proper normal fields contradicts length minimality. Piecewise smooth fields are allowed; alternatively, smoothing the corner in the first-derivative integral norm preserves the strict negative inequality. □

Remark 2.3.17 (The cylinder example). The converse is false. A flat cylinder has no conjugate points, but a geodesic winding around it need not be shortest between its endpoints. Absence of conjugate points controls the index form, not competition from curves in different homotopy classes.

2.3.4 The Index Lemma

Theorem 2.3.18 (Nonnegativity without interior conjugate points)**.** *Suppose no $\gamma(t)$ with $a < t < b$ is conjugate to $\gamma(a)$. For every proper normal field V,*

$$I(V, V) \geq 0,$$

with equality exactly when V is a Jacobi field. If the endpoint is not conjugate either, equality holds only for $V = 0$.

Proof. Shift a to 0 and write $b = L$. Choose a parallel orthonormal frame of the normal spaces. In this frame let $A(t)$ be the matrix whose columns are the Jacobi fields with zero initial value and initial derivatives the frame vectors. Thus

$$A'' + \mathcal{R}A = 0, \qquad A(0) = 0, \quad A'(0) = \mathrm{id},$$

where $\mathcal{R}_{ij} = R(E_j, \dot\gamma, \dot\gamma, E_i)$ is symmetric. The hypothesis makes $A(t)$ invertible for $0 < t < L$. Differentiating the Wronskian shows

$$\frac{d}{dt}(A^*A' - A'^*A) = 0, \qquad A^*A' = A'^*A.$$

For $V = Af$ on $(0, L)$, set $X = A'f$ and $Y = Af'$, so $D_tV = X + Y$. The Wronskian identity and the Jacobi equation imply

$$\frac{d}{dt}\langle V, X\rangle = |X|^2 + 2\langle X, Y\rangle - \langle \mathcal{R}V, V\rangle,$$
$$|D_tV|^2 - \langle \mathcal{R}V, V\rangle = \frac{d}{dt}\langle V, X\rangle + |Y|^2.$$

For example, the cross-term identity is

$$\langle Af, A'f'\rangle = \langle A'f, Af'\rangle.$$

Integrate piecewise; the boundary terms at internal partition points cancel because $f = A^{-1}V$ is continuous there.

We justify the endpoint limits, including a possibly conjugate endpoint. Near 0, the ODE gives $A(t) = t\,\mathrm{id} + O(t^3)$. Since $V(0) = 0$ and V has a bounded one-sided derivative, f and X stay bounded, so $\langle V, X\rangle \to 0$. The same conclusion at L is immediate if $A(L)$ is invertible.

If $A(L)$ is singular, let $K = \ker A(L)$. The Wronskian identity gives

$$A'(L)K \perp \operatorname{im} A(L).$$

The map $A'(L)|_K$ is injective by uniqueness of Jacobi fields with zero final data. Since the dimensions agree, it maps K isomorphically onto $(\operatorname{im} A(L))^\perp$. Split the domain as $K \oplus K^\perp$ and the range as $(\operatorname{im} A(L))^\perp \oplus \operatorname{im} A(L)$. Taylor expansion in these blocks shows that the first diagonal block vanishes linearly with invertible leading coefficient, while the second is invertible at L. Thus

$$\|A(t)^{-1}\| = O((L - t)^{-1}).$$

As $V(L) = 0$, we have $V(t) = O(L - t)$. Again f and X are bounded and the boundary term tends to zero. This endpoint argument avoids assuming that a Jacobi frame is invertible at a conjugate endpoint.

Taking limits in the integrated identity now yields

$$I(V, V) = \int_0^L |Af'|^2\,dt \geq 0.$$

Equality forces $f' = 0$ on every smooth piece of $(0, L)$. Continuity makes f one constant vector, so $V = Af$ is a Jacobi field. Conversely, a proper Jacobi field has zero index by integration by parts. If $A(L)$ is invertible, the condition $V(L) = 0$ forces that constant to be zero. □

Corollary 2.3.19 (Index lemma)**.** *Under the same absence of interior conjugate points, let J be a normal Jacobi field and V a continuous piecewise smooth normal field with the same endpoint values. Then*

$$I(V, V) \geq I(J, J).$$

If the endpoint is not conjugate to the initial point, equality holds only for $V = J$.

Solution of the exercise in the lecture. Put $W = V - J$. It is proper, and integration by parts gives $I(J, W) = 0$. Hence

$$I(V, V) = I(J, J) + I(W, W) \geq I(J, J).$$

The equality statement follows from Theorem 2.3.18. If the endpoint is conjugate,

equality instead means that $V - J$ is a proper Jacobi field. □

Remark 2.3.20 (Prescribed endpoint values). If $\gamma(b)$ is not conjugate to $\gamma(a)$, there is a unique Jacobi field with any prescribed values at a, b. Start with a solution having the first value; adding a zero-initial-value solution adjusts the terminal value through an invertible linear map. For normal endpoint values the resulting field is normal, since its tangential component is affine and zero at both endpoints.

2.3.5 Cartan–Hadamard and Killing–Hopf

Theorem 2.3.21 (Cartan–Hadamard)**.** *Let (M, g) be connected and complete, with nonpositive sectional curvature. For every $p \in M$, the exponential map*

$$\exp_p : T_pM \longrightarrow M$$

is a smooth covering map. If M is simply connected, it is a diffeomorphism.

Proof. If a Jacobi field J vanishes at two times, then

$$\frac{d^2}{dt^2}|J|^2 = 2|D_tJ|^2 - 2R(J, \dot\gamma, \dot\gamma, J) \geq 0.$$

The nonnegative convex function $|J|^2$ is zero at both endpoints, hence zero between them. Uniqueness then makes J identically zero. There are no conjugate points, and $d\exp_p$ is nonsingular everywhere, including at 0 where it is the identity.

Consequently $h = \exp_p^* g$ is a smooth Riemannian metric on all of T_pM, and $\exp_p : (T_pM, h) \to (M, g)$ is a local isometry. The source must still be proved complete; completeness of the target alone would not suffice.

Let $r(v) = |v|_{g_p}$. The Gauss lemma gives $|dr|_h = 1$ off the origin, and radial lines have their Euclidean lengths. Therefore every curve from 0 to v has h-length at least $r(v)$, while the radial line attains it. Thus

$$d_h(0, v) = |v|_{g_p}.$$

Closed h-balls centered at 0 are exactly Euclidean closed balls. They are compact, since the smooth metric h has the usual topology. A Cauchy sequence is bounded, so it has a convergent subsequence in such a ball and hence converges. The source is complete.

The complete-local-isometry covering theorem from the geodesics section now applies. Finally T_pM is connected and simply connected. A connected covering of a simply connected target is one-sheeted, proving the last assertion. □

Corollary 2.3.22. *A compact simply connected manifold of dimension $n \geq 2$ admits no metric of nonpositive sectional curvature. In particular this applies to spheres*

and complex projective spaces.

Proof. A metric on a compact manifold is complete. Cartan–Hadamard would identify the manifold with $\mathbb{R}^n$, contradicting compactness. □

Corollary 2.3.23. *The universal cover of a connected complete manifold of non-positive sectional curvature is diffeomorphic to $\mathbb{R}^n$. Consequently $\pi_k(M) = 0$ for every $k \geq 2$.*

Proof. The lifted metric is complete and has the same sectional curvatures. Apply the theorem upstairs. Covering maps induce isomorphisms of higher homotopy groups, and $\mathbb{R}^n$ is contractible. □

Theorem 2.3.24 (Killing–Hopf)**.** *Let M be a connected, simply connected, complete Riemannian manifold of dimension $n \geq 2$. If its sectional curvature is the constant c, then it is isometric to the corresponding standard model:*

$$\mathbb{R}^n \quad (c = 0), \qquad \mathbb{S}^n(1/\sqrt{c}) \quad (c > 0), \qquad \mathbb{H}^n_c \quad (c < 0).$$

Proof. For $c \leq 0$, exponential maps on both the model and M are global diffeomorphisms by Cartan–Hadamard. Choose a linear isometry of tangent spaces and compose the two exponential maps as in the local-isometry corollary. The polar metric formula holds for every radius, so this map is a global isometry.

For $c > 0$, write $a = 1/\sqrt{c}$ and choose a north pole N on $\mathbb{S}^n(a)$. Its exponential map is a diffeomorphism from the tangent ball of radius πa onto the complement of the south pole $-N$. For $p \in M$ and a linear isometry $L : T_N\mathbb{S}^n(a) \to T_pM$, define

$$F_1 = \exp_p \circ L \circ \exp_N^{-1} \quad \text{on } \mathbb{S}^n(a) \setminus \{-N\}.$$

Completeness makes $\exp_p$ defined throughout this ball, and the Jacobi formulas show that F_1 is a local isometry there.

Choose $Q \neq N, -N$ and construct a second such map

$$F_2 = \exp_{F_1(Q)} \circ (dF_1)_Q \circ \exp_Q^{-1} \quad \text{on } \mathbb{S}^n(a) \setminus \{-Q\}.$$

The two maps agree in value and differential at Q. Local isometries on a connected domain are determined by this first-order data: they preserve geodesics and hence agree on a normal neighborhood; the set of points where both their values and differentials agree is open and closed. The overlap, a sphere minus two points, is connected for $n \geq 2$. Thus $F_1 = F_2$ on the overlap.

They patch to a local isometry $F : \mathbb{S}^n(a) \to M$. The sphere is complete, so F is a covering map. Since M is simply connected and the source is connected, it is a one-sheeted covering and hence a global isometry. □

2.4 Comparison Theorems

Nonpositive curvature does not force a trivial fundamental group: flat tori have fundamental group $\mathbb{Z}^n$, and closed hyperbolic surfaces of genus $g \geq 2$ have

$$\pi_1(\Sigma_g) = \langle a_1, b_1, \ldots, a_g, b_g \mid [a_1, b_1] \cdots [a_g, b_g] = 1 \rangle.$$

Here $[a, b] = aba^{-1}b^{-1}$. We now compare the growth of radial geodesics, first through the Hessian of distance and then through Jacobi fields. This will give more precise geometric and topological restrictions.

2.4.1 Hessians and the Index Form

Definition 2.4.1 (Hessian)**.** For a smooth function f, its Hessian is the covariant derivative of df:

$$\operatorname{Hess} f(X, Y) = (\nabla_X df)(Y) = X(Yf) - (\nabla_X Y)f = \langle \nabla_X \nabla f, Y \rangle.$$

Here $\langle \nabla f, X \rangle = df(X)$. Torsion-freeness implies that the Hessian is symmetric, since

$$\operatorname{Hess} f(X, Y) - \operatorname{Hess} f(Y, X) = [X, Y]f - (\nabla_X Y - \nabla_Y X)f = 0.$$

Thus this convention agrees with the reversed order of the two arguments used in the manuscript.

In coordinates,

$$(\operatorname{Hess} f)_{ij} = \partial_i \partial_j f - \Gamma_{ij}^k \partial_k f.$$

In Euclidean Cartesian coordinates this is the ordinary Hessian matrix.

Let U be a normal ball about p, with radial function $r(\exp_p v) = |v|$. On a sufficiently small such ball, $r = d_g(p, \cdot)$. More generally all the following identities hold on the normal domain before the cut locus, which is studied in Section 3.1. The function r^2 is smooth at p, while r is smooth away from p.

Lemma 2.4.2. *On the punctured normal domain,*

$$\nabla r = \partial_r, \qquad |\nabla r| = 1, \qquad \operatorname{Hess} r(\partial_r, X) = 0.$$

Proof. The Gauss lemma makes the unit radial vector orthogonal to angular directions. Since $dr(\partial_r) = 1$ and dr vanishes on angular vectors, it follows that $\nabla r = \partial_r$. Radial curves are unit-speed geodesics, so $\nabla_{\partial_r} \partial_r = 0$, proving the Hessian identity. Symmetry gives the same vanishing in the other slot. □

Proposition 2.4.3 (Hessian as an index form)**.** *Let $\gamma : [0, \ell] \to M$ be a unit-speed radial geodesic before the cut locus, with $q = \gamma(\ell)$. For $X, Y \perp \dot{\gamma}(\ell)$, let J_X, J_Y be*

the normal Jacobi fields with

$$J_X(0) = J_Y(0) = 0, \qquad J_X(\ell) = X, \quad J_Y(\ell) = Y.$$

Then

$$\operatorname{Hess} r(X, Y) = \langle D_t J_X(\ell), Y \rangle = I(J_X, J_Y).$$

Proof. There are no conjugate points along this segment, including its endpoint, so the prescribed-endpoint fields exist uniquely. Varying the initial unit direction realizes J_X by radial unit-speed geodesics. Torsion-freeness gives

$$D_t J_X = \nabla_{J_X} \partial_r.$$

At ℓ its inner product with Y is $\operatorname{Hess} r(X, Y)$. Integration by parts in the index form cancels the Jacobi equation and leaves precisely this terminal boundary term, since the initial value is zero. □

Remark 2.4.4 (A local radial branch). Along a geodesic with no conjugate point to its origin, one may also use the radial coordinate obtained from a local inverse of $\exp_p$. The same formula holds even when that branch is not the global distance. In that setting we denote its normal Hessian by $H(t)$, characterized by

$$H(t)(X, Y) = \langle D_t J_X(t), Y \rangle, \quad J_X(0) = 0, \quad J_X(t) = X.$$

This distinction lets us compare Jacobi fields without silently assuming global minimization.

2.4.2 Hessian Comparison

Theorem 2.4.5 (Comparison along two radial geodesics)**.** *Let $\gamma_i : [0, \ell] \to M_i$, $i = 1, 2$, be unit-speed geodesics on manifolds of the same dimension $n \geq 2$. Assume there are no points conjugate to $\gamma_i(0)$ at times $0 < t \leq \ell$. Suppose, at each time, that*

$$K_1(\Pi_1) \leq K_2(\Pi_2)$$

for every pair of two-planes Π_i containing $\dot{\gamma}_i$. If $X_i \perp \dot{\gamma}_i(\ell)$ and $|X_1| = |X_2|$, then

$$H_1(\ell)(X_1, X_1) \geq H_2(\ell)(X_2, X_2).$$

On minimizing normal domains these are the Hessians of distance.

Proof. Choose a parallel isometry P_t between the normal spaces along the two curves with $P_\ell X_1 = X_2$. Let J_1 be the zero-initial-value Jacobi field ending at X_1 and

transfer it to $Z = P_t J_1$. Then $Z(0) = 0$, $Z(\ell) = X_2$, and

$$|Z| = |J_1|, \qquad |D_t Z| = |D_t J_1|.$$

The curvature hypothesis gives $I_2(Z, Z) \le I_1(J_1, J_1)$: the derivative terms are equal, while the curvature term is subtracted. By the index lemma, the Jacobi field J_2 with the same endpoints minimizes I_2. Therefore

$$H_2(X_2, X_2) = I_2(J_2, J_2) \le I_2(Z, Z) \le I_1(J_1, J_1) = H_1(X_1, X_1).$$

At times when a field vanishes, the curvature inequality is automatic. This proves the result without dividing by its norm. □

Recall s_c from Section 2.3, and put

$$a_c(t) = \frac{s_c'(t)}{s_c(t)}, \qquad b_c = \begin{cases} \pi/\sqrt{c}, & c > 0, \\ \infty, & c \le 0. \end{cases}$$

All comparisons involving a_c below are restricted to $0 < t < b_c$.

Corollary 2.4.6 (Comparison with a space form)**.** *Let $r = d_g(p, \cdot)$, and let q be before the cut locus, with $0 < r(q) < b_c$. A radial sectional-curvature bound along the minimizing segment to q gives*

$$\begin{aligned} K \le c \quad &\Longrightarrow \quad \operatorname{Hess} r \ge a_c(r)(g - dr \otimes dr), \\ K \ge c \quad &\Longrightarrow \quad \operatorname{Hess} r \le a_c(r)(g - dr \otimes dr). \end{aligned}$$

These are inequalities of quadratic forms.

Proof. For a normal terminal vector X in the model, its Jacobi field is

$$J(t) = \frac{s_c(t)}{s_c(\ell)} P_t X,$$

where $P_t X$ is the parallel transport of the terminal vector backwards along the segment. Hence $H_c(X, X) = a_c(\ell)|X|^2$. Apply the preceding theorem in the appropriate order. Finally split an arbitrary vector into radial and normal components and use the vanishing of the radial Hessian terms. □

2.4.3 Convexity, Fixed Points, and Torsion

Definition 2.4.7. A Cartan–Hadamard manifold is a connected, simply connected, complete Riemannian manifold with $K \le 0$.

Its exponential map at any point is a global diffeomorphism. Any two points are joined by a unique geodesic, which minimizes; r_p is smooth off p and r_p^2 is smooth everywhere.

Proposition 2.4.8 (Convexity of squared distance)**.** *On a Cartan–Hadamard manifold,*

$$f_p(x) = \tfrac{1}{2} d_g(p,x)^2, \qquad \operatorname{Hess} f_p \geq g.$$

In particular f_p is strictly convex along every nonconstant affinely parametrized geodesic, and metric balls are geodesically convex.

Proof. Away from p, the product rule and comparison with $c = 0$ give

$$\operatorname{Hess} f_p = dr_p \otimes dr_p + r_p \operatorname{Hess} r_p \geq dr_p \otimes dr_p + (g - dr_p \otimes dr_p) = g.$$

The inequality extends to p by smoothness (in normal coordinates its Hessian at p is exactly g_p). Along a geodesic σ,

$$(f_p \circ \sigma)'' = \operatorname{Hess} f_p(\dot\sigma, \dot\sigma) \geq |\dot\sigma|^2.$$

Convexity keeps the value below the larger endpoint value, proving convexity of both open and closed balls. □

Lemma 2.4.9 (Smallest enclosing ball)**.** *A nonempty compact subset S of a Cartan–Hadamard manifold has a unique smallest enclosing closed ball. Its center is called the circumcenter (the manuscript's* 1*-center).*

Proof. Set $F(x) = \max_{z \in S} d_g(x,z)$ and $r = \inf_x F(x)$. The function F is continuous, indeed 1-Lipschitz. Fix $z_0 \in S$ and choose x_j with $F(x_j) \to r$. Since $d_g(x_j, z_0) \leq F(x_j)$, the centers stay in a bounded closed ball, which is compact by completeness. A convergent subsequence gives a center x with $F(x) = r$.

If S is a singleton the assertion is immediate. Otherwise suppose $x \neq y$ are two centers of radius r, and let m be the midpoint of their geodesic. Integrating the bound in Proposition 2.4.8 along this unit-speed segment of length $d = d_g(x,y)$ gives, for every $z \in S$,

$$d_g(z,m)^2 \leq \tfrac{1}{2} d_g(z,x)^2 + \tfrac{1}{2} d_g(z,y)^2 - \tfrac{1}{4} d^2 \leq r^2 - \tfrac{1}{4} d^2 < r^2.$$

Thus one uniformly smaller radius encloses S, a contradiction. The displayed quantitative inequality follows by subtracting $t^2/2$ from f_z along the segment and applying ordinary convexity at the midpoint. □

Theorem 2.4.10 (Cartan's fixed point theorem)**.** *A compact Lie group acting smoothly by isometries on a Cartan–Hadamard manifold has a common fixed point.*

Proof. The orbit $S = Gx_0$ is nonempty and compact. If x is its unique circumcenter, every $\phi \in G$ takes its smallest enclosing ball to another ball of the same radius enclosing $\phi S = S$. Uniqueness gives $\phi x = x$ for all ϕ. □

Theorem 2.4.11 (Cartan's torsion theorem)**.** *If M is connected and complete with $K \leq 0$, then $\pi_1(M)$ is torsion-free. Equivalently, it has no nontrivial finite subgroup.*

Proof. The universal cover $\widetilde{M}$, with the lifted metric, is complete and Cartan–Hadamard. The deck group, identified with $\pi_1(M)$, acts by isometries. Any finite subgroup is a compact zero-dimensional Lie group, so it has a common fixed point by the fixed point theorem. But a deck transformation fixing one point is the identity: lift paths from that point and use uniqueness of lifts. Thus the subgroup is trivial. A finite-order element would generate such a subgroup. □

Remark 2.4.12. The flat torus example shows that torsion-free does not mean cyclic. The manuscript also names Preissmann's theorem, but supplies no proof: on a closed manifold of strictly negative sectional curvature, every nontrivial abelian subgroup of the fundamental group is infinite cyclic. This stronger statement is recorded as an external result, not as a consequence of the fixed point argument.

2.4.4 Rauch Comparison and Conjugate Times

Theorem 2.4.13 (Rauch comparison, the lecture's form)**.** *Let $\gamma_i : [0, \ell] \to M_i$ be unit-speed geodesics, with no point conjugate to their respective origins for $0 < t < \ell$. Suppose all radial sectional curvatures satisfy $K_1 \geq K_2$ at corresponding times. Let J_i be Jacobi fields with*

$$J_i(0) = 0, \quad |D_t J_1(0)| = |D_t J_2(0)|, \quad \langle D_t J_1(0), \dot{\gamma}_1(0) \rangle = \langle D_t J_2(0), \dot{\gamma}_2(0) \rangle.$$

Then $|J_1(t)| \leq |J_2(t)|$ for $0 \leq t \leq \ell$.

Proof. The tangential components are $bt\dot{\gamma}_i$, with the same constant b. Subtracting them leaves normal Jacobi fields with equal initial derivative norms. If those norms vanish, both normal fields vanish. Otherwise the normal fields are nonzero on $(0, \ell)$ by the no-conjugate-point assumption.

Put $f_i = |J_i|^2$ for these normal fields. The radial Hessian identity and Theorem 2.4.5, applied to unit terminal vectors, give

$$\frac{f_1'}{2f_1} = H_1\left(\frac{J_1}{|J_1|}, \frac{J_1}{|J_1|}\right) \leq H_2\left(\frac{J_2}{|J_2|}, \frac{J_2}{|J_2|}\right) = \frac{f_2'}{2f_2}.$$

Thus f_2/f_1 is nondecreasing. Since $J_i(t) = tD_tJ_i(0) + O(t^2)$ in parallel frames, its limit at 0 is 1. Therefore $f_2 \geq f_1$. Adding the equal tangential squared norms proves the result on the interior; continuity includes both endpoints. □

Corollary 2.4.14 (Bounds on the first conjugate time)**.** *Assume $n \geq 2$ and the indicated bounds on radial sectional curvatures along a unit-speed geodesic starting at p.*

1. *If $K \leq k$ with $k > 0$, there is no point conjugate to p before $\pi/\sqrt{k}$.*
2. *If $K \geq k$ with $k > 0$ and the geodesic exists through $\pi/\sqrt{k}$, a conjugate point occurs at or before that time.*

Proof. For the first claim, suppose a first conjugate time $t_0 < \pi/\sqrt{k}$ exists. Compare a nonzero Jacobi field vanishing at $0, t_0$ with the constant-k model, first for $t < t_0$. Rauch gives $|J(t)| \geq s_k(t)|D_t J(0)|$. Passing to t_0 contradicts $J(t_0) = 0$. A first conjugate time exists if any occur on this compact segment: nonsingularity holds near 0, and the singular set of the differential is closed.

For the second claim, either a conjugate point occurs earlier or comparison is valid up to the model's first zero $b = \pi/\sqrt{k}$. Any nonzero normal field with $J(0) = 0$ then satisfies $|J(t)| \leq s_k(t)|D_t J(0)|$ for $t < b$, so $J(b) = 0$. The endpoint itself is conjugate. □

2.4.5 From Differential to Triangle Comparison

Proposition 2.4.15 (Expansion in normal coordinates)**.** *Suppose $\exp_{p_i}$ are diffeomorphisms on tangent balls $B_r(0)$, and the radial curvatures of M_1 are at least those of M_2 at corresponding radii. For a linear isometry $L : T_{p_1}M_1 \to T_{p_2}M_2$, put*

$$F = \exp_{p_2} \circ L \circ \exp_{p_1}^{-1}.$$

Then $|dF(X)| \geq |X|$ on the first normal ball. If both image balls are geodesically convex, then

$$d_{M_2}(F(x), F(y)) \geq d_{M_1}(x, y).$$

Proof. Write $q = \exp_{p_1}(av)$ with $|v| = 1$, $a > 0$, and $W = (d\exp_{p_1})_{av}^{-1} X$. Along the two unit-speed radial geodesics use

$$J_1(t) = (d\exp_{p_1})_{tv}(tW/a), \qquad J_2(t) = (d\exp_{p_2})_{tLv}(tLW/a).$$

Their initial derivatives are W/a and LW/a, with equal lengths and radial components. Rauch gives $|X| = |J_1(a)| \leq |J_2(a)| = |dF(X)|$. At the center $dF = L$. Notice the factor $1/a$ in both initial derivatives.

For the distance assertion, take a minimizing curve in the second convex ball and pull it back by F^{-1}. The differential estimate says its pullback is no longer. Taking the distance between the original endpoints proves the inequality. Merely applying the estimate to a curve in the first ball would give the wrong logical direction for an infimum of lengths. □

Definition 2.4.16 (Geodesic and comparison triangles)**.** A geodesic triangle consists of three vertices and three specified geodesic segments joining them. Unless stated otherwise, our comparison triangles in M use minimizing sides. A comparison triangle in the simply connected two-dimensional space form M_c^2 has the same three side lengths, with minimizing model sides.

For $c \le 0$, strict triangle inequalities determine a nondegenerate model triangle uniquely up to isometry. For $c > 0$, one also requires perimeter less than $2\pi/\sqrt{c}$; this implies each side is shorter than $\pi/\sqrt{c}$. Degenerate cases are interpreted by limits. These restrictions matter: an arbitrary triple of lengths need not define a spherical comparison triangle.

Theorem 2.4.17 (Toponogov angle comparison; external input)**.** *Let M be complete of dimension at least two, with $K \ge c$. A triangle with minimizing sides has angles at least the corresponding angles of its comparison triangle in M_c^2, whenever that comparison triangle is nondegenerate. For $c > 0$, the perimeter is at most $2\pi/\sqrt{c}$; the angle formulation is first stated with strict perimeter inequality, and limiting configurations are handled by continuity.*

Remark 2.4.18 (What is proved here)*.* The manuscript states a local triangle comparison and mentions the stronger Toponogov theorem. The preceding differential comparison explains the local mechanism. Global Toponogov requires an additional argument and is used here as a theorem, as in do Carmo [3]; it is not proved by the local expansion proposition alone. The angle comparison will be used in Section 3.1.

2.4.6 Supplement: Laplacian Comparison

The following short supplement retains the Laplacian comparison from the earlier typeset version; it is not a transcription of an additional proof in these manuscript pages. We use $\Delta f = \operatorname{tr}_g \operatorname{Hess} f = \operatorname{div} \nabla f$.

Theorem 2.4.19. *Let M^n be complete, $n \ge 2$, and* $\operatorname{Ric} \ge (n-1)c\,g$*. At a point before the cut locus of p, with $0 < r = d_g(p, \cdot) < b_c$,*

$$\Delta r \le (n-1)\frac{s_c'(r)}{s_c(r)}.$$

Proof. Along the minimizing unit-speed segment of length $\ell = r$, choose parallel orthonormal normal fields $E_1, \dots, E_{n-1}$. Let J_i be the Jacobi field ending at $E_i(\ell)$ and vanishing initially. The trial field $V_i(t) = s_c(t)E_i(t)/s_c(\ell)$ has the same endpoints.

The Hessian identity and index lemma give

$$\begin{aligned}\Delta r &= \sum_i I(J_i, J_i) \le \sum_i I(V_i, V_i) \\ &\le \frac{n-1}{s_c(\ell)^2} \int_0^\ell \left(s_c'(t)^2 - c s_c(t)^2\right) dt \\ &= (n-1)\frac{s_c'(\ell)}{s_c(\ell)}.\end{aligned}$$

The curvature terms sum to the radial Ricci curvature. The final equality uses $(s_c s_c')' = s_c'^2 - c s_c^2$ and $s_c(0) = 0$. A Ricci lower bound does not imply a sectional-curvature lower bound; tracing Hessian comparison alone would therefore not prove this theorem. □

Remark 2.4.20 (The closing metric-geometry aside). The manuscript ends with the convex graphs $z = \sqrt{x^2 + y^2 + 1/j}$ and their conical limit as $j \to \infty$, followed by a question about Hausdorff and Gromov–Hausdorff convergence. This is a motivation for studying nonsmooth metric limits, not a proved convergence theorem in the lecture. The distinction is that Hausdorff distance compares subsets of one ambient metric space, whereas Gromov–Hausdorff distance compares metric spaces through isometric embeddings into common ambient spaces. For these noncompact graphs one must specify a local or pointed notion of convergence.

Chapter 3

Further Topics

3.1 Sphere Theorems and Cut Locus

The guiding question of this section is:

> When does the geometry of a closed, simply connected manifold force it to be a sphere?

Here *closed* means compact and without boundary. In dimension two, a closed simply connected surface is a sphere; the corresponding assertion in dimension three is the Poincaré theorem. Simple connectivity alone is insufficient in higher dimensions: $\mathbb{CP}^2$ and $\mathbb{S}^2 \times \mathbb{S}^2$ are closed simply connected four-manifolds that are not spheres. The generalized Poincaré theorem concerns the stronger assumption of being a *homotopy sphere*; its conclusion is a homeomorphism, not in general a diffeomorphism. We therefore seek geometric restrictions, specifically bounds on sectional curvature.

Theorem 3.1.1 (Topological sphere theorem, Berger–Klingenberg)**.** *Let (M^n, g), $n \geq 2$, be a closed simply connected Riemannian manifold. If its sectional curvature satisfies*

$$\frac{1}{4} < K \leq 1,$$

then M is homeomorphic to $\mathbb{S}^n$.

Remark 3.1.2 (Sharpness and the smooth conclusion). The lower inequality must be strict: for $m \geq 2$, the Fubini–Study metric on $\mathbb{CP}^m$, suitably rescaled, satisfies $1/4 \leq K \leq 1$, whereas $\mathbb{CP}^m$ is not a sphere. The differentiable sphere theorem of Brendle–Schoen [2] strengthens the conclusion to a diffeomorphism under *strict pointwise quarter-pinching*:

$$0 < K_{\min}(x), \qquad K_{\max}(x) < 4K_{\min}(x) \quad \text{for every } x \in M.$$

The global bound in Theorem 3.1.1 implies this pointwise condition; the converse need not hold after a single constant rescaling.

Following the handwritten notes and do Carmo [3], we prepare the geometry of the cut locus before proving the topological conclusion.

3.1.1 Cut Points and Cut Locus

Throughout this subsection (M, g) is complete and connected. For $p \in M$ and $v \in T_pM$ with $|v|_g = 1$, write

$$\gamma_v(t) = \exp_p(tv), \qquad t \geq 0.$$

For small t, $d_g(p, \gamma_v(t)) = t$. If a segment ceases to minimize, every longer segment also fails to minimize: a shorter competitor for the initial segment can be concatenated with the remaining part of γ_v.

Definition 3.1.3 (Cut time, cut point, and cut locus)**.** The **cut time** is

$$c_p(v) = \sup\{t > 0 : d_g(p, \gamma_v(t)) = t\} \in (0, \infty].$$

If $c_p(v) < \infty$, then $\gamma_v(c_p(v))$ is the **cut point** of p along γ_v. The **cut locus** of p is

$$\mathrm{Cut}(p) = \{\exp_p(c_p(v)v) : |v|_g = 1,\ c_p(v) < \infty\}.$$

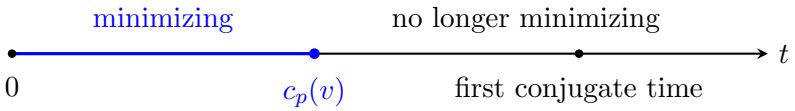
When $c_p(v) = \infty$, the ray has no cut point.

Continuity of distance shows that the segment up to a finite cut time is still minimizing. Thus the minimizing times are exactly $[0, c_p(v)]$, or all of $[0, \infty)$.

A cut point records a global failure of minimization; a conjugate point records a singularity of the exponential map. If the first conjugate time is finite, then

$$\text{cut time} \ \leq \ \text{first conjugate time}.$$

Equality can occur, and cut points may also occur without any conjugate point.



Schematic when the cut and first conjugate times are finite and distinct.

Proposition 3.1.4 (Characterization of cut points)**.** *If $q = \gamma_v(t_0)$ is the cut point of p along γ_v, then at least one of the following holds:*

1. *q is conjugate to p along γ_v;*
2. *a second unit-speed minimizing geodesic $\sigma : [0, t_0] \to M$ joins p to q.*

The alternatives are not mutually exclusive. Conversely, if either condition holds at time $t_0 > 0$, then $c_p(v) \leq t_0$.

Proof. Choose $\varepsilon_j \downarrow 0$. By completeness, there is a unit-speed minimizing geodesic σ_j from p to $\gamma_v(t_0 + \varepsilon_j)$, with length

$$s_j = d_g(p, \gamma_v(t_0 + \varepsilon_j)) < t_0 + \varepsilon_j, \qquad s_j \longrightarrow t_0.$$

After taking a subsequence, $w_j = \dot\sigma_j(0)$ converges to a unit vector w. Smooth dependence on initial data gives

$$\exp_p(t_0 w) = q.$$

The limiting geodesic has length $t_0 = d_g(p, q)$, so it minimizes. If $w \neq v$, this proves the second alternative. If $w = v$ and q is not conjugate along γ_v, the inverse function theorem makes $\exp_p$ injective on a neighborhood of $t_0 v$. For large j, both $s_j w_j$ and $(t_0 + \varepsilon_j)v$ belong to that neighborhood and have the same image. Their equality contradicts $s_j < t_0 + \varepsilon_j$.

Conversely, a conjugate point at t_0 is an interior conjugate point of every longer segment, so the second variation criterion rules out minimization beyond t_0. If a second minimizer reaches $\gamma_v(t_0)$, follow it and then extend along γ_v. The resulting curve has a corner: equality of terminal velocities would force the two geodesics to coincide by uniqueness of the geodesic equation. Smoothing this corner shortens the curve, so again no longer segment minimizes. □

Corollary 3.1.5 (Symmetry and normal coordinates)**.** *We have $q \in \mathrm{Cut}(p)$ if and only if $p \in \mathrm{Cut}(q)$. If $x \notin \mathrm{Cut}(p) \cup \{p\}$, there is a unique minimizing geodesic from p to x, and $r_p(x) = d_g(p, x)$ is smooth near x.*

Proof. Reverse a minimizing geodesic from p to a cut point q. Conjugacy is symmetric under reversal, and two distinct minimizers remain distinct after reversal. Proposition 3.1.4, together with minimization up to the endpoint, gives cut time $d_g(p, q)$ in the reverse direction. Outside the cut locus, the same proposition excludes both a second minimizer and singularity of $d\exp_p$. The inverse function theorem and Gauss lemma then give smooth polar coordinates and a smooth distance function. □

Example 3.1.6 (Model cut loci)**.**

- In $\mathbb{R}^n$, and more generally on a complete simply connected manifold with $K \leq 0$, the cut locus is empty by Cartan–Hadamard.
- On the round sphere, $\mathrm{Cut}(p) = \{-p\}$; the antipode is also conjugate to p.
- On the flat cylinder $\mathbb{S}^1 \times \mathbb{R}$, the cut locus is the generator opposite p. There are two shortest routes around the cylinder and no conjugate points.
- On the standard $\mathbb{RP}^n$, the cut locus is the projective hyperplane $\mathbb{RP}^{n-1}$ orthogonal to a representative of p.
- On a flat square torus, the boundary of a fundamental square centered at p represents the cut locus. With opposite edges identified, it is a wedge of two circles, $\mathbb{S}^1 \vee \mathbb{S}^1$.

3.1.2 Continuity of Cut Time and Injectivity Radius

The unit tangent bundle and the cut-time function are

$$T^1M = \{(p, v) \in TM : |v|_g = 1\}, \qquad c : T^1M \to (0, \infty], \quad c(p, v) = c_p(v).$$

We equip $(0,\infty]$ with the extended-real topology: $t_j \to \infty$ means that $t_j > A$ eventually for every $A > 0$. This is *not* the one-point compactification of $(0,\infty)$: approaching 0 does not mean approaching ∞.

Proposition 3.1.7 (Continuity of cut time)**.** *On a complete Riemannian manifold, $c : T^1M \to (0,\infty]$ is continuous.*

Proof. Let $(p_j, v_j) \to (p, v)$, set $c_j = c(p_j, v_j)$ and $c_0 = c(p, v)$, and write $\gamma_j(t) = \exp_{p_j}(tv_j)$ and $\gamma(t) = \exp_p(tv)$.

Upper semicontinuity. If $c_0 < \infty$ and $t > c_0$, then

$$d_g(p, \gamma(t)) < t.$$

Continuity of distance and of the geodesic flow gives $d_g(p_j, \gamma_j(t)) < t$ for large j, hence $c_j < t$. Letting $t \downarrow c_0$ yields $\limsup c_j \le c_0$. For $c_0 = \infty$ this inequality is automatic.

Lower semicontinuity. Suppose a subsequence satisfies $c_j \to a < c_0$. A common normal neighborhood near p gives a positive lower bound on these cut times, so $a > 0$. At each cut point, apply Proposition 3.1.4. If conjugacy occurs on a subsequence, continuity of $d\exp$ gives a singular differential at av, contradicting $a < c_0$.

Otherwise choose second minimizing directions $w_j \neq v_j$ at p_j, with

$$\exp_{p_j}(c_j w_j) = \exp_{p_j}(c_j v_j).$$

Using a local trivialization, pass to a subsequence with $w_j \to w$. If $w \neq v$, the limit gives a second minimizer of length a to $\gamma(a)$, again impossible before the cut time. If $w = v$, consider the map

$$F : TM \longrightarrow M \times M, \qquad F(x, z) = (x, \exp_x z).$$

Since $a < c_0$, its differential is invertible at (p, av). Thus F is injective on one neighborhood of this point. Both $(p_j, c_j w_j)$ and $(p_j, c_j v_j)$ lie there for large j and have the same image, a contradiction. Therefore $\liminf c_j \ge c_0$, including when $c_0 = \infty$. □

Corollary 3.1.8 (Closedness of the cut locus)**.** *The set $\mathrm{Cut}(p)$ is closed. It is compact if M is compact.*

Proof. Suppose $q_j \in \mathrm{Cut}(p)$ and $q_j \to q$. Write $q_j = \exp_p(t_j v_j)$ with $t_j = c_p(v_j) = d_g(p, q_j)$. Passing to a subsequence, $v_j \to v$, while $t_j \to d_g(p, q) < \infty$. Continuity of cut time gives $c_p(v) = d_g(p, q)$, so $q = \exp_p(c_p(v)v) \in \mathrm{Cut}(p)$. Compactness follows when M is compact. □

Definition 3.1.9 (Injectivity radius)**.** The injectivity radius at p and the global injectivity radius are

$$\operatorname{inj}(p) = \inf_{|v|_g=1} c_p(v), \qquad \operatorname{inj}(M) = \inf_{p\in M} \operatorname{inj}(p).$$

Equivalently, $\operatorname{inj}(p) = d_g(p, \operatorname{Cut}(p))$, with distance to the empty set interpreted as ∞.

The map $\exp_p$ is a diffeomorphism from $B_r(0) \subset T_pM$ onto $B_r(p)$ for $r \le \operatorname{inj}(p)$. When the cut locus is nonempty, a closest cut point exists by Hopf–Rinow and closedness. If M is compact, continuity on the compact unit tangent bundle gives

$$\operatorname{inj}(M) = \min_{(p,v)\in T^1M} c(p, v) > 0.$$

Proposition 3.1.10 (Closest cut points)**.** *Let $q \in \operatorname{Cut}(p)$ satisfy $\ell = d_g(p, q) = \operatorname{inj}(p)$. Either q is conjugate to p along some minimizing geodesic, or there are exactly two unit-speed minimizing geodesics $\gamma, \sigma : [0, \ell] \to M$ from p to q, and*

$$\dot\gamma(\ell) = -\dot\sigma(\ell).$$

Proof. Assume no minimizing geodesic from p to q has conjugate endpoints. The cut-point characterization gives at least two minimizers. Write their terminal unit vectors as u_1, u_2. If $u_1 \neq -u_2$, then $W = -(u_1 + u_2)$ satisfies

$$\langle W, u_i\rangle = -1 - \langle u_1, u_2\rangle < 0, \qquad i = 1, 2.$$

Move the endpoint along $q_s = \exp_q(sW)$. The two local inverse branches of $\exp_p$ give distinct geodesic variations joining p to q_s. By the first variation formula their lengths $L_i(s)$ satisfy

$$L_i(0) = \ell, \qquad L_i'(0) = \langle W, u_i\rangle < 0.$$

For small $s > 0$, both lengths are less than ℓ. Every geodesic from p of length less than $\operatorname{inj}(p)$ is the unique minimizer to its endpoint, so these two distinct branches are impossible. Thus any two terminal vectors are opposite. There can be no third: its terminal vector would equal that of one of the first two, forcing equality of the geodesics. This proves the claim. □

Theorem 3.1.11 (Klingenberg's characterization)**.** *If M is closed and $\ell = \operatorname{inj}(M)$, then either some minimizing geodesic of length ℓ has conjugate endpoints, or there is a smooth closed geodesic of length 2ℓ.*

Proof. Choose (p, v) attaining the global minimum and put $q = \exp_p(\ell v)$. Suppose

the first alternative does not occur anywhere. Proposition 3.1.10 gives two minimizers from p to q whose terminal velocities are opposite. The reversed curves are also minimizers, and cut-locus symmetry gives $p \in \mathrm{Cut}(q)$. Since

$$\ell \le \mathrm{inj}(q) \le d_g(q,p) = \ell,$$

p is a closest cut point of q. Applying the proposition at q shows that the original initial velocities are opposite as well. Joining one geodesic to the reverse of the other is therefore smooth at *both* q and p, producing a closed geodesic of length 2ℓ. □

3.1.3 Proof of the Topological Sphere Theorem

Assume henceforth the hypotheses of Theorem 3.1.1. Compactness allows us to choose a constant δ and a radius ρ such that

$$\frac{1}{4} < \delta < K \le 1, \qquad \frac{\pi}{2\sqrt{\delta}} < \rho < \pi. \tag{3.1}$$

We follow the five steps of the handwritten proof.

Step 1: a uniform injectivity radius.

Lemma 3.1.12 (Klingenberg's estimate)**.** *Under the above assumptions,* $\mathrm{inj}(M) \ge \pi$.

Remark 3.1.13 (Input from Morse theory). As in the lecture notes, we use this estimate as an external theorem; its full Morse-theoretic proof is not reproduced here. Rauch comparison (Theorem 2.4.13) and $K \le 1$ exclude conjugate points before time π. The additional global argument in Klingenberg's estimate excludes the short closed geodesic alternative under the pinching and simple-connectivity hypotheses. The estimate does not follow from Rauch comparison alone; see do Carmo [3].

Step 2: directions at a diameter pair. Choose $p, q \in M$ with $D = d_g(p,q) = \mathrm{diam}(M)$.

Lemma 3.1.14 (Berger's lemma)**.** *For every $w \in T_qM$ there is a unit-speed minimizing geodesic γ from q to p with $\langle w, \dot\gamma(0)\rangle \ge 0$.*

Proof. Let $\mathcal{U} \subset T_qM$ be the set of initial unit vectors of minimizers from q to p. It is nonempty and compact. The first variation formula for distance gives the one-sided directional derivative

$$\left.\frac{d}{ds}\right|_{s=0+} d_g(p, \exp_q(sw)) = -\max_{u\in\mathcal{U}} \langle w, u\rangle.$$

This formula allows multiple minimizers: the minimum length selects the smallest first variation, equivalently the negative of the displayed maximum. It follows by applying first variation to minimizing segments and taking convergent subsequences of their initial directions. Since q maximizes $d_g(p,\cdot)$, the derivative is nonpositive. Hence the maximum is nonnegative, as required. □

Step 3: two normal balls cover the manifold.

Proposition 3.1.15 (The covering proposition)**.** *For ρ as in (3.1),*

$$M = B_\rho(p) \cup B_\rho(q).$$

Proof. Suppose r lies outside both balls, and set $a = d_g(p,r)$ and $b = d_g(q,r)$. Then $a, b \geq \rho$ and $D \geq a, b$. Choose a minimizer from q to r. By Berger's lemma, some minimizer from q to p makes an angle $\theta \leq \pi/2$ with it. Choose κ with $\delta < \kappa \leq K$ everywhere. The positive-curvature perimeter bound in Theorem 2.4.17 gives perimeter at most $2\pi/\sqrt{\kappa} < 2\pi/\sqrt{\delta}$, so the curvature-$\delta$ comparison triangle is admissible. Toponogov's angle comparison gives $\theta \geq \widetilde{\theta}$. The spherical cosine rule and $\cos\theta \leq \cos\widetilde{\theta}$ therefore yield

$$\cos(\sqrt{\delta}\,a) \geq \cos(\sqrt{\delta}\,b)\cos(\sqrt{\delta}\,D) + \sin(\sqrt{\delta}\,b)\sin(\sqrt{\delta}\,D)\cos\theta.$$

Bonnet–Myers gives $a, b, D \leq \pi/\sqrt{\kappa} < \pi/\sqrt{\delta}$. A degenerate triangle follows by a limit of the same comparison inequality. Since $b, D \geq \rho > \pi/(2\sqrt{\delta})$, both cosine factors on the right are negative, so their product is positive. The remaining term is nonnegative. Thus $\cos(\sqrt{\delta}\,a) > 0$, implying $a < \pi/(2\sqrt{\delta}) < \rho$, a contradiction. □

The strict choice $\rho < \pi \leq \mathrm{inj}(M)$ is crucial: the closed radius-ρ balls lie inside normal neighborhoods. Moreover, if $d_g(p,x) = d_g(q,x)$, the covering proposition forces both distances to be strictly less than ρ.

Step 4: construct the equator using all radial directions. Set

$$E = \{x \in M : d_g(p,x) = d_g(q,x)\}.$$

Lemma 3.1.16 (Radial equator lemma)**.** *For every unit vector $v \in T_pM$, the segment $\gamma_v(t) = \exp_p(tv)$, $0 \leq t \leq \rho$, meets E at exactly one time $h_p(v) \in (0,\rho)$. The function h_p is continuous, and*

$$v \longmapsto \exp_p(h_p(v)v)$$

is a homeomorphism from the unit sphere in T_pM onto E. The analogous statement

holds with p and q interchanged.

Proof. Define $f_v(t) = t - d_g(q, \gamma_v(t))$. Lemma 3.1.12 gives $d_g(p, \gamma_v(t)) = t$ on $[0, \rho]$. Hence

$$f_v(0) = -D < 0, \qquad f_v(\rho) > 0,$$

where the second inequality follows because $\gamma_v(\rho) \notin B_\rho(p)$ and therefore belongs to $B_\rho(q)$. The triangle inequality also gives, for $s < t$,

$$d_g(q, \gamma_v(t)) - d_g(q, \gamma_v(s)) \leq t - s,$$

so f_v is nondecreasing and has a zero.

At a zero t, both distances are $t < \rho < \mathrm{inj}(M)$, and the point is distinct from p, q. Thus both distance functions are smooth there, and

$$f_v'(t) = 1 - \langle \nabla r_q, \dot{\gamma}_v(t) \rangle > 0.$$

Indeed, equality would imply that the two unit terminal velocities agree. The two minimizing geodesics also have the same length t, so uniqueness of the geodesic equation backwards would imply $p = q$. This is impossible. A nondecreasing function with two zeros is zero on the interval between them, contradicting this positive derivative. The zero is therefore unique.

For continuity, if $v_j \to v$, any convergent subsequence of $h_p(v_j) \in [0, \rho]$ has limit t satisfying $f_v(t) = 0$. The strict endpoint signs and uniqueness imply $t = h_p(v)$, proving continuity. Every point of E has distance less than ρ from p, so normal coordinates give a unique radial direction to it. The displayed map is therefore a continuous bijection from a compact sphere to the Hausdorff space E, hence a homeomorphism. □

Step 5: glue two closed balls.

Completion of the proof of Theorem 3.1.1. Define

$$D_p = \{x : r_p(x) \leq r_q(x)\}, \qquad D_q = \{x : r_q(x) \leq r_p(x)\}.$$

The covering proposition implies $D_p \subset B_\rho(p)$ and $D_q \subset B_\rho(q)$. By the radial equator lemma,

$$D_p = \{\exp_p(tv) : |v| = 1,\ 0 \leq t \leq h_p(v)\}.$$

In polar coordinates the map from the closed Euclidean unit ball to D_p is

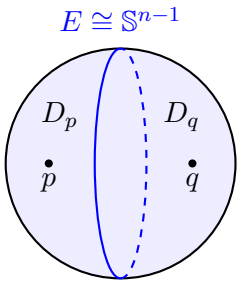
$$H_p(sv) = \exp_p(s\, h_p(v) v), \quad 0 < s \leq 1, \qquad H_p(0) = p.$$

It is continuous at the origin because h_p is bounded. Uniqueness of normal coordi-

nates makes it a bijection, so compactness makes it a homeomorphism. Its boundary is mapped onto E. The same construction gives $D_q \cong \overline{\mathbb{D}}^n$, and

$$M = D_p \cup D_q, \qquad D_p \cap D_q = \partial D_p = \partial D_q = E.$$

Thus M is two closed n-balls glued by a boundary homeomorphism φ. Every homeomorphism of $\mathbb{S}^{n-1}$ extends radially to the ball by $sv \mapsto s\varphi(v)$, so this gluing is homeomorphic to the standard gluing of two hemispheres. Consequently $M \cong \mathbb{S}^n$. □



$$D_p \cong \overline{\mathbb{D}}^n \quad \text{and} \quad D_q \cong \overline{\mathbb{D}}^n$$
$$D_p \cap D_q = \partial D_p = \partial D_q = E$$
$$M \cong \mathbb{S}^n$$

Topological schematic of the two radial domains and their common boundary.

Remark 3.1.17. The equator is parametrized by *all* initial unit directions at p, not just by minimizing geodesics connecting p to q. The final gluing argument is topological: the radial extension of a boundary homeomorphism need not be smooth at the center. A diffeomorphism to the standard sphere requires the stronger differentiable sphere theorem.

3.2 Further Topics in Riemannian Geometry

We discuss Ricci flow on closed manifolds, meaning compact manifolds without boundary. For the isotropic-curvature results and the Brendle–Schoen theorem below, assume dimension $n \geq 4$. This final lecture introduces the proof strategy; the analytic convergence theorems are used as external results.

This lecture explains the analytic ideas behind the differentiable sphere theorem. The main theme is that one should organize curvature assumptions as convex cones of algebraic curvature tensors and then study how those cones behave under Ricci flow.

3.2.1 Algebraic Curvature Tensors

Let $V = \mathbb{R}^n$ be equipped with its standard inner product. For isotropic-curvature statements, assume $n \geq 4$. In this section, the algebraic tensor uses the curvature-operator convention

$$R(X, Y, Z, W) = g(R^{\text{geom}}(X, Y)W, Z),$$

where R^{geom} is the curvature operator defined earlier. This reverses the last two slots of the earlier covariant tensor, so $R(X, Y, X, Y) = K(X \wedge Y)$ for orthonormal X, Y.

Definition 3.2.1 (Algebraic curvature tensor)**.** A $(0, 4)$-tensor $R \in \otimes^4 V^*$ is called an **algebraic curvature tensor** if it satisfies the same symmetries as the Riemann curvature tensor:

$$R(X, Y, Z, W) = -R(Y, X, Z, W) = -R(X, Y, W, Z) = R(Z, W, X, Y)$$

and the first Bianchi identity

$$R(X, Y, Z, W) + R(Y, Z, X, W) + R(Z, X, Y, W) = 0.$$

The vector space of all algebraic curvature tensors on V is denoted by $\text{Curv}(V)$.

The curvature tensor can also be viewed as a symmetric bilinear form on $\Lambda^2 V$ by setting

$$\mathcal{R}(X \wedge Y, Z \wedge W) = R(X, Y, Z, W).$$

Thus curvature conditions can often be interpreted as positivity conditions on $\mathcal{R}$. The elementary implication chain from the blackboard is

$$\mathcal{R} \geq 0 \implies K \geq 0 \implies \text{Ric} \geq 0 \implies \text{Scal} \geq 0.$$

The first implication tests decomposable bivectors, and the last two take traces. In the diagram below, NIC denotes nonnegative isotropic curvature; strict PIC is defined separately. All solid arrows concern weak, nonnegative conditions.

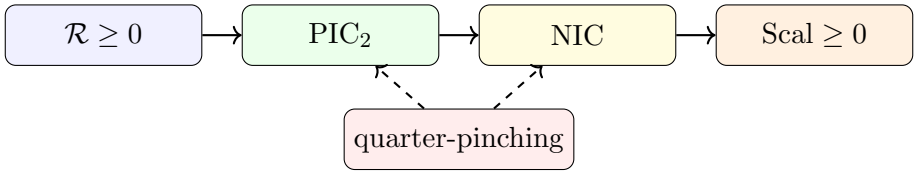
Definition 3.2.2 (Positive curvature operator)**.** An algebraic curvature tensor R has **positive curvature operator** if

$$\mathcal{R}(\varphi, \varphi) > 0 \qquad \text{for every } 0 \neq \varphi \in \Lambda^2 V.$$

Remark 3.2.3. If $\varphi = e_1 \wedge e_2$ is decomposable and e_1, e_2 are orthonormal, then

$$\mathcal{R}(\varphi, \varphi) = R(e_1, e_2, e_1, e_2) = K(\operatorname{span}\{e_1, e_2\}).$$

Hence positive curvature operator implies positive sectional curvature. It is much stronger than $K > 0$ because it tests all elements of $\Lambda^2 V$, not only decomposable ones.



Ricci flow preserves suitable cones and improves the pinching.

3.2.2 Complex Sectional Curvature and Isotropic Curvature

To describe the curvature cones used in the differentiable sphere theorem, we complexify the tangent space. Let

$$V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}.$$

There are two natural extensions of the inner product. The complex bilinear extension is useful for defining isotropic vectors, while the Hermitian product

$$\langle z, w \rangle_{\mathbb{C}} = z \cdot \overline{w}$$

is positive definite and satisfies $\langle z, z \rangle_{\mathbb{C}} \geq 0$.

Definition 3.2.4 (Complex sectional curvature)**.** Let $\Pi \subset V_{\mathbb{C}}$ be a complex two-plane and let z, w be a complex basis for Π. Extending R complex-multilinearly, define

$$K_{\mathbb{C}}(\Pi) = \frac{R(z, w, \overline{z}, \overline{w})}{|z|^2 |w|^2 - |\langle z, w \rangle_{\mathbb{C}}|^2}.$$

This normalized value is independent of the basis: a complex basis change with matrix A multiplies $z \wedge w$ by $\det A$, so both numerator and denominator are multiplied

by $|\det A|^2$. The numerator is real by the pair symmetry of R and complex conjugation. The denominator is positive by the Hermitian Gram determinant. Thus the numerator alone suffices for sign tests, as in the handwritten shorthand.

Definition 3.2.5 (Totally isotropic plane)**.** A complex two-plane $\Pi \subset V_{\mathbb{C}}$ is called **totally isotropic** if the complex bilinear form vanishes on Π, equivalently

$$u \cdot v = 0 \qquad \text{for all } u, v \in \Pi.$$

Definition 3.2.6 (Positive isotropic curvature)**.** An algebraic curvature tensor $R \in \mathrm{Curv}(V)$ satisfies **positive isotropic curvature**, abbreviated PIC, if

$$K_{\mathbb{C}}(\Pi) > 0$$

for every totally isotropic complex two-plane $\Pi \subset V_{\mathbb{C}}$.

If e_1, e_2, e_3, e_4 are orthonormal real vectors and

$$z = e_1 + ie_2, \qquad w = e_3 + ie_4,$$

then $\Pi = \mathrm{span}_{\mathbb{C}}\{z, w\}$ is totally isotropic. A direct computation gives

$$4K_{\mathbb{C}}(\Pi) = R_{1313} + R_{2424} + R_{1414} + R_{2323} - 2R_{1234}.$$

Conversely, choose a Hermitian-orthogonal basis of a totally isotropic plane and rescale its two vectors to have squared norm 2. Writing real and imaginary parts gives the displayed form with four real orthonormal vectors: the isotropic equations give equal lengths and orthogonality within each pair, and the bilinear and Hermitian orthogonality equations give all cross-orthogonalities. Thus the PIC condition can be tested by requiring

$$R_{1313} + R_{2424} + R_{1414} + R_{2323} - 2R_{1234} > 0$$

for every orthonormal four-frame e_1, e_2, e_3, e_4.

Lemma 3.2.7 (Quarter-pinching implies PIC)**.** *Suppose $R \in \mathrm{Curv}(V)$ satisfies*

$$\frac{1}{4} < K(\sigma) \leq 1$$

for every real two-plane $\sigma \subset V$. Then R has positive isotropic curvature.

Proof using Berger's algebraic estimate. For an orthonormal four-frame, the sectional terms satisfy

$$R_{1313}, R_{2424}, R_{1414}, R_{2323} > \frac{1}{4}.$$

Berger's algebraic estimate, $|R_{1234}| \leq \frac{2}{3}(K_{\max} - K_{\min})$ for four orthonormal vectors, is used here as an input. Under the normalization $\frac{1}{4} < K \leq 1$ it gives

$$|R_{1234}| \leq \frac{1}{2}.$$

Hence

$$R_{1313} + R_{2424} + R_{1414} + R_{2323} - 2R_{1234} > 0.$$

This is precisely the isotropic curvature inequality. □

Lemma 3.2.8 (Positive curvature operator implies PIC)**.** *If $\mathcal{R}$ is positive definite, then R has positive isotropic curvature.*

Proof. For the above z, w, write $z \wedge w = \alpha + i\beta$, where

$$\alpha = e_1 \wedge e_3 - e_2 \wedge e_4, \qquad \beta = e_1 \wedge e_4 + e_2 \wedge e_3.$$

Symmetry cancels the imaginary cross terms, giving

$$R(z, w, \overline{z}, \overline{w}) = \mathcal{R}(\alpha, \alpha) + \mathcal{R}(\beta, \beta) > 0.$$

Both real bivectors are nonzero. The same computation with weak inequalities proves that a nonnegative curvature operator has nonnegative complex sectional curvature for every complex two-plane. □

Theorem 3.2.9 (Micallef–Moore; external input)**.** *Let (M^n, g) be compact and simply connected, with $n \geq 4$. If M has positive isotropic curvature, then M is homeomorphic to $\mathbb{S}^n$.*

Remark 3.2.10. This theorem is a topological sphere theorem. It uses the higher-dimensional Poincaré conjecture after showing, via stable minimal two-spheres, that the relevant homotopy groups vanish. It does not by itself identify the smooth structure.

Definition 3.2.11 (PIC_2 condition)**.** Let

$$\pi : V \times \mathbb{R}^2 \longrightarrow V$$

be the projection, and let $\pi^* R$ be the pullback algebraic curvature tensor on $V \times \mathbb{R}^2$. We say that R satisfies the PIC_2 **condition** if $\pi^* R$ has nonnegative isotropic curvature. This is equivalent to nonnegative complex sectional curvature on V. The interior of this cone is described by strictly positive complex sectional curvature on V; it must not be defined by requiring the product tensor to have strictly positive isotropic curvature on every plane, since the flat factors force zero values.

Remark 3.2.12. For $n \geq 4$, the PIC_2 cone can also be tested by

$$R_{1313} + \lambda^2 R_{1414} + \mu^2 R_{2323} + \lambda^2\mu^2 R_{2424} - 2\lambda\mu R_{1234} \geq 0$$

for every orthonormal four-frame and $\lambda, \mu \in [-1, 1]$; strict inequalities characterize its interior. This algebraic equivalence and the implication from strict quarter-pinching are inputs from Brendle–Schoen [2], not consequences merely of the four-frame PIC inequality.

Since V sits inside $V \times \mathbb{R}^2$, the implication

$$\mathrm{PIC}_2 \Longrightarrow \text{nonnegative isotropic curvature}$$

is immediate. The key point in Brendle–Schoen's work is that the Ricci flow preserves this stronger condition and improves it toward constant curvature.

3.2.3 The Heat Equation as a Model

Before introducing Ricci flow, it is useful to recall the heat equation on a compact Riemannian manifold. For $f \in C^\infty(M)$, write

$$\Delta_g f = \mathrm{tr}_g(\mathrm{Hess}\, f) = g^{ij}\left(\frac{\partial^2 f}{\partial x^i \partial x^j} - \Gamma_{ij}^k \frac{\partial f}{\partial x^k}\right).$$

Proposition 3.2.13 (Convergence of heat flow)**.** *Let (M, g) be closed and connected, and let $f(t, x)$ solve*

$$\frac{\partial f}{\partial t} = \Delta_g f.$$

Then $f(t, \cdot)$ converges smoothly to the constant function

$$\overline{f} = \frac{1}{\mathrm{Vol}(M)} \int_M f(0, x)\, d\mu_g.$$

Proof idea. First, integration of the heat equation gives

$$\frac{d}{dt} \int_M f(t, x)\, d\mu_g = \int_M \Delta_g f\, d\mu_g = 0.$$

The preserved integral must be divided by $\mathrm{Vol}(M)$ to obtain the constant limit. The metric here is fixed. The Laplacian is self-adjoint and nonpositive in the sense that

$$\int_M h \Delta_g f\, d\mu_g = -\int_M \langle \nabla h, \nabla f \rangle\, d\mu_g.$$

Hence $-\Delta_g$ has an orthonormal eigenbasis

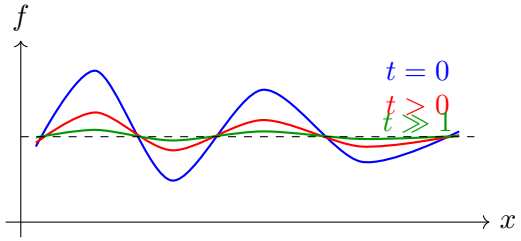
$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots, \qquad -\Delta_g \phi_j = \lambda_j \phi_j.$$

Expanding $f(0,x) = c + \sum_{j\geq 1} a_j \phi_j(x)$ gives

$$f(t,x) = c + \sum_{j\geq 1} a_j e^{-\lambda_j t} \phi_j(x),$$

so all nonconstant modes decay exponentially. For smooth initial data, elliptic spectral estimates give convergence in every C^k norm, hence smooth convergence. The spectral theorem and these regularity estimates are analytic inputs. □

The maximum principle gives the same intuition: at a spatial maximum, $\Delta_g f \leq 0$, so the maximum decreases; at a spatial minimum, $\Delta_g f \geq 0$, so the minimum increases.



heat flow smooths and converges to the average

For a nonlinear heat-type equation

$$\frac{\partial f}{\partial t} = \Delta_g f + F(f),$$

there is a diffusion term and a forcing term. Ricci flow should be viewed as a nonlinear heat equation for the metric, modulo the degeneracy coming from diffeomorphism invariance.

3.2.4 Ricci Flow

Definition 3.2.14 (Ricci flow)**.** Let (M, g_0) be a Riemannian manifold. A smooth one-parameter family of Riemannian metrics $g(t)$ is called a **Ricci flow** if

$$\frac{\partial}{\partial t} g(t) = -2\,\mathrm{Ric}_{g(t)}, \qquad g(0) = g_0.$$

Example 3.2.15 (Einstein metrics)**.** Suppose g_E is Einstein, so

$$\mathrm{Ric}_{g_E} = \lambda g_E$$

for some constant $\lambda \in \mathbb{R}$. The Ricci flow starting at g_E is

$$g(t) = (1 - 2\lambda t) g_E.$$

For a positive constant a, the Levi-Civita connection and the $(0,2)$ Ricci tensor are unchanged under $g_E \mapsto a g_E$. Substituting $g(t) = a(t) g_E$ gives $a' = -2\lambda$ and $a(0) = 1$. The formula is a Riemannian metric only while $1 - 2\lambda t > 0$. Thus a positive Einstein metric shrinks until $T = 1/(2\lambda)$, a Ricci-flat metric is stationary, and a negative Einstein metric expands for all forward time.

Theorem 3.2.16 (Hamilton–DeTurck short-time existence)**.** *If (M, g_0) is closed, then the Ricci flow with initial metric g_0 has a unique smooth solution on a maximal time interval $[0, T)$, where $0 < T \leq \infty$. If $T < \infty$, then*

$$\limsup_{t \nearrow T} \| \operatorname{Rm}_{g(t)} \|_{C^0} = +\infty.$$

Remark 3.2.17. Diffeomorphism invariance makes the equation weakly parabolic. A fixed pullback preserves solutions. A pullback by a family of maps adds a Lie derivative term. DeTurck's trick uses this term to give a strictly parabolic system.

Proposition 3.2.18 (Scalar curvature evolution)**.** *Along Ricci flow,*

$$\frac{\partial}{\partial t} \operatorname{Scal} = \Delta \operatorname{Scal} + 2 |\operatorname{Ric}|^2.$$

In particular,

$$\frac{\partial}{\partial t} \operatorname{Scal} \geq \Delta \operatorname{Scal} + \frac{2}{n} \operatorname{Scal}^2 .$$

Proof idea. The first formula follows by differentiating the curvature under Ricci flow and tracing. The inequality is the Cauchy–Schwarz estimate

$$|\operatorname{Ric}|^2 \geq \frac{1}{n} \operatorname{Scal}^2 .$$

For the finite-time consequence assume M is closed and set $s_0 = \min_M \operatorname{Scal}_{g(0)} > 0$. The scalar ODE solution is

$$\phi' = \frac{2}{n} \phi^2, \quad \phi(0) = s_0, \qquad \phi(t) = \frac{s_0}{1 - 2 s_0 t / n}.$$

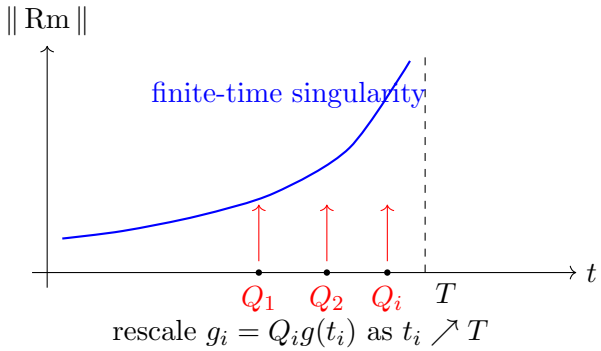
With $u = \operatorname{Scal} - \phi$, the evolution inequality gives

$$\partial_t u \geq \Delta_{g(t)} u + \frac{2}{n} (\operatorname{Scal} + \phi) u.$$

On any compact time interval on which both quantities exist, the maximum principle and $u(0) \geq 0$ imply $u \geq 0$. If the smooth flow existed past $n/(2s_0)$, this lower bound would contradict bounded scalar curvature there. Consequently its maximal time satisfies

$$T \leq \frac{n}{2 s_0} < \infty.$$

The curvature extension criterion gives curvature blow-up at T. This bounds the maximal time; it does not assert equality with the ODE blow-up time. □



3.2.5 Curvature Evolution and Invariant Cones

The curvature tensor evolves by a heat equation plus a quadratic reaction term. In a moving orthonormal frame, one can write the equation schematically as

$$\frac{\partial}{\partial t} R = \Delta R + Q(R).$$

With the slot convention fixed at the start of this section and indices in the moving orthonormal frame, the reaction term is

$$\begin{aligned} Q(R)_{ijkl} = &\sum_{p,q} R_{ijpq} R_{klpq} \\ &+ 2 \sum_{p,q} \big(R_{ipkq} R_{jplq} - R_{iplq} R_{jpkq} \big). \end{aligned}$$

This is the component version of the quadratic term on the blackboard; see [2]. The frame convention matters: ordinary components in fixed coordinates also acquire metric-contraction terms. The associated finite-dimensional equation is $\dot{R} = Q(R)$. The term ΔR is the diffusion part, while $Q(R)$ is the purely algebraic ODE part.

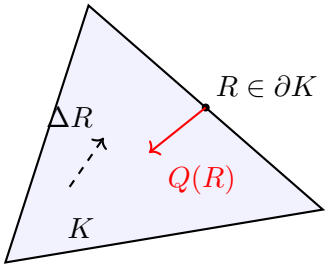
Theorem 3.2.19 (Hamilton's maximum principle for curvature cones)**.** *Work on a closed manifold and identify the metric bundles using moving orthonormal frames. Let Ω_x be closed convex subsets of $\mathrm{Curv}(T_x M)$, invariant under parallel translation. Suppose that for every boundary point $R \in \partial\Omega_x$, the vector $Q(R)$ lies in the closed tangent cone of Ω_x at R, allowing tangency. If*

$$R_{g(0)}(x) \in \Omega_x \qquad \text{for all } x \in M,$$

then along Ricci flow

$$R_{g(t)}(x) \in \Omega_x \qquad \text{for all } t \geq 0$$

as long as the flow exists.



closed convex set
$Q(R)$ points inward or is tangent
$\Longrightarrow$ set is preserved

This reduces many curvature-preservation questions to finite-dimensional algebra: find a closed convex cone $K \subset \mathrm{Curv}(V)$ such that $Q(R)$ points into K along ∂K.

3.2.6 Pinching Cones and the Differentiable Sphere Theorem

Definition 3.2.20 (Volume-normalized Ricci flow)**.** The normalized equation on a closed n-manifold is

$$\partial_t g = -2\,\mathrm{Ric} + \frac{2}{n}\overline{S}\,g, \qquad \overline{S} = \frac{\int_M \mathrm{Scal}\; d\mu_g}{\mathrm{Vol}_g(M)}.$$

The density variation is $\partial_t d\mu_g = (-\mathrm{Scal} + \overline{S}) d\mu_g$, whose integral is zero. Thus the total volume is preserved. This equation is obtained from unnormalized flow by a time-dependent constant scaling and a change of time variable.

Theorem 3.2.21 (Brendle–Schoen; external input [2])**.** *Let (M^n, g_0) be closed, simply connected, with $n \geq 4$, and strictly pointwise quarter-pinched. That is, for each point $p \in M$ there is a number $a(p) > 0$ such that*

$$\frac{a(p)}{4} < K(\sigma) \leq a(p) \qquad \textit{for every } \sigma \subset T_pM.$$

Then the volume-normalized Ricci flow starting at g_0 exists for all forward time and converges smoothly to a metric of constant positive sectional curvature. Consequently M is diffeomorphic to $\mathbb{S}^n$.

Proof scheme. The proof may be organized as follows.

1. Start Ricci flow from the quarter-pinched metric g_0.
2. Strict pointwise quarter-pinching places the initial curvature in the interior of the PIC_2 cone.
3. Brendle–Schoen's invariant cone estimates show that this condition is preserved and improves under the curvature ODE.

4. Positive initial scalar curvature forces finite maximal time for the unnormalized flow. The pinching and convergence theorem supplies smooth convergence after volume normalization and time reparametrization. Equivalently, suitably normalized slices $a_i g(t_i)$, with $t_i \nearrow T$, approach a round metric. The normalization factors are not arbitrary, and neither subsequential compactness nor roundness follows from finite-time blow-up alone.

5. By the Killing–Hopf classification, a simply connected complete manifold with constant positive sectional curvature is isometric to a round sphere after scaling.

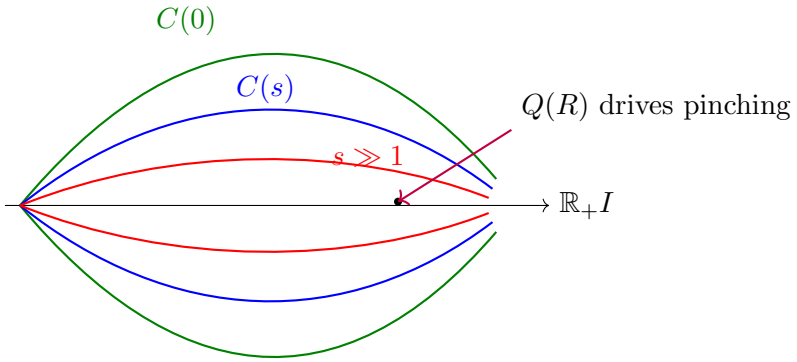
□

The Böhm–Wilking pinching construction gives a geometric way to visualize the last part of the argument. The space $\mathrm{Curv}(V)$ contains a distinguished ray

$$\mathbb{R}_+ I,$$

where I is the curvature tensor of the round metric of sectional curvature 1. Pinching means that the curvature tensor is forced closer and closer to this ray.

Remark 3.2.22 (Pinching families: a schematic picture). In the pinching constructions used in Ricci-flow convergence theorems, suitable invariant curvature cones become progressively narrower around the positive constant-curvature ray. After normalizing the scalar curvature, this expresses convergence toward the round tensor I. The diagram below illustrates this mechanism only: it is not a precise statement of a pinching-family theorem. The admissible initial cones, strict positivity assumptions, and invariant pinching sets require the full hypotheses of the relevant convergence theorem. In particular, an arbitrary compact subset of a weakly nonnegative cone need not satisfy these hypotheses.



nested invariant cones squeeze curvature toward constant curvature

Thus Ricci flow upgrades the topological conclusion to a diffeomorphism by producing a round metric.

Appendix A

Topological Background for Riemannian Geometry

This appendix collects the topology used in the main text. The manifold axioms support local-to-global constructions; compactness controls geodesic limits; covering spaces explain the global conclusions of Cartan–Hadamard and Synge. Throughout, a manifold is Hausdorff, second countable, finite-dimensional, and without boundary unless stated otherwise.

A.1 Open Sets, Separation, and Countability

Definition A.1.1 (Topology and continuity)**.** A topology $\mathcal{T}$ on a set X contains $\emptyset$ and X, is closed under arbitrary unions, and is closed under finite intersections. Its elements are open sets; their complements are closed. A map $f : X \to Y$ is continuous if $f^{-1}(V)$ is open for every open $V \subset Y$. A homeomorphism is a continuous bijection with continuous inverse.

The subspace topology on $A \subset X$ consists of sets $A \cap U$ with U open in X. The closure $\overline{A}$ is the smallest closed set containing A, and $\partial A = \overline{A} \setminus \mathrm{int}(A)$.

Definition A.1.2 (Separation conditions)**.** For distinct points $x, y \in X$:

- T_0 means that some open set contains one of x, y but not the other.
- T_1 means that there is an open set containing x but not y, and an open set containing y but not x. Equivalently, every singleton is closed.
- T_2, or Hausdorff, means that x, y have disjoint open neighborhoods.

A space is regular if a point and a closed set not containing it have disjoint open neighborhoods. It is normal if two disjoint closed sets have disjoint open neighborhoods. We use the conventions $T_3 = T_1 + \text{regular}$ and $T_4 = T_1 + \text{normal}$.

Thus $T_4 \Rightarrow T_3 \Rightarrow T_2 \Rightarrow T_1 \Rightarrow T_0$. These are properties of a topology, not additional axioms defining a topology. In particular, T_1 must not be confused with the stronger Hausdorff condition.

Definition A.1.3 (Bases and countability)**.** A basis $\mathcal{B}$ is a collection of open sets such that every open set is a union of members of $\mathcal{B}$. A local basis at x consists of neighborhoods B of x such that every neighborhood of x contains one of them. A space is first countable if every point has a countable local basis, and second countable if it has a countable basis for the whole topology.

Second countability implies first countability. We avoid calling these conditions C_1, C_2, since that notation can be confused with differentiability classes.

Proposition A.1.4 (Consequences used in the notes)**.** *On a Hausdorff space limits of convergent sequences are unique, and compact subsets are closed. A second countable space is Lindelöf: every open cover has a countable subcover. On a first countable space, a point lies in $\overline{A}$ precisely when it is the limit of a sequence in A.*

Proof. Disjoint neighborhoods exclude two distinct limits. To separate a point from a compact set, separate it from each point of that set and use a finite subcover. For the Lindelöf assertion, choose, for each basis element contained in some member of the cover, one such member; the chosen countable family still covers. For the last assertion choose a nested countable local basis B_j at x and points $a_j \in A \cap B_j$. □

Remark A.1.5 (Continuous versus smooth separation). Urysohn's lemma says that disjoint closed subsets of a normal space can be separated by a continuous function with values in $[0, 1]$. It does not produce a smooth function. Smooth cutoffs require charts and the construction in Appendix B.

A.2 Compactness, Completeness, and Local Finiteness

Definition A.2.1 (Compactness and properness)**.** A space is compact if every open cover has a finite subcover. A map is proper if the inverse image of every compact set is compact. A metric space is proper if every closed bounded set is compact, and complete if every Cauchy sequence converges.

In metric spaces, compactness is equivalent to sequential compactness, and also to completeness plus total boundedness. Total boundedness means that for every $\varepsilon > 0$ finitely many radius-ε balls cover the space. Completeness alone does not imply compactness, as $\mathbb{R}^n$ shows.

Proposition A.2.2 (Compact-to-Hausdorff principle)**.** *A continuous bijection from a compact space to a Hausdorff space is a homeomorphism.*

Proof. A closed subset of the domain is compact; its image is compact and therefore closed. The map is thus closed, so its inverse is continuous. □

This is the principle used to identify the equator and the two closed radial domains in Section 3.1.

Definition A.2.3 (Locally finite families and refinements)**.** A family $\{A_i\}$ is locally finite if every point has a neighborhood meeting only finitely many A_i. An open cover $\{V_j\}$ refines $\{U_i\}$ if each V_j lies in some U_i. A space is paracompact if every open cover has a locally finite open refinement.

Pointwise finiteness is weaker than local finiteness. For example, the singletons $\{1/j\} \subset \mathbb{R}$ form a pointwise finite family, but every neighborhood of 0 meets infinitely many of them. Local finiteness, not merely pointwise finiteness, makes a sum of smooth functions smooth.

Proposition A.2.4 (Compact exhaustions of manifolds)**.** *Every manifold M has compact sets K_j satisfying*

$$K_j \subset \operatorname{int} K_{j+1}, \qquad M = \bigcup_{j \geq 1} \operatorname{int} K_j.$$

Proof. Charts give relatively compact open neighborhoods: pull back Euclidean balls whose closures stay inside a chart. Second countability gives a countable cover $\{U_j\}$ by such neighborhoods. Start with $K_1 = \overline{U}_1$. Having chosen K_j, cover the compact set $K_j \cup \overline{U}_{j+1}$ by finitely many relatively compact open neighborhoods and let K_{j+1} be the union of their compact closures. The union of those open neighborhoods lies in $\operatorname{int} K_{j+1}$, giving the required inclusion and exhaustion. □

Appendix B uses this exhaustion to prove paracompactness directly, without first constructing a Riemannian metric. This avoids a circular argument when partitions of unity are then used to construct that metric. The Urysohn metrization theorem also applies: a regular second countable Hausdorff space is metrizable.

Remark A.2.5 (What Hopf–Rinow adds). For a connected Riemannian manifold, metric completeness is equivalent to geodesic completeness and implies properness of the Riemannian distance. Thus closed bounded subsets are compact. These conclusions use the finite-dimensional Riemannian setting; they are not general facts about complete metric spaces.

A.3 Connectedness, Quotients, and Covering Spaces

Definition A.3.1 (Paths and simple connectivity)**.** A path is a continuous map $[0, 1] \to X$. A space is path connected if any two points can be joined by a path.

It is simply connected if it is path connected and every loop can be continuously contracted to a point. The fundamental group $\pi_1(X, p)$ consists of based loops modulo homotopy fixing the base point, with multiplication given by concatenation.

Charts give neighborhoods that are path connected. Thus path components of a manifold are open, and a connected manifold is path connected. Using coordinate balls along a path produces a piecewise smooth path, as required in the definition of Riemannian distance.

Definition A.3.2 (Quotient topology)**.** If $\sim$ is an equivalence relation on X, the quotient topology on $X/\sim$ is defined by: V is open if and only if its inverse image under the quotient map is open. A map out of the quotient is continuous precisely when its composition with the quotient map is continuous.

This construction underlies $\mathbb{RP}^n = \mathbb{S}^n/\{\pm 1\}$ and the Hopf quotient $\mathbb{CP}^m = \mathbb{S}^{2m+1}/\mathbb{S}^1$. A quotient need not be a manifold in general. A free proper smooth Lie group action gives a smooth quotient and a submersion; an invariant Riemannian metric then descends through horizontal tangent spaces, as in the discussion of Riemannian submersions.

Definition A.3.3 (Covering map)**.** A surjective continuous map $\pi : \widetilde{X} \to X$ is a covering map if every $x \in X$ has an open neighborhood U for which $\pi^{-1}(U)$ is a disjoint union of open sets, each mapped homeomorphically onto U. Such a neighborhood is evenly covered.

Paths and homotopies lift uniquely after fixing the initial lift. Every connected manifold has a simply connected universal cover. Smooth charts lift to this cover, and a Riemannian metric lifts as $\widetilde{g} = \pi^* g$, making π a local isometry. If the base metric is complete, the lifted metric is complete: geodesics project to complete geodesics and extend by path lifting.

Definition A.3.4 (Deck transformation)**.** A deck transformation of $\pi : \widetilde{M} \to M$ is a diffeomorphism $h : \widetilde{M} \to \widetilde{M}$ with $\pi \circ h = \pi$. For a connected universal cover its deck group is isomorphic to $\pi_1(M)$ after choosing a base point and a convention for path concatenation. With the lifted metric,

$$h^* \widetilde{g} = h^* \pi^* g = (\pi \circ h)^* g = \widetilde{g},$$

so deck transformations are isometries. A deck transformation fixing one point is the identity: it and the identity map give lifts of every path from that point with the same initial value. This free action is the fact used in Cartan's torsion theorem in Section 2.4.

Proposition A.3.5 (Consequences of covering theory)**.** *A connected covering of a simply connected manifold has one sheet, hence is a homeomorphism (a diffeomorphism for a smooth covering). Covering maps induce isomorphisms on π_k for $k \geq 2$. A subgroup of index two in $\pi_1(M)$ corresponds to a connected double cover.*

The first statement follows by lifting paths from a base point; simple connectivity makes the resulting lift independent of the path. The higher-homotopy statement follows by lifting maps and homotopies from simply connected spheres. For a nonorientable connected manifold the orientation double cover is connected; Synge's theorem can show that this cover is simply connected, yielding $\pi_1(M) \cong \mathbb{Z}/2\mathbb{Z}$ in the even-dimensional positive-curvature case.

A local diffeomorphism is not automatically a covering map: for example, $(0, 4\pi) \to \mathbb{S}^1$, $t \mapsto e^{it}$, is surjective and locally invertible but not a covering. The completeness hypothesis in Cartan–Hadamard supplies essential global information. Once the universal cover is diffeomorphic to $\mathbb{R}^n$, it is contractible, and hence $\pi_k(M) = 0$ for every $k \geq 2$.

A.4 Homeomorphism, Homotopy, and Gluing Balls

A map $f : X \to Y$ is a homotopy equivalence if there is a map $g : Y \to X$ such that $g \circ f$ and $f \circ g$ are homotopic to the respective identity maps. Homotopy equivalence is weaker than homeomorphism, which in turn is weaker than diffeomorphism for smooth manifolds. A homotopy sphere is a closed manifold homotopy equivalent to a sphere; simple connectivity alone does not make a manifold a homotopy sphere.

Proposition A.4.1 (Two-ball gluing)**.** *If a space is obtained by gluing two closed n-balls along a homeomorphism of their boundary spheres, then it is homeomorphic to $\mathbb{S}^n$.*

Proof. Any boundary homeomorphism $\varphi : \mathbb{S}^{n-1} \to \mathbb{S}^{n-1}$ extends to the closed ball by $sv \mapsto s\varphi(v)$ and $0 \mapsto 0$. Changing coordinates on one ball by this extension reduces the gluing to the identity, which produces the two hemispheres of $\mathbb{S}^n$. □

This radial extension is continuous but need not be smooth at the center. Accordingly, the argument in Section 3.1 proves a topological sphere theorem, and does not by itself settle the smooth structure.

Appendix B

Partitions of Unity and Local-to-Global Constructions

Partitions of unity justify several constructions used earlier: the existence of Riemannian metrics, localization of sections and connections, and integration in charts. This appendix completes the construction from compact exhaustions and Euclidean bump functions. It follows the standard approach in differential geometry; see Merry [6].

B.1 Supports and Smooth Cutoff Functions

Definition B.1.1 (Support and cutoff)**.** For a function $f : M \to \mathbb{R}$, its support is

$$\operatorname{supp} f = \overline{\{x : f(x) \neq 0\}}.$$

A cutoff for $K \subset U$, with U open, is a smooth function $\chi : M \to [0,1]$ equal to 1 on a neighborhood of K and satisfying $\operatorname{supp} \chi \subset U$. We write $C_c^\infty(U)$ for smooth functions with compact support contained in U.

Lemma B.1.2 (Euclidean bump function)**.** *For $0 < r < R$, there is a smooth function $\beta : \mathbb{R}^n \to [0,1]$ equal to 1 on $\overline{B}_r(0)$ and zero outside $B_R(0)$.*

Proof. Set $\eta(t) = e^{-1/t}$ for $t > 0$ and $\eta(t) = 0$ for $t \leq 0$. All right derivatives at 0 vanish, so η is smooth. Then

$$\beta(x) = \frac{\eta(R^2 - |x|^2)}{\eta(R^2 - |x|^2) + \eta(|x|^2 - r^2)}$$

has the stated properties, since the denominator never vanishes. Its support is $\overline{B}_R(0)$; to obtain support inside a prescribed open ball, choose R strictly smaller than that ball's radius. □

Proposition B.1.3 (Compact-set cutoff)**.** *If $K \subset U \subset M$, K is compact, and U is open, then a cutoff $\chi \in C_c^\infty(U)$ exists.*

Proof. For each point of K, choose nested coordinate balls with compact closures in U. Pull back the Euclidean bump function and extend it by zero; its support stays inside the chart, so this extension is smooth. Choose finitely many such functions $\beta_1, \dots, \beta_N$ whose regions of value 1 cover K. Then

$$\chi = 1 - \prod_{i=1}^{N} (1 - \beta_i)$$

is smooth, lies in $[0, 1]$, and is 1 on an open neighborhood of K. Its support is contained in the finite union of the compact supports of the β_i, which lies in U. □

Remark B.1.4. This proof uses only charts and compactness, not a partition of unity. It can therefore be used in the construction of partitions below. Merely requiring $f = 0$ outside U does not ensure $\operatorname{supp} f \subset U$: the closure in the definition of support matters.

B.2 A Locally Finite Refinement by Coordinate Neighborhoods

Proposition B.2.1 (Nested locally finite refinement)**.** *For any open cover $\{U_a\}_{a \in A}$ of M, there are countably many open sets V_i, W_i and indices $a(i)$ such that*

$$M = \bigcup_i V_i, \qquad \overline{V}_i \subset W_i, \qquad \overline{W}_i \text{ is compact and contained in } U_{a(i)},$$

and the family $\{W_i\}$ is locally finite. The sets may be chosen inside coordinate neighborhoods.

Proof. Take a compact exhaustion $K_j \subset \operatorname{int} K_{j+1}$ from Proposition A.2.4, and set $K_0 = K_{-1} = \emptyset$. The compact shells

$$C_j = K_j \setminus \operatorname{int} K_{j-1}$$

cover M. Each point of C_j has nested coordinate neighborhoods $V \Subset W$ with

$$\overline{W} \subset U_a \cap \left(\operatorname{int} K_{j+1} \setminus K_{j-2}\right)$$

for some a. This is possible because the set on the right is an open neighborhood of the point. Choose finitely many such V covering each C_j and retain their corresponding W.

The resulting countable family covers M. To check local finiteness, fix $x \in \operatorname{int} K_N$. Neighborhoods chosen for shells with $j \geq N+2$ avoid $K_{j-2} \supset K_N$, and therefore do not meet $\operatorname{int} K_N$. Only finitely many earlier shells remain, with finitely many neighborhoods for each. Thus $\{W_i\}$ is locally finite. □

Here $V \Subset W$ means that $\overline{V}$ is compact and contained in W. In particular, every manifold is paracompact. This direct proof is sufficient for the notes and avoids a detour through the more general theorem that every metric space is paracompact.

B.3 Construction of a Subordinate Partition

Definition B.3.1 (Smooth partition of unity)**.** A smooth partition of unity on M is a family $\{\lambda_i\}$ of smooth functions $M \to [0,1]$ whose supports form a locally finite family and for which $\sum_i \lambda_i = 1$. It is subordinate to $\{U_i\}$ when $\operatorname{supp} \lambda_i \subset U_i$ for every i.

The sum is finite on a neighborhood of each point, not necessarily a finite sum globally.

Theorem B.3.2 (Existence of partitions of unity)**.** *Every open cover of a smooth manifold admits a subordinate smooth partition of unity. After passing to a countable locally finite refinement, the partition can be chosen with compact supports.*

Proof. Choose V_i, W_i as in Proposition B.2.1. By Proposition B.1.3, choose $\beta_i \in C_c^\infty(W_i)$ with $0 \leq \beta_i \leq 1$ and $\beta_i = 1$ near $\overline{V}_i$. The locally finite sum

$$b(x) = \sum_i \beta_i(x)$$

is smooth and strictly positive because the V_i cover M. Hence

$$\lambda_i(x) = \frac{\beta_i(x)}{b(x)}$$

defines a smooth partition with compact support in $W_i \subset U_{a(i)}$.

To index a partition by the original cover, set

$$\mu_a = \sum_{i : a(i) = a} \lambda_i.$$

These sums are locally finite, and $\sum_a \mu_a = 1$. A locally finite union of closed sets is closed; consequently

$$\operatorname{supp} \mu_a \subset \bigcup_{i : a(i) = a} \operatorname{supp} \lambda_i \subset U_a.$$

The grouped supports are still locally finite, though they need not be compact. This proves both forms of the theorem. □

Corollary B.3.3 (Cutoffs for a closed set)**.** *For a closed set $K \subset U$ with U open, there exists a smooth cutoff $\chi : M \to [0,1]$ supported in U and equal to 1 near K. Compact support is guaranteed when K is compact, but cannot be required in general for a noncompact K.*

Proof. Use a partition $\mu_U + \mu_{M\setminus K} = 1$ subordinate to the two-set cover $\{U, M \setminus K\}$. Since the closed support of $\mu_{M\setminus K}$ misses K, that function vanishes on a neighborhood of K. Thus $\chi = \mu_U$ works. □

B.4 Applications to Metrics, Sections, and Connections

Proposition B.4.1 (Existence of a Riemannian metric)**.** *Every smooth manifold admits a Riemannian metric.*

Proof. Take a coordinate cover $\{U_i\}$, pull back the Euclidean inner product to obtain a metric g_i on each U_i, and choose a subordinate partition $\{\lambda_i\}$. Extend $\lambda_i g_i$ by zero outside U_i and set

$$g = \sum_i \lambda_i g_i.$$

Each extension is smooth because $\operatorname{supp} \lambda_i \subset U_i$, and the sum is locally finite. For $0 \neq v \in T_pM$, at least one coefficient $\lambda_i(p)$ is positive, so $g_p(v,v) = \sum_i \lambda_i(p)(g_i)_p(v,v) > 0$. □

Proposition B.4.2 (Extension near a point)**.** *Let s be a smooth section of a vector bundle E over an open set U, and let $K \subset U$ be compact. There is a global smooth section agreeing with s near K.*

Proof. Choose $\chi \in C_c^\infty(U)$ equal to 1 near K. Define the extension to be χs on U and zero outside U. Near every point outside the support it vanishes identically, so it is smooth across the boundary of U. □

This explains the cutoff argument proving locality of a connection. It also justifies replacing local vector fields by global ones when a calculation concerns only a neighborhood of a fixed point. Extension of a section is not generally unique.

Proposition B.4.3 (Existence of connections)**.** *Every smooth vector bundle admits a connection.*

Proof. Choose local trivializations and their coordinatewise flat connections $\nabla^{(i)}$. With a subordinate partition define

$$\nabla_X s = \sum_i \lambda_i \nabla^{(i)}_X s,$$

where the weighted local terms are extended by zero. Local finiteness gives smoothness and linearity. The Leibniz rule follows from

$$\nabla_X(fs) = \sum_i \lambda_i \big(X(f)s + f\nabla_X^{(i)} s\big) = X(f)s + f\nabla_X s.$$

The identity $\sum_i \lambda_i = 1$ is essential here. □

Partitions of unity preserve properties defined by convex combinations, such as positivity of a metric. They do not preserve arbitrary nonlinear constraints: averaging local metrics does not generally preserve a curvature bound, and averaging local solutions of a nonlinear PDE does not give a solution.

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